

CONTAINERIZATION

&

DEVOPS

LAB FILE



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Lab Experiments – Containerization & DevOps

This section contains all the **laboratory experiments** performed as part of the **Containerization and DevOps** coursework.

Each experiment is designed to provide hands-on exposure to core DevOps concepts, tools, and practices through practical implementation.

The experiments are organized sequentially and focus on building a strong foundation in containerization, automation, and deployment workflows.

List of Experiments

The table below provides an overview of the lab experiments included in this repository:

Experiment No.	Title	Result
Experiment-1	VM vs Container - Utilization	The experiment demonstrates that containers are significantly more lightweight and resource-efficient compared to virtual machines
Experiment-2	Docker Installation, Configuration, and Running Images	Docker images were successfully pulled, containers executed, and lifecycle commands performed.
Experiment-3	Docker Deploy NGINX Using Different Base Images and Comparing Image Layers	Use Official nginx on production deployment whereas Alpine-Based Image with Microservices.
Experiment-4	Containerizing Applications with Dockerfile	Containerizing prevents from dependency issue or version outages
Experiment-5	Docker - Volumes, Environment Variables, Monitoring & Networks	Essential Docker features for building, configuring, and managing production-ready containerized applications.
Experiment-6	Docker Run vs Docker-Compose	Compose is an easy way to create containers
Experiment-7	CI/CD via Jenkins	Automate Build via Jenkins

Experiment No.	Title	Result
Experiment-9	Ansible	Automate Infrastructure Orchestration & Updates
Experiment-10	Working with SonarQube	Generate Code Quality Report via SonarQube
Experiment-11	Orchestration with Docker Swarm	Compose defines the application. Swarm runs it reliably.
Experiment-12	Kubernetes	Kubernetes automatically decides where to run your app

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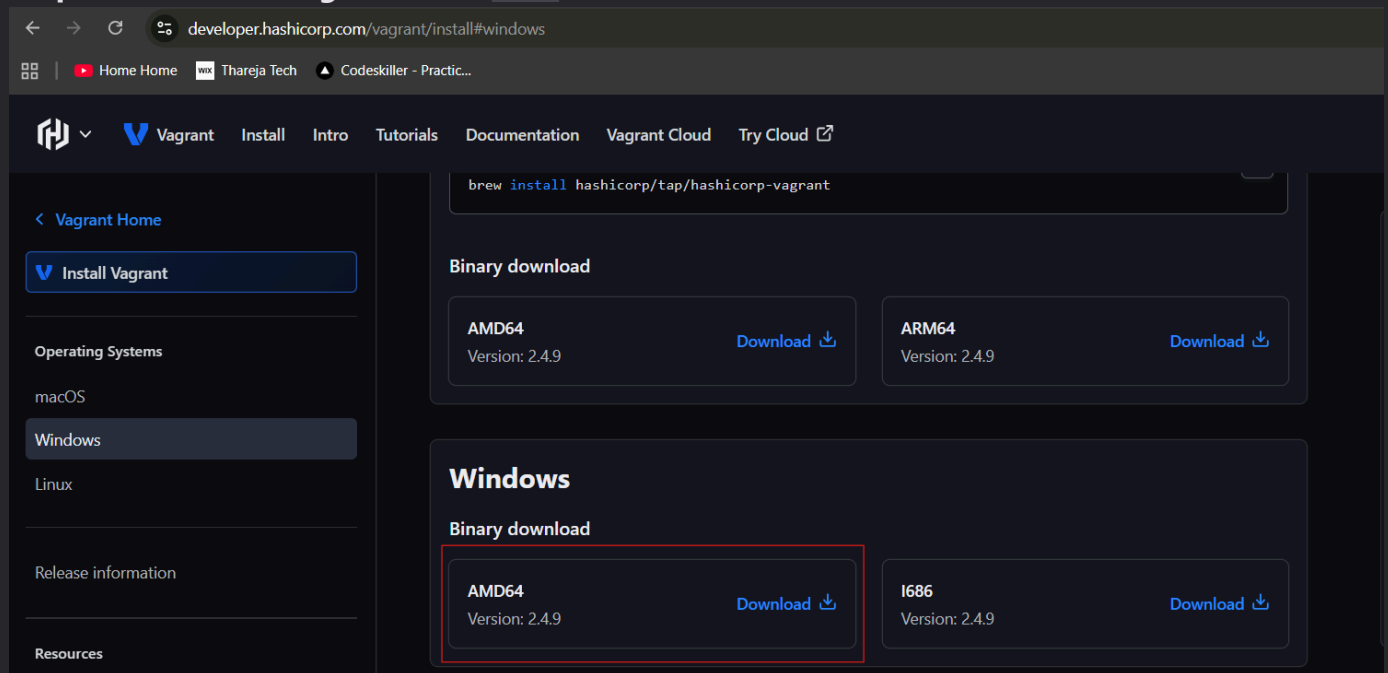
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Experiment 1 : VM vs Container

PART-A:- VIRTUAL MACHINE (WINDOWS)

Step 1: Download Virtual Box from [here](#)

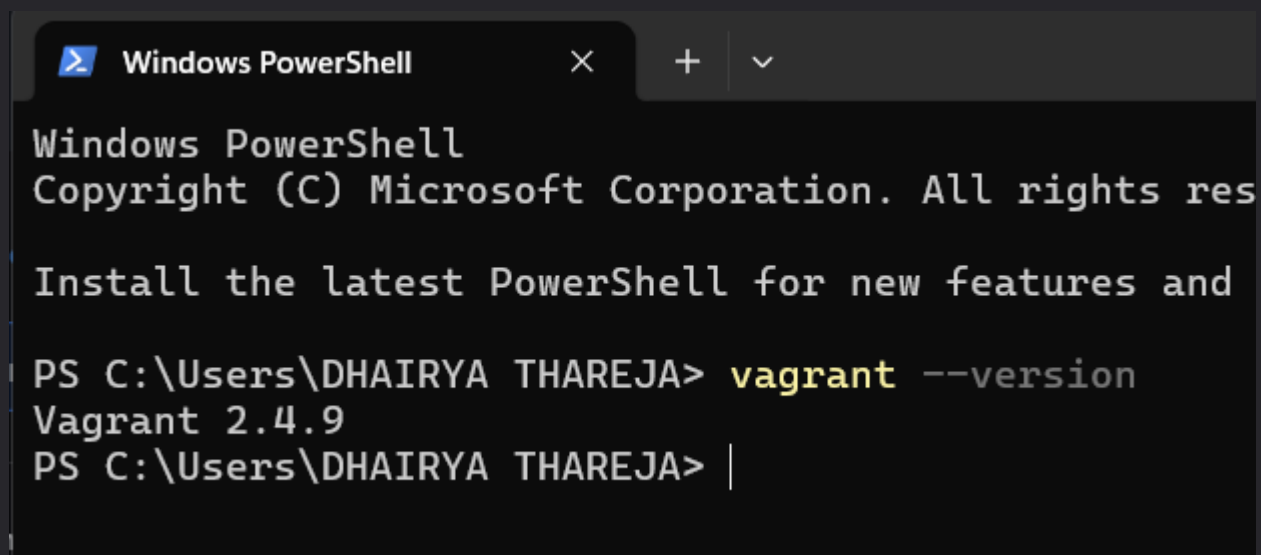
Step 2: Download Vagrant from [here](#)



The screenshot shows the Vagrant website's installation page for Windows. The browser address bar displays 'developer.hashicorp.com/vagrant/install#windows'. The page features a sidebar on the left with navigation links: 'Vagrant Home', 'Install Vagrant', 'Operating Systems' (with sub-links for macOS, Windows, and Linux), 'Release information', and 'Resources'. The main content area includes a terminal snippet: `brew install hashicorp/tap/hashicorp-vagrant`. Below this, there are two sections for binary downloads. The first section, 'Binary download', offers downloads for 'AMD64' (Version: 2.4.9) and 'ARM64' (Version: 2.4.9). The second section, 'Windows', also offers binary downloads for 'AMD64' (Version: 2.4.9) and 'I686' (Version: 2.4.9). The 'AMD64' download link in the Windows section is highlighted with a red rectangle.

Step 3: To verify the installation we will check the version via following command

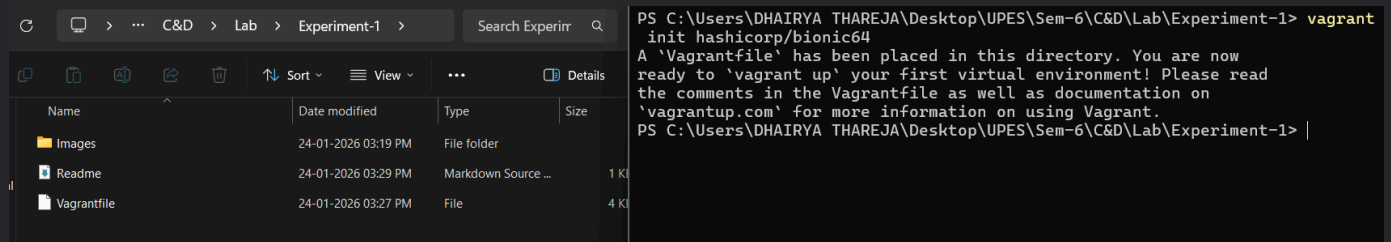
```
vagrant --version
```



The screenshot shows a Windows PowerShell terminal window. The title bar reads 'Windows PowerShell'. The terminal output displays the Windows PowerShell copyright notice and a message to install the latest PowerShell. The user then enters the command `vagrant --version`, and the output shows 'Vagrant 2.4.9'.

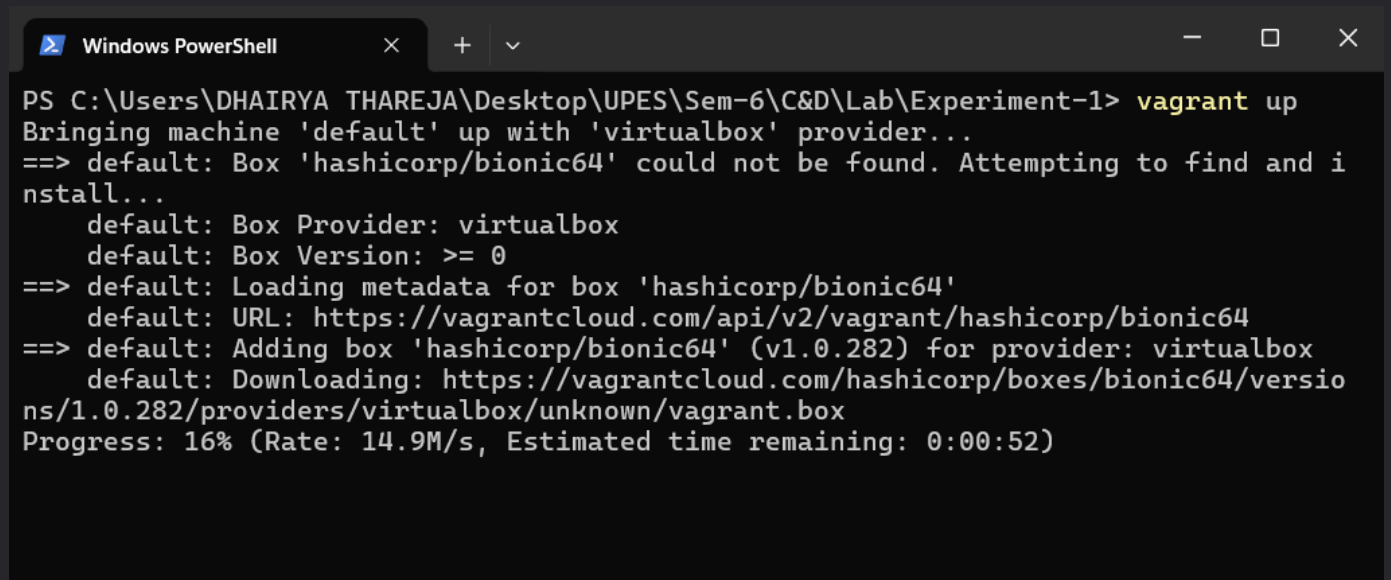
Step 4: Initialize Vagrant with Ubuntu box:

```
vagrant init hashicorp/bionic64
```



Step 5: Start the VM:

```
vagrant up
```



Step 6: Access the VM:

```
vagrant ssh
```

```
vagrant@vagrant: ~  
PS C:\Users\DHAIRYA THAREJA\Desktop\UPES\Sem-6\C&D\Lab\Experiment-1> vagrant ssh  
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 4.15.0-58-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
System information as of Sat Jan 24 10:06:28 UTC 2026  
  
System load:  0.06          Processes:            93  
Usage of /:   2.5% of 61.80GB Users logged in:      0  
Memory usage: 11%          IP address for eth0: 10.0.2.15  
Swap usage:   0%  
  
* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s  
  just raised the bar for easy, resilient and secure K8s cluster deployment.  
  
  https://ubuntu.com/engage/secure-kubernetes-at-the-edge  
  
0 packages can be updated.  
0 updates are security updates.  
  
vagrant@vagrant:~$ |
```

Step 7: Install Nginx inside VM

```
sudo apt update  
sudo apt install -y nginx  
sudo systemctl start nginx
```

Step 8: Verify Nginx

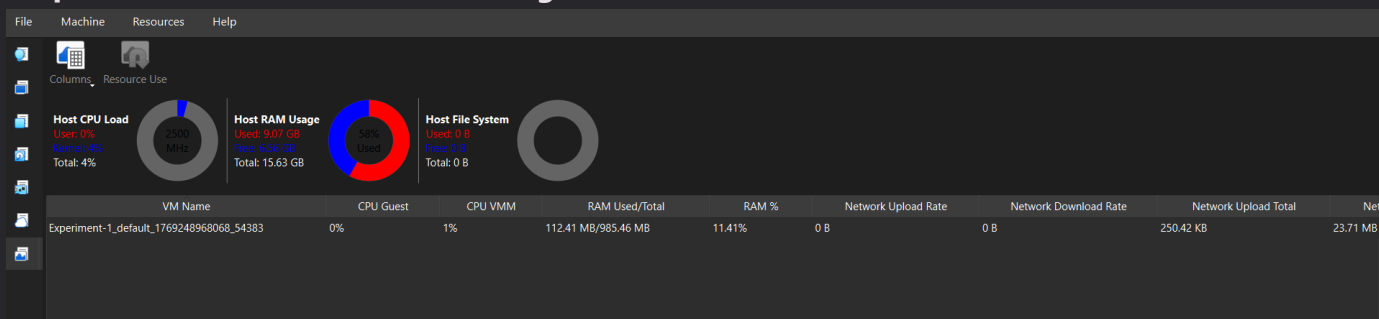
```
curl localhost
```

```
vagrant@vagrant: ~$ curl localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
  body {
    width: 35em;
    margin: 0 auto;
    font-family: Tahoma, Verdana, Arial, sans-serif;
  }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
vagrant@vagrant:~$ |
```

Step 9: Utilization Matrix In Running State

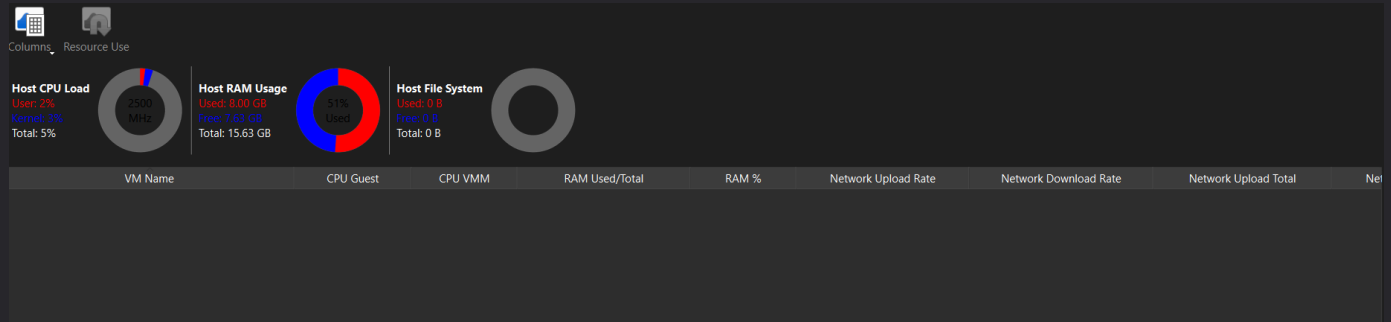


Step 10: Stop VM

```
vagrant halt
```

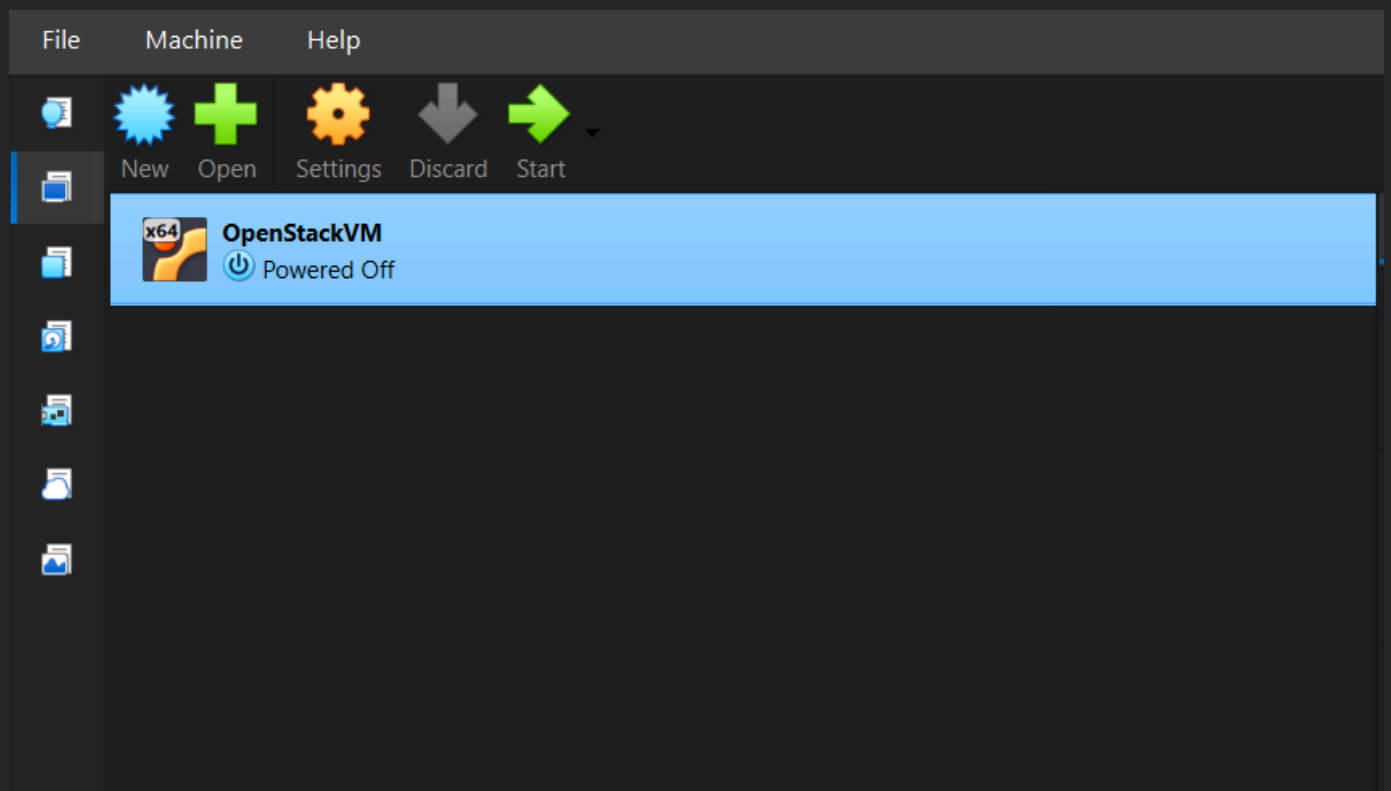
```
Windows PowerShell
PS C:\Users\DHAIRYA THAREJA\Desktop\UPES\Sem-6\C&D\Lab\Experiment-1> vagrant halt
==> default: Attempting graceful shutdown of VM...
PS C:\Users\DHAIRYA THAREJA\Desktop\UPES\Sem-6\C&D\Lab\Experiment-1> |
```

Step 11: Utilization Matrix In Stop State



Step 12: Remove VM

```
vagrant destroy
```



PART-B:- CONTAINERS USING WSL (WINDOWS)

Step 1: Install WSL

```
wsl --install
```

Reboot the system after installation.

Step 2: Install Ubuntu on WSL

```
wsl --install -d Ubuntu
```

```
Windows PowerShell
PS C:\Users\DHAIRYA THAREJA\Desktop\UPES\Sem-6\C&D\Lab\Experiment-1> wsl --install -d Ubuntu
A distribution with the supplied name already exists. Use --name to chose a different name.
Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS
PS C:\Users\DHAIRYA THAREJA\Desktop\UPES\Sem-6\C&D\Lab\Experiment-1> |
```

Step 3: Install Docker Engine inside WSL

```
sudo apt update
sudo apt install -y docker.io
sudo systemctl start docker
sudo usermod -aG docker $USER
```

Logout and login again to apply group changes.

Step 4: Verify Docker Installation

```
docker --version
```

```
dhairya@LAPTOP-H7SV3T8B: x + v
PS C:\Users\DHAIRYA THAREJA\Desktop\UPES\Sem-6\C&D\Lab\Experiment-1> wsl
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Lab/Experiment-1$ sudo docker --version
[sudo] password for dhairya:
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Lab/Experiment-1$ |
```

Step 5: Pull Ubuntu Image

```
docker pull ubuntu
```

```
dhairya@LAPTOP-H7SV3T8B: x + v
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Lab/Experiment-1$ sudo docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
```

Step 6: Run Ubuntu Container with Nginx

```
docker run -d -p 8080:80 --name nginx-container nginx
```

```
dhairya@LAPTOP-H7SV3T8B: x + v
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Lab/Experiment-1$ sudo docker run -d -p 8080:80 --name Exp1-Container nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
700146c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Waiting
10b68cfefe1: Download complete
57f0dd1bef2: Download complete
eaf8753feae0: Download complete
```

Step 7: Verify Nginx in Container

```
curl localhost:8080
```

```
dhairya@LAPTOP-H7SV3T8B: x + v
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Lab/Experiment-1$ curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Lab/Experiment-1$ |
```

Step 8: Container Observation Commands

```
docker stats

free -h
```

```
dhairya@LAPTOP-H7SV3T8B: x + v
CONTAINER ID   NAME          CPU %       MEM USAGE / LIMIT   MEM %      NET I/O       BLOCK I/O     PIDS
b156e18855a0   Exp1-Container 0.00%       13.2MiB / 7.57GiB   0.17%      2.65kB / 2.68kB  8.19kB / 12.3kB 17
```

Parameters to Compare

Parameter	Virtual Machine	Container
Boot Time	High	Very Low
RAM Usage	High	Low
CPU Overhead	Higher	Minimal

Parameter	Virtual Machine	Container
Disk Usage	Larger	Smaller
Isolation	Strong	Moderate

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Experiment 2:- Docker Installation, Configuration, and Running Images

Step 1: Check Docker Version

```
docker --version
```

```
dhairya@LAPTOP-H7SV3T8B: x + v
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ sudo docker --version
[sudo] password for dhairya:
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$
```

Step 2: Run Container with Port Mapping

```
docker run -d -p 8080:80 nginx
```

```
dhairya@LAPTOP-H7SV3T8B: x + v
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ sudo docker run -d -p 8080:80 nginx
[sudo] password for dhairya:
4aaebcb6e8ae7df4086ba92a2136640fb2af9aaaaa830756b97c67f058ec7eab
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ |
```

Step 3: Verify Running Containers

```
docker ps
```

```
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
4aaebcb6e8ae	nginx	"/docker-entrypoint..."	About a minute ago	Up About a minute	0.0.0.0:8080->80/tcp, [::]:80

```
80->80/tcp lucid_sinoussi
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ |
```

Step 4: Stop Container

```
docker stop <container_id>
```



```
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ sudo docker stop 4aa
4aa
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ docker ps -a
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.50/containers/json?all=1": dial unix /var/run/docker.sock: connect: permission denied
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ sudo docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
4aaebcb6e8ae	nginx	"/docker-entrypoint..."	10 minutes ago	Exited (0) 2 minutes ago	
lucid_sinoussi					

Step 5: Remove Container

```
docker rm <container_id>
```

```
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ sudo docker rm 4aa
4aa
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ sudo docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
b156e18855a0	nginx	"/docker-entrypoint..."	2 hours ago	Exited (0) 2 hours ago	

Container

Step 5: Remove Image

```
docker rmi nginx
```

```
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ sudo docker rmi nginx
Untagged: nginx:latest
Untagged: nginx@sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Deleted: sha256:4af177a024eb8a1e43f4fb6c66735bb8260115cb5925a64f51673219bd97c144
Deleted: sha256:74821a6b71dfac14eaeb80c7c9e48130391306557f75228278054afe105cde75
Deleted: sha256:ae46c5037a8cc17f658f78b0e6523c9363ca800bf51f91f58cbc3aad8ffe53b5
Deleted: sha256:ec81248cdbcb6b02bb24b1c32d4f59194a4a42aab3c81a42631efa1af7cba4f69
Deleted: sha256:774cc13f29089d7d0ebaaa291c5ac2d12841f2d70ab07a21c470d3cbdc11c059
Deleted: sha256:f1155703f3479bf6b295f5e17684156f82f815ca0798af8a2d08cf994411a055
Deleted: sha256:a455653294af11c547d0b3d2d769fd184f357cd5c0decabd1f89ffe9b380985aa
Deleted: sha256:e50a58335e1366e2581fe61794c1651afe2fe04e881e795aa166f24f4fc78d92
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$ sudo docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
dhairya-app	1.0	d4f512c030d2	27 hours ago	773MB
java-test	1.0	9f8e38d4121d	3 days ago	773MB
myrepo/test-app	1.0	07da462e253f	3 days ago	922MB
ubuntu	latest	493218ed0f40	2 weeks ago	78.1MB
ubuntu	22.04	65c77cbc27c2	3 weeks ago	77.9MB

```
dhairya@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA$
```

Result Docker images were successfully pulled, containers executed, and lifecycle commands performed.

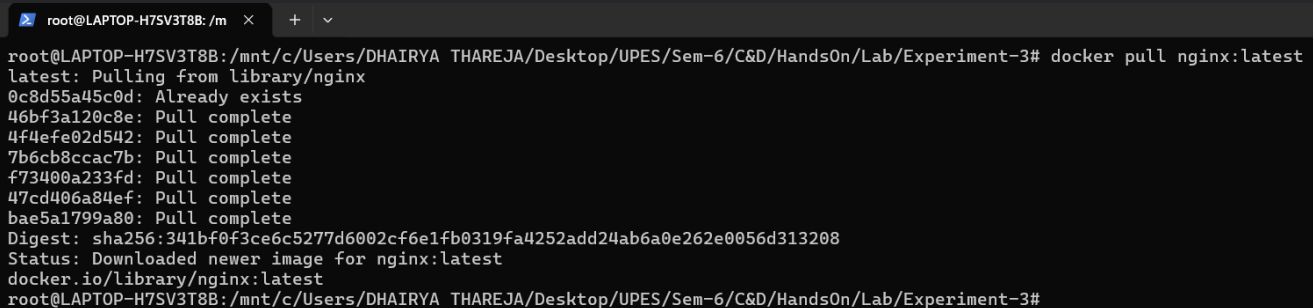
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Experiment 3:- Docker Deploy NGINX Using Different Base Images and Comparing Image Layers

PART 1: DEPLOY NGINX USING OFFICIAL IMAGE (RECOMMENDED APPROACH)

Step 1: Pull the Image

```
docker pull nginx:latest
```



```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker pull nginx:latest
latest: Pulling from library/nginx
0c8d55a45c0d: Already exists
46bf3a120c8e: Pull complete
4f4efe02d542: Pull complete
7b6cb8ccac7b: Pull complete
f73400a233fd: Pull complete
47cd406a84ef: Pull complete
bae5a1799a80: Pull complete
Digest: sha256:341bf0f3ce6c5277d6002cf6e1fb0319fa4252add24ab6a0e262e0056d313208
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3#
```

Step 2: Run the Container

```
docker run -d --name nginx-official -p 8080:80 nginx
```



```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker run -d --name nginx-official -p 8080:80 nginx
b6b96bc48f76d819a16cb2915673d5ae0f2b3acbc93e6377f16e34f754c423a6
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3#
```

Step 3: Verify

```
curl http://localhost:8080
```

```
root@LAPTOP-H7SV3T8B: /m  × + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# curl http://localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# |
```

You should see the **NGINX** welcome page.

Key Observations

```
docker images nginx
```

```
root@LAPTOP-H7SV3T8B: /m  × + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker images nginx
REPOSITORY   TAG       IMAGE ID       CREATED        SIZE
nginx        latest    5cdef4ac3335   10 days ago   161MB
nginx        <none>    248d2326f351   12 days ago   161MB
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# |
```

- Image is **pre-optimized**
- Minimal configuration required
- Uses Debian-based OS internally

PART 2: CUSTOM NGINX USING UBUNTU BASE IMAGE

Step 1: Create Dockerfile

```
FROM ubuntu:22.04

RUN apt-get update && \
    apt-get install -y nginx && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80
```

```
CMD ["nginx", "-g", "daemon off;"]
```

Step 2: Build Image

```
docker build -t nginx-ubuntu .
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  ▾

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker build -t nginx-ubuntu .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  223.7kB
Step 1/4 : FROM ubuntu:22.04
--> 65c77cbc27c2
Step 2/4 : RUN apt-get update &&          apt-get install -y nginx &&          apt-get clean &&          rm -rf /var/lib/apt/lists/*
--> Running in d1db9a088ad7
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
Get:3 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [62.6 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3708 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
```

Step 3: Run Container

```
docker run -d --name nginx-ubuntu -p 8081:80 nginx-ubuntu
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  ▾  -  🗑  ×

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker run -d --name nginx-ubuntu -p 8081:80 nginx-ubuntu
e023e4c8cf8b5d70fa25407488c5a40ee08ec73ab6b438c4bd24bf1714dbb8b9
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# |
```

Observations

```
docker images nginx-ubuntu
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  ▾

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker images nginx-ubuntu
REPOSITORY      TAG          IMAGE ID      CREATED        SIZE
nginx-ubuntu     latest       5b7b84b69d9b  42 seconds ago  134MB
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# |
```

- Much **larger image size**
- More layers
- Full OS utilities available

PART 3: CUSTOM NGINX USING ALPINE BASE IMAGE

Step 1: Create Dockerfile

```
FROM alpine:latest

RUN apk add --no-cache nginx

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Step 2: Build Image

```
docker build -t nginx-alpine .
```

```
root@LAPTOP-H7SV3T8B: /m  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker build -t nginx-alpine .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  375.3kB
Step 1/4 : FROM alpine:latest
latest: Pulling from library/alpine
589002ba0eae: Already exists
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
--> a40c03cbb81c
Step 2/4 : RUN apk add --no-cache nginx
--> Running in f5e1967815b4
(1/2) Installing pcre2 (10.47-r0)
(2/2) Installing nginx (1.28.2-r0)
     Executing nginx-1.28.2-r0.pre-install
     Executing nginx-1.28.2-r0.post-install
Executing busybox-1.37.0-r30.trigger
OK: 9.9 MiB in 18 packages
INFO[2026-02-15T19:06:19.123136334Z] ignoring event          container=f5e1967815b4e024848b7c241700d5d6de84f97e203c0585ff35d75e4f17f3d
5 module=libcontainerd namespace=moby topic=/tasks/delete type="events.TaskDelete" id=f5e1967815b4e024848b7c241700d5d6de84f97e203c0585ff35d75e4f17f3d5 names
INFO[2026-02-15T19:06:19.123066760Z] shim disconnected      id=f5e1967815b4e024848b7c241700d5d6de84f97e203c0585ff35d75e4f17f3d5 names
pace=moby
WARN[2026-02-15T19:06:19.123205390Z] cleaning up after shim disconnected id=f5e1967815b4e024848b7c241700d5d6de84f97e203c0585ff35d75e4f17f3d5 names
pace=moby
```

Step 3: Run Container

```
docker run -d --name nginx-alpine -p 8082:80 nginx-alpine
```

```
root@LAPTOP-H7SV3T8B: /m  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker run -d --name nginx-alpine -p 8082:80 nginx-alpine
0e4bf2b6dba7b7e035317d01549114f9b8059a1df050dd9968c94ea7112992ae5
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# |
```

Observations

```
docker images nginx-alpine
```

```
root@LAPTOP-H7SV3T8B: /m  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker images nginx-alpine
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
nginx-alpine        latest             3d3790f65176       49 seconds ago     10.4MB
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# |
```

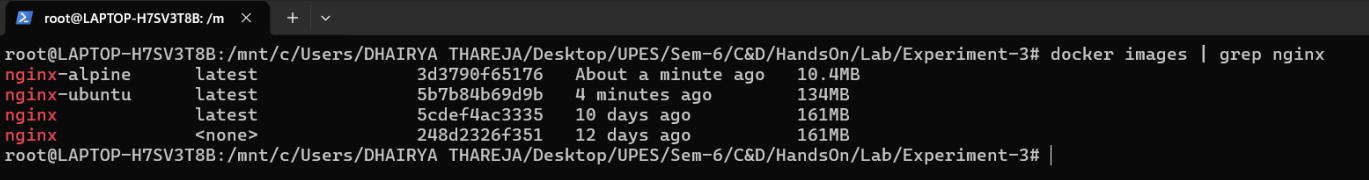
- Extremely **small image**
- Fewer packages

- Faster pull and startup time

PART 4: IMAGE SIZE AND LAYER COMPARISON

Compare Sizes

```
docker images | grep nginx
```



Typical result (approx):

Image Type	Size
nginx:latest	161 MB
nginx-ubuntu	134 MB
nginx-alpine	10.4 MB

Inspect Layers

```
docker history nginx
docker history nginx-ubuntu
docker history nginx-alpine
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker history nginx
IMAGE          CREATED          CREATED BY          SIZE      COMMENT
5cdef4ac3335   10 days ago     CMD ["nginx" "-g"  "daemon off;"]  0B        buildkit.dockerfile.v0
<missing>      10 days ago     STOPIGNAL SIGQUIT  0B        buildkit.dockerfile.v0
<missing>      10 days ago     EXPOSE map[80/tcp:{}]  0B        buildkit.dockerfile.v0
<missing>      10 days ago     ENTRYPOINT ["/docker-entrypoint.sh"]  0B        buildkit.dockerfile.v0
<missing>      10 days ago     COPY 30-tune-worker-processes.sh /docker-ent...  4.62kB    buildkit.dockerfile.v0
<missing>      10 days ago     COPY 20-envsubst-on-templates.sh /docker-ent...  3.02kB    buildkit.dockerfile.v0
<missing>      10 days ago     COPY 15-local-resolvers.envsh /docker-entryp...  389B      buildkit.dockerfile.v0
<missing>      10 days ago     COPY 10-listen-on-ipv6-by-default.sh /docker...  2.12kB    buildkit.dockerfile.v0
<missing>      10 days ago     COPY docker-entrypoint.sh / # buildkit  1.62kB    buildkit.dockerfile.v0
<missing>      10 days ago     RUN /bin/sh -c set -x  && groupadd --syst...  82.2MB    buildkit.dockerfile.v0
<missing>      10 days ago     ENV DYNPKG_RELEASE=1~trixie  0B        buildkit.dockerfile.v0
<missing>      10 days ago     ENV PKG_RELEASE=1~trixie  0B        buildkit.dockerfile.v0
<missing>      10 days ago     ENV ACME_VERSION=0.3.1  0B        buildkit.dockerfile.v0
<missing>      10 days ago     ENV NJS_RELEASE=1~trixie  0B        buildkit.dockerfile.v0
<missing>      10 days ago     ENV NJS_VERSION=0.9.5  0B        buildkit.dockerfile.v0
<missing>      10 days ago     ENV NGINX_VERSION=1.29.5  0B        buildkit.dockerfile.v0
<missing>      10 days ago     LABEL maintainer=NGINX Docker Maintainers <d...  0B        buildkit.dockerfile.v0
<missing>      13 days ago     # debian.sh --arch 'amd64' out/ 'trixie' '@1...  78.6MB    debuerreotype 0.17
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker history nginx-ubuntu
IMAGE          CREATED          CREATED BY          SIZE      COMMENT
5b7b84b69d9b   5 minutes ago   /bin/sh -c #(nop)  CMD ["nginx" "-g" "daemon...  0B        buildkit.dockerfile.v0
0cf25afdfb80   5 minutes ago   /bin/sh -c #(nop)  EXPOSE 80  0B        buildkit.dockerfile.v0
d02d5c45fd52   5 minutes ago   /bin/sh -c apt-get update && apt-get ins...  55.7MB    buildkit.dockerfile.v0
65c77cbc27c2   5 weeks ago     /bin/sh -c #(nop)  CMD ["/bin/bash"]  0B        buildkit.dockerfile.v0
<missing>      5 weeks ago     /bin/sh -c #(nop)  ADD file:b499000226bd9a7c5...  77.9MB    buildkit.dockerfile.v0
<missing>      5 weeks ago     /bin/sh -c #(nop)  LABEL org.opencontainers....  0B        buildkit.dockerfile.v0
<missing>      5 weeks ago     /bin/sh -c #(nop)  LABEL org.opencontainers....  0B        buildkit.dockerfile.v0
<missing>      5 weeks ago     /bin/sh -c #(nop)  ARG LAUNCHPAD_BUILD_ARCH  0B        buildkit.dockerfile.v0
<missing>      5 weeks ago     /bin/sh -c #(nop)  ARG RELEASE  0B        buildkit.dockerfile.v0
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# docker history nginx-alpine
IMAGE          CREATED          CREATED BY          SIZE      COMMENT
3d3790f65176   2 minutes ago   /bin/sh -c #(nop)  CMD ["nginx" "-g" "daemon...  0B        buildkit.dockerfile.v0
6082afb7d745a   2 minutes ago   /bin/sh -c #(nop)  EXPOSE 80  0B        buildkit.dockerfile.v0
11falaa2ae99    2 minutes ago   /bin/sh -c apk add --no-cache nginx  2MB       buildkit.dockerfile.v0
a40c03cbb81c    2 weeks ago     CMD ["/bin/sh"]  0B        buildkit.dockerfile.v0
<missing>      2 weeks ago     ADD alpine-minirootfs-3.23.3-x86_64.tar.gz /...  8.44MB    buildkit.dockerfile.v0
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3#

```

Observations:

- Ubuntu has many filesystem layers
- Alpine has minimal layers
- Official NGINX image is optimized but heavier than Alpine

PART 5: FUNCTIONAL TASKS USING NGINX

Task 1: Serve Custom HTML Page

```

mkdir html

echo "<h1>Hello from Docker NGINX</h1>" > html/index.html

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# mkdir html
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# echo "<h1>Hello from Docker NGINX</h1>" > html/index.html
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# ls
Dockerfile  README.md  Ubuntu_Dockerfile
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3# cd html/
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3/html# ls
index.html
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-3/html#

```

Run:

```

docker run -d \

-p 8083:80 \

```

```
-v $(pwd)/html:/usr/share/nginx/html \
nginx
```

Task 2: Reverse Proxy (Conceptual)

NGINX can:

- Forward traffic to backend services
- Load balance multiple containers
- Terminate SSL

Example use cases:

- Frontend for microservices
- API gateway
- Static file server

PART 6: COMPARISON SUMMARY

Image Comparison Table

Feature	Official NGINX	Ubuntu + NGINX	Alpine + NGINX
Image Size	Medium	Large	Very Small
Ease of Use	Very Easy	Medium	Medium
Startup Time	Fast	Slow	Very Fast
Debugging Tools	Limited	Excellent	Minimal
Security Surface	Medium	Large	Small
Production Ready	Yes	Rarely	Yes

PART 7: WHEN TO USE WHAT

Use Official NGINX Image When:

- Production deployment
- Standard web hosting
- Reverse proxy / load balancer

Use Ubuntu-Based Image When:

- Learning Linux + NGINX internals
- Heavy debugging needed
- Custom system-level dependencies

Use Alpine-Based Image When:

- Microservices
- CI/CD pipelines
- Cloud and Kubernetes workloads



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Built by Dhairya Thareja 🚀

This site uses [Just the Docs](#), a documentation theme for Jekyll.

Experiment 4: Docker Essentials

Dockerfile .dockerignore tagging publishing



CONTAINERIZING APPLICATIONS WITH DOCKERFILE

Step-1: Create a Directory for Python application

```
mkdir my-flask-app  
cd my-flask-app
```

```
root@LAPTOP-H7SV3T8B: /m x + v  
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4# mkdir my-flask-app  
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4# cd my-flask-app  
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app#
```

Step-2:- Create an `app.py`

```
from flask import Flask  
app = Flask(__name__)  
  
@app.route('/')  
def hello():  
    return "Hello from Docker!"  
  
@app.route('/health')  
def health():  
    return "OK"  
  
if __name__ == '__main__':  
    app.run(host='0.0.0.0', port=5000)
```

```
root@LAPTOP-H7SV3T8B: /m x + v
GNU nano 7.2 app.py
from flask import Flask
app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello from Docker!"

@app.route('/health')
def health():
    return "OK"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

Step-3:- Create ****requirements.txt****

```
Flask==2.3.3
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# nano requirements.txt
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# cat requirements.txt
Flask==2.3.3
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app#
```

Step-4: Create ****Dockerfile****

```
# Use Python base image
FROM python:3.9-slim

# Set working directory
WORKDIR /app

# Copy requirements file
COPY requirements.txt .

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY app.py .

# Expose port
EXPOSE 5000

# Run the application
CMD ["python", "app.py"]
```

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# nano Dockerfile
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# cat Dockerfile
# Use Python base image
FROM python:3.9-slim

# Set working directory
WORKDIR /app

# Copy requirements file
COPY requirements.txt .

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Copy application code
COPY app.py .

# Expose port
EXPOSE 5000

# Run the application
CMD ["python", "app.py"]
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Step-5:- Create a **.dockerignore**

```
# Python files
__pycache__/
*.pyc
*.pyo
*.pyd

# Environment files
.env
.venv
env/
venv/

# IDE files
.vscode/
.idea/

# Git files
.git/
.gitignore

# OS files
.DS_Store
Thumbs.db

# Logs
*.log
logs/

# Test files
```

```
tests/  
test_*.py
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# nano .dockerignore  
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# cat .dockerignore  
# Python files  
__pycache__/  
*.pyc  
*.pyo  
*.pyd  
  
# Environment files  
.env  
.venv  
env/  
env/  
  
# IDE files  
.vscode/  
.idea/  
  
# Git files  
.git/  
.gitignore  
  
# OS files  
.DS_Store  
Thumbs.db  
  
# Logs  
*.log  
logs/  
  
# Test files  
tests/  
test_*.py  
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Why `.dockerignore` is Important

- Prevents unnecessary files from being copied
- Reduces image size
- Improves build speed
- Increases security

Step-6: Build image from Dockerfile

```
docker build -t my-flask-app .
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker build -t my-flask-app .  
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.  
Install the buildx component to build images with BuildKit:  
https://docs.docker.com/go/buildx/  
  
Sending build context to Docker daemon  5.12kB  
Step 1/7 : FROM python:3.9-slim  
3.9-slim: Pulling from library/python  
38513bd72563: Pulling fs layer  
b3ec39b36ae8: Pulling fs layer  
fc7443084902: Pulling fs layer  
ea56f685404a: Pulling fs layer  
ea56f685404a: Waiting  
b3ec39b36ae8: Verifying Checksum  
b3ec39b36ae8: Download complete  
fc7443084902: Verifying Checksum  
fc7443084902: Download complete  
ea56f685404a: Verifying Checksum  
ea56f685404a: Download complete  
38513bd72563: Verifying Checksum  
38513bd72563: Download complete  
38513bd72563: Pull complete  
b3ec39b36ae8: Pull complete  
fc7443084902: Pull complete  
ea56f685404a: Pull complete  
Digest: sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
```

Step-7:- Verify Image

```
docker images
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
my-flask-app         latest             5d9c9dc2e420       36 seconds ago    132MB
nginx-alpine         latest             3d3790f65176       5 days ago        10.4MB
nginx-ubuntu         latest             5b7b84b69d9b       5 days ago        134MB
multistage           1.0               e41104ca49d1       10 days ago       900kB
singlestage          1.0               e59489fea0a3       10 days ago       334MB
sapid-checker        500118984         d958530ec807       2 weeks ago       161MB
nginx                latest             5cdef4ac3335       2 weeks ago       161MB
python               3.11.14-alpine3.23 2dbdacd9e496       2 weeks ago       54.3MB
nginx                <none>            248d2326f351       2 weeks ago       161MB
dhairya-app          1.0               d4f512c030d2       3 weeks ago       773MB
java-test            1.0               9f8e38d4121d       3 weeks ago       773MB
alpine               latest             a40c03cbb81c       3 weeks ago       8.44MB
myrepo/test-app      1.0               07da462e253f       3 weeks ago       922MB
ubuntu               latest             493218ed0f40       5 weeks ago       78.1MB
ubuntu               22.04             65c77cbc27c2       6 weeks ago       77.9MB
python               3.9-slim          085da638e1b8       3 months ago      122MB
busybox              latest             af3f0f48a24e       17 months ago     4.43MB
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app#
```

Step-8: Add Tag in Image

```
docker build -t my-flask-app:1.0 .
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker build -t my-flask-app:1.0 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   5.12kB
Step 1/7 : FROM python:3.9-slim
--> 085da638e1b8
Step 2/7 : WORKDIR /app
--> Using cache
--> 5ba3b2f3a775
Step 3/7 : COPY requirements.txt .
--> Using cache
--> babffdd372f2
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
--> Using cache
--> 21e156d72ba1
Step 5/7 : COPY app.py .
--> Using cache
--> b3f0d7cf3d32
Step 6/7 : EXPOSE 5000
--> Using cache
--> 280a33431b44
Step 7/7 : CMD ["python", "app.py"]
--> Using cache
--> 5d9c9dc2e420
Successfully built 5d9c9dc2e420
Successfully tagged my-flask-app:1.0
```

Step-9:- Add multiple tags

```
docker build -t my-flask-app:latest -t my-flask-app:1.0 .
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker build -t my-flask-app:latest -t my-flask-app:1.0 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon   5.12kB
Step 1/7 : FROM python:3.9-slim
--> 085da638e1b8
Step 2/7 : WORKDIR /app
--> Using cache
--> 5ba3b2f3a775
Step 3/7 : COPY requirements.txt .
--> Using cache
--> babffdd372f2
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
--> Using cache
--> 21e156d72ba1
Step 5/7 : COPY app.py .
--> Using cache
--> b3f0d7cf3d32
Step 6/7 : EXPOSE 5000
--> Using cache
--> 280a33431b44
Step 7/7 : CMD ["python", "app.py"]
--> Using cache
--> 5d9c9dc2e420
Successfully built 5d9c9dc2e420
Successfully tagged my-flask-app:latest
Successfully tagged my-flask-app:1.0
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app#
```

Step-10:- Add Tag with custom registry

```
docker build -t username/my-flask-app:1.0 .
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker build -t username/my-flask-app:1.0 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  5.12kB
Step 1/7 : FROM python:3.9-slim
--> 085da638e1b8
Step 2/7 : WORKDIR /app
--> Using cache
--> 5ba3b2f3a775
Step 3/7 : COPY requirements.txt .
--> Using cache
--> babffdd372f2
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
--> Using cache
--> 21e156d72ba1
Step 5/7 : COPY app.py .
--> Using cache
--> b3f0d7cf3d32
Step 6/7 : EXPOSE 5000
--> Using cache
--> 280a33431b44
Step 7/7 : CMD ["python", "app.py"]
--> Using cache
--> 5d9c9dc2e420
Successfully built 5d9c9dc2e420
Successfully tagged username/my-flask-app:1.0
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app#
```

Step-11:- Tag existing image

```
docker tag my-flask-app:latest my-flask-app:v1.0
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker tag my-flask-app:latest my-flask-app:v1.0
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app#
```

Step-12: View Image Details

```
docker images
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
my-flask-app	1.0	5d9c9dc2e420	2 minutes ago	132MB
my-flask-app	latest	5d9c9dc2e420	2 minutes ago	132MB
my-flask-app	v1.0	5d9c9dc2e420	2 minutes ago	132MB
username/my-flask-app	1.0	5d9c9dc2e420	2 minutes ago	132MB
nginx-alpine	latest	3d3790f65176	5 days ago	10.4MB
nginx-ubuntu	latest	5b7b84b69d9b	5 days ago	134MB
multistage	1.0	e41104ca49d1	10 days ago	900kB
singlestage	1.0	e59489fea0a3	10 days ago	334MB
apid-checker	500118984	d958530ec807	2 weeks ago	161MB
nginx	latest	5cdef4ac3335	2 weeks ago	161MB
python	3.11.14-alpine3.23	2dbdacd9e496	2 weeks ago	54.3MB
nginx	<none>	248d2326f351	2 weeks ago	161MB
dhairya-app	1.0	d4f512c030d2	3 weeks ago	773MB
java-test	1.0	9f8e38d4121d	3 weeks ago	773MB
alpine	latest	a40c03cbb81c	3 weeks ago	8.44MB
myrepo/test-app	1.0	07da462e253f	3 weeks ago	922MB
ubuntu	latest	493218ed0f40	5 weeks ago	78.1MB
ubuntu	22.04	65c77cbc27c2	6 weeks ago	77.9MB
python	3.9-slim	085da638e1b8	3 months ago	122MB
busybox	latest	af3f0f48a24e	17 months ago	4.43MB

Step-13:- Show image history

```
docker history my-flask-app
```

```

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker history my-flask-app
IMAGE          CREATED          CREATED BY          SIZE      COMMENT
5d9c9dc2e420   2 minutes ago   /bin/sh -c #(nop)  CMD ["python" "app.py"]   0B
280a33431b44   2 minutes ago   /bin/sh -c #(nop)  EXPOSE 5000              0B
b3f0d7cf3d32   2 minutes ago   /bin/sh -c #(nop)  COPY file:444a3904a5c3d221... 228B
21e156d72ba1   2 minutes ago   /bin/sh -c pip install --no-cache-dir -r req... 10.5MB
babffdd372f2   2 minutes ago   /bin/sh -c #(nop)  COPY file:f095db6d544b2123... 13B
5ba3b2f3a775   2 minutes ago   /bin/sh -c #(nop)  WORKDIR /app             0B
085da638e1b8   3 months ago    CMD ["python3"]        0B      buildkit.dockerfile.v0
<missing>      3 months ago    RUN /bin/sh -c set -eux; for src in idle3 p... 36B      buildkit.dockerfile.v0
<missing>      3 months ago    RUN /bin/sh -c set -eux; savedAptMark="$(a... 39.5MB    buildkit.dockerfile.v0
<missing>      3 months ago    ENV PYTHON_SHA256=00e07d7c0f2f0cc002432d1ee8... 0B        buildkit.dockerfile.v0
<missing>      3 months ago    ENV PYTHON_VERSION=3.9.25 0B        buildkit.dockerfile.v0
<missing>      3 months ago    ENV GPG_KEY=E3FF2839C048B25C084DEBE9B26995E3... 0B        buildkit.dockerfile.v0
<missing>      3 months ago    RUN /bin/sh -c set -eux; apt-get update; a... 3.81MB    buildkit.dockerfile.v0
<missing>      3 months ago    ENV LANG=C.UTF-8        0B        buildkit.dockerfile.v0
<missing>      3 months ago    ENV PATH=/usr/local/bin:/usr/local/sbin:/usr... 0B        buildkit.dockerfile.v0
<missing>      4 months ago    # debian.sh --arch 'amd64' out/ 'trixie' '@1... 78.6MB    debuerreotype 0.16
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |

```

Step-14:- Inspect image details

```
docker inspect my-flask-app
```

```

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker inspect my-flask-app
[
  {
    "Id": "sha256:5d9c9dc2e420552928017b728f6abdddf4496e81316416c59641f67ced5cbaf6",
    "RepoTags": [
      "my-flask-app:1.0",
      "my-flask-app:latest",
      "my-flask-app:v1.0",
      "username/my-flask-app:1.0"
    ],
    "RepoDigests": [],
    "Parent": "sha256:280a33431b442948c19f476ee0a881ddb5bf37027c3012aa8643667806b40e1",
    "Comment": "",
    "Created": "2026-02-21T09:04:12.373470136Z",
    "DockerVersion": "28.2.2",
    "Author": "",
    "Config": {
      "ExposedPorts": {
        "5000/tcp": {}
      },
      "Env": [
        "PATH=/usr/local/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "LANG=C.UTF-8",
        "GPG_KEY=E3FF2839C048B25C084DEBE9B26995E310250568",
        "PYTHON_VERSION=3.9.25",
        "PYTHON_SHA256=00e07d7c0f2f0cc002432d1ee84d2a40dae404a99303e3f97701c10966c91834"
      ],
      "Cmd": [
        "python",
        "app.py"
      ],
      "WorkingDir": "/app"
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 132482301,
    "GraphDriver": {
      "Data": {
        "LowerDir": "/var/lib/docker/overlay2/7fed05053322dd0472b95ebb585184b5041dab439fb52f286b10511415dd4ad9/diff:/var/lib/docker/overlay2/b7ba3261cb1ceb684ed227ca3289959ac9bd108470ba928c6057c469561f29e9/diff:/var/lib/docker/overlay2/a9563a41e24ce5875fa7645a00f05abad503571fd027934fdfe0e5a53fab375/dif
f:/var/lib/docker/overlay2/bd0af826834c91e15c6407b3379a7a7fe16fd7138f4ed9484ca200dfddab6b7/diff:/var/lib/docker/overlay2/e5329b85324931266f29ded9140f99ee65

```

Step-15: Run Container

```
docker run -d -p 5000:5000 --name flask-container my-flask-app
```

```

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker run -d -p 5000:5000 --name flask-cont
ainer my-flask-app
afedc2af4191a524e7a34c3b7c151d1be681fd6649b1c949393a338429a9506
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |

```

Step-16:- Test the application

```
curl http://localhost:5000
```



```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# curl http://localhost:5000
Hello from Docker!root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Step-17:- View running containers

```
docker ps
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
afedc2af4191   my-flask-app   "python app.py"         50 seconds ago Up 50 seconds  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flask-container
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Step-18:- View container logs

```
docker logs flask-container
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker logs flask-container
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
172.17.0.1 - - [21/Feb/2026 09:08:51] "GET / HTTP/1.1" 200 -
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Step-19:- Stop container

```
docker stop flask-container
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker stop flask-container
INFO[2026-02-21T09:10:15.797715016Z] Container failed to exit within 10s of signal 15 - using the force container=afedc2af4191a524e7a34c3b7c151d1be681f1d6649b1c949393a338429a9506
INFO[2026-02-21T09:10:15.816057074Z] shim disconnected id=afedc2af4191a524e7a34c3b7c151d1be681f1d6649b1c949393a338429a9506 names=pace=moby
INFO[2026-02-21T09:10:15.816081949Z] ignoring event container=afedc2af4191a524e7a34c3b7c151d1be681f1d6649b1c949393a338429a9506
6 module=libcontainerd namespace=moby topic=/tasks/delete type="events.TaskDelete"
WARN[2026-02-21T09:10:15.816122304Z] cleaning up after shim disconnected id=afedc2af4191a524e7a34c3b7c151d1be681f1d6649b1c949393a338429a9506 names=pace=moby
INFO[2026-02-21T09:10:15.816131124Z] cleaning up dead shim namespace=moby
flask-container
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Step-20:- Start stopped container

```
docker start flask-container
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker start flask-container
flask-container
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Step-21:- Remove container

```
docker rm flask-container
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker rm flask-container
Error response from daemon: cannot remove container "flask-container": container is running: stop the container before removing or force remove
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# S|
```

Step-22:- Remove container forcefully

Since Container is in Running State, so we will remove it forcefully...

```
docker rm -f flask-container
```

```
Windows PowerShell: root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker rm -f flask-container
[46Z] shim disconnected id=3bd7d0f7b3972b2d46272ccdee06a73024078ac888a7cd36ed2aba79781b5b66 names
[30Z] cleaning up after shim disconnected id=3bd7d0f7b3972b2d46272ccdee06a73024078ac888a7cd36ed2aba79781b5b66 names
pace=moby
INFO[2026-02-21T09:14:16.322869016Z] cleaning up dead shim namespace=moby
INFO[2026-02-21T09:14:16.322853386Z] ignoring event container=3bd7d0f7b3972b2d46272ccdee06a73024078ac888a7cd36ed2aba79781b5b6
6 module=libcontainerd namespace=moby topic=/tasks/delete type="events.TaskDelete"
flask-container
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Multi-stage Builds

Why Multi-stage Builds?

- Smaller final image size
- Better security (remove build tools)
- Separate build and runtime environments

Step-23: Create a Simple `**Dockerfile.multistage:**`

```
# STAGE 1: Builder stage

FROM python:3.9-slim AS builder

WORKDIR /app

# Copy requirements
COPY requirements.txt .

# Install dependencies in virtual environment
RUN python -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip install --no-cache-dir -r requirements.txt

# STAGE 2: Runtime stage

FROM python:3.9-slim
```

```
WORKDIR /app
```

```
# Copy virtual environment from builder
```

```
COPY --from=builder /opt/venv /opt/venv
```

```
ENV PATH="/opt/venv/bin:$PATH"
```

```
# Copy application code
```

```
COPY app.py .
```

```
# Create non-root user
```

```
RUN useradd -m -u 1000 appuser
```

```
USER appuser
```

```
# Expose port
```

```
EXPOSE 5000
```

```
# Run application
```

```
CMD ["python", "app.py"]
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4# nano Dockerfile.multistage
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4# cat Dockerfile.multistage
# STAGE 1: Builder stage
FROM python:3.9-slim AS builder

WORKDIR /app

# Copy requirements
COPY requirements.txt .

# Install dependencies in virtual environment
RUN python -m venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"
RUN pip install --no-cache-dir -r requirements.txt

# STAGE 2: Runtime stage
FROM python:3.9-slim

WORKDIR /app

# Copy virtual environment from builder
COPY --from=builder /opt/venv /opt/venv
ENV PATH="/opt/venv/bin:$PATH"

# Copy application code
COPY app.py .

# Create non-root user
RUN useradd -m -u 1000 appuser
USER appuser

# Expose port
EXPOSE 5000

# Run application
CMD ["python", "app.py"]
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4# |
```

Step-24:- Build regular image

```
docker build -t flask-regular .
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker build -t flask-regular .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  6.656kB
Step 1/7 : FROM python:3.9-slim
--> 085da638e1b8
Step 2/7 : WORKDIR /app
--> Using cache
--> 5ba3b2f3a775
Step 3/7 : COPY requirements.txt .
--> Using cache
--> babffdd372f2
Step 4/7 : RUN pip install --no-cache-dir -r requirements.txt
--> Using cache
--> 21e156d72ba1
Step 5/7 : COPY app.py .
--> Using cache
--> b3f0d7cf3d32
Step 6/7 : EXPOSE 5000
--> Using cache
--> 280a33431b44
Step 7/7 : CMD ["python", "app.py"]
--> Using cache
--> 5d9c9dc2e420
Successfully built 5d9c9dc2e420
Successfully tagged flask-regular:latest
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app#
```

Step-25:- Build multi-stage image

```
docker build -f Dockerfile.multistage -t flask-multistage .
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker build -f Dockerfile.multistage -t flask-multistage .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  6.656kB
Step 1/15 : FROM python:3.9-slim AS builder
--> 085da638e1b8
Step 2/15 : WORKDIR /app
--> Using cache
--> 5ba3b2f3a775
Step 3/15 : COPY requirements.txt .
--> Using cache
--> babffdd372f2
Step 4/15 : RUN python -m venv /opt/venv
--> Running in 6da0f43c0449
INFO[2026-02-21T09:24:26.815494086Z] shim disconnected id=6da0f43c044909a13db9534e3c669434dc8fd64a63cdd78af656155acbbb9583 namespace=moby
WARN[2026-02-21T09:24:26.815547312Z] cleaning up after shim disconnected id=6da0f43c044909a13db9534e3c669434dc8fd64a63cdd78af656155acbbb9583 namespace=moby
WARN[2026-02-21T09:24:26.815552642Z] cleaning up dead shim namespace=moby
INFO[2026-02-21T09:24:26.815758154Z] ignoring event container=6da0f43c044909a13db9534e3c669434dc8fd64a63cdd78af656155acbbb958
3 module=libcontainerd namespace=moby topic=/tasks/delete type=*events.TaskDelete"
--> Removed intermediate container 6da0f43c0449
--> 47180deeb530
Step 5/15 : ENV PATH="/opt/venv/bin:$PATH"
--> Running in 43b28d59d4ea
INFO[2026-02-21T09:24:27.677500996Z] Layer sha256:46c13b1aaab270602499a8a0400a05aeea11bd87b41f1b48ab9dbe9f8f5bf87b cleaned up
--> Removed intermediate container 43b28d59d4ea
--> 6e04c8b34606
Step 6/15 : RUN pip install --no-cache-dir -r requirements.txt
--> Running in 018965fa5942
Collecting Flask==2.3.3
  Downloading flask-2.3.3-py3-none-any.whl (96 kB)
    _____ 96.1/96.1 kB 696.5 kB/s eta 0:00:00
```

Step-26:- Compare sizes

```
docker images | grep flask-
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker images | grep flask-
flask-multistage          latest          78183d125751   28 seconds ago   145MB
username/my-flask-app    1.0            5d9c9dc2e420   20 minutes ago   132MB
flask-regular            latest         5d9c9dc2e420   20 minutes ago   132MB
my-flask-app             1.0            5d9c9dc2e420   20 minutes ago   132MB
my-flask-app             latest         5d9c9dc2e420   20 minutes ago   132MB
my-flask-app             v1.0           5d9c9dc2e420   20 minutes ago   132MB
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app#
```

Publishing to Docker Hub

Step-27:- Login to Docker Hub

```
docker login
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker login
Authenticating with existing credentials... [Username: dhairyathareja]

Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Step-28:- Tag image for Docker Hub

```
docker tag my-flask-app:latest username/my-flask-app:1.0

docker tag my-flask-app:latest username/my-flask-app:latest
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker tag my-flask-app:latest dhairyatharej
a/my-flask-app:1.0
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker tag my-flask-app:latest dhairyatharej
a/my-flask-app:latest
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Step-29:- Push to Docker Hub

```
docker push username/my-flask-app:1.0
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker push dhairyathareja/my-flask-app:1.0
The push refers to repository [docker.io/dhairyathareja/my-flask-app]
f83712a1d7c9: Pushed
5759a6d34fe1: Pushed
43e87d7605ae: Pushed
bb924d16437b: Pushed
c8f6b54339a8: Mounted from library/python
298992e09a03: Mounted from library/python
4f237755fbae: Mounted from library/python
d7c97cb6f1fe: Mounted from library/python
1.0: digest: sha256:feb912340599bfa298f5785e3260fd011dacbf1de1b7daf317283a135d215e73 size: 1990
```

Step-30:- Push Latest Image

```
docker push username/my-flask-app:latest
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker push dhairyathareja/my-flask-app:late
st
The push refers to repository [docker.io/dhairyathareja/my-flask-app]
f83712a1d7c9: Layer already exists
5759a6d34fe1: Layer already exists
43e87d7605ae: Layer already exists
bb924d16437b: Layer already exists
c8f6b54339a8: Layer already exists
298992e09a03: Layer already exists
4f237755fbae: Layer already exists
d7c97cb6f1fe: Layer already exists
latest: digest: sha256:feb912340599bfa298f5785e3260fd011dacbf1de1b7daf317283a135d215e73 size: 1990
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# INFO[2026-02-21T09:39:53.885297417Z] Network
```

Step-31:- Pull from Docker Hub (on another machine)

```
docker pull username/my-flask-app:latest
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker pull dhairyathareja/my-flask-app:late
st
latest: Pulling from dhairyathareja/my-flask-app
Digest: sha256:feb912340599bfa298f5785e3260fd011dacbf1de1b7daf317283a135d215e73
Status: Image is up to date for dhairyathareja/my-flask-app:latest
docker.io/dhairyathareja/my-flask-app:latest
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Step-32:- Run the pulled image

```
docker run -d -p 5000:5000 username/my-flask-app:latest
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# docker run -d -p 5000:5000 dhairyathareja/my
-flask-app:latest
dce689aa79e34413496625ed83320a2b5cd04a29eff350e516bdd52487d2be31
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-flask-app# |
```

Node.js Application Example (Quick Version)**

Step-33:- Create a directory Node.js Application

```
mkdir my-node-app
cd my-node-app
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4# mkdir my-node-app
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4# cd my-node-app
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# |
```

Step-34:- Create a `app.js`

```
const express = require('express');

const app = express();

const port = 3000;

app.get('/', (req, res) => {
  res.send('Hello from Node.js Docker!');
});

app.get('/health', (req, res) => {
  res.json({ status: 'healthy' });
});
```

```
app.listen(port, () => {  
    console.log(`Server running on port ${port}`);  
});
```

```
root@LAPTOP-H7SV3T8B: /m  × + ▾  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# nano app.js  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# cat app.js  
const express = require('express');  
const app = express();  
const port = 3000;  
  
app.get('/', (req, res) => {  
    res.send('Hello from Node.js Docker!');  
});  
  
app.get('/health', (req, res) => {  
    res.json({ status: 'healthy' });  
});  
  
app.listen(port, () => {  
    console.log(`Server running on port ${port}`);  
});  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# |
```

Step-35:- Create a **package.json**

```
{  
  "name": "node-docker-app",  
  "version": "1.0.0",  
  "main": "app.js",  
  "dependencies": {  
    "express": "^4.18.2"  
  }  
}
```

```
root@LAPTOP-H7SV3T8B: /m  × + ▾  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# nano package.json  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# cat package.json  
{  
  "name": "node-docker-app",  
  "version": "1.0.0",  
  "main": "app.js",  
  "dependencies": {  
    "express": "^4.18.2"  
  }  
}  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# |
```

Step-36:- Create a Node.js Dockerfile

```
FROM node:18-alpine  
  
WORKDIR /app  
  
COPY package*.json ./  
  
RUN npm install --only=production  
  
COPY app.js .
```

```
CMD ["node", "app.js"]
```

Step-37:- Build image

```
root@LAPTOP-H7SV3T8B: /m + -
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# docker build -t my-node-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 4.096kB
Step 1/7 : FROM node:18-alpine
18-alpine: Pulling from library/node
f18232174bc9: Pulling fs layer
dd71dde834b5: Pulling fs layer
1e5a4c89cee5: Pulling fs layer
25ff2da83641: Pulling fs layer
25ff2da83641: Waiting
f18232174bc9: Download complete
f18232174bc9: Pull complete
1e5a4c89cee5: Verifying Checksum
1e5a4c89cee5: Download complete
25ff2da83641: Verifying Checksum
dd71dde834b5: Verifying Checksum
dd71dde834b5: Download complete
dd71dde834b5: Pull complete
1e5a4c89cee5: Pull complete
25ff2da83641: Pull complete
Digest: sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e
Status: Downloaded newer image for node:18-alpine
--> ee77c6cd7c18
Step 2/7 : WORKDIR /app
--> Running in 4976e97455ac
--> Removed intermediate container 4976e97455ac
--> b44fa3cb89e9
Step 3/7 : COPY package*.json ./
--> 6f0d5dca0240
Step 4/7 : RUN npm install --only=production
--> Running in fe9a37f7ae96
```

Step-38:- Run container

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# docker run -d -p 3000:3000 --name node-container my-node-app
9d67fc4cf462f8b8d1582eca99c1554dff9c667ad507c0c716de11835c1d73d5
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app#
```

Step-39:- Test on Terminal


```
curl http://localhost:3000
```

```
root@LAPTOP-H7SV3T8B: /m  x  +  -  x
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# curl http://localhost:3000
Hello from Node.js Docker!root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-4/my-node-app# |
```

Step-40:- Test on Browser

localhost:3000

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Hello from Node.js Docker!

DOCKERHUB

View Pushed Image on [DockerHub](#)

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Notifications

Billing

Usage

Pulls

Storage

Repositories

All repositories within the dhairythareja namespace.

Search by repository name

All content

Create a repository

Name	Last Pushed	Contains	Visibility	Scout
dhairythareja/my-node-app	4 minutes ago	IMAGE	Public	Inactive
dhairythareja/my-flask-app	1 day ago	IMAGE	Public	Inactive

1-2 of 2

CONCLUSION

ESSENTIAL DOCKER COMMANDS CHEATSHEET

Command	Purpose	Example
<code>docker build</code>	Build image	<code>docker build -t myapp .</code>
<code>docker run</code>	Run container	<code>docker run -p 3000:3000 myapp</code>
<code>docker ps</code>	List containers	<code>docker ps -a</code>
<code>docker images</code>	List images	<code>docker images</code>
<code>docker tag</code>	Tag image	<code>docker tag myapp:latest myapp:v1</code>

Command	Purpose	Example
<code>docker login</code>	Login to Dockerhub using username and password or token	<code>echo "token" docker login -u username -password-stdin</code>
<code>docker push</code>	Push to registry	<code>docker push username/myapp</code>
<code>docker pull</code>	Pull from registry	<code>docker pull username/myapp</code>
<code>docker rm</code>	Remove container	<code>docker rm container-name</code>
<code>docker rmi</code>	Remove image	<code>docker rmi image-name</code>
<code>docker logs</code>	View logs	<code>docker logs container-name</code>
<code>docker exec</code>	Execute command	<code>docker exec -it container-name bash</code>

COMMON WORKFLOW SUMMARY

Development Workflow:

1. Create Dockerfile and .dockerignore

2. Build image

```
docker build -t myapp .
```

3. Test Locally

```
docker run -p 8080:8080 myapp
```

4. Tag for production

```
docker tag myapp:latest myapp:v1.0
```

5. Push to registry

```
docker push myapp:v1.0
```

Production Workflow:

1. Pull from registry

```
docker pull myapp:v1.0
```

2. Run in production

```
docker run -d -p 80:8080 --name prod-app myapp:v1.0
```

```
# 3. Monitor
```

```
docker logs -f prod-app
```

Key Takeaways

- 1 **Dockerfile** defines how to build your image
- 2 **.dockerignore** excludes unnecessary files (important for security and performance)
- 3 **Tagging** helps version and organize images
- 4 **Multi-stage builds** create smaller, more secure production images
- 5 **Docker Hub** allows sharing and distributing images
- 6 **Always test** images locally before publishing

CLEANUP

```
# Remove all stopped containers
```

```
docker container prune
```

```
# Remove unused images
```

```
docker image prune
```

```
# Remove everything unused
```

```
docker system prune -a
```

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Experiment 5: Docker - Volumes, Environment Variables, Monitoring & Networks

HANDSON

Step-1: Run a test container

```
# Create a container that writes data

docker run -it --name test-container ubuntu /bin/bash

# Inside container:

echo "Hello World" > /data/message.txt

cat /data/message.txt # Shows "Hello World"

exit

# Restart container

docker start test-container

docker exec test-container cat /data/message.txt

# ERROR: File doesn't exist! Data was lost.
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -it --name test-container ubuntu /bin/bash
root@3135bb818e9c:/# echo "Hello World" > message.txt
root@3135bb818e9c:/# cat message.txt
Hello World
root@3135bb818e9c:/# exit
exit
INFO[2026-04-06T12:28:24.129469686Z] ignoring event          container=3135bb818e9c6f3b4103e14c472fa1c4c15e9662ced6cea2450c15bcd02aba7
d module=libcontainerd namespace=moby topic=/tasks/delete type="*events.TaskDelete"
shim disconnected          id=3135bb818e9c6f3b4103e14c472fa1c4c15e9662ced6cea2450c15bcd02aba7d namespace=moby
WARN[2026-04-06T12:28:24.129619690Z] cleaning up after shim disconnected id=3135bb818e9c6f3b4103e14c47
2fa1c4c15e9662ced6cea2450c15bcd02aba7d namespace=moby
INFO[2026-04-06T12:28:24.129762818Z] cleaning up dead shim
namespace=moby
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker start test-container
test-container
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker exec test-cont
ainer cat /data/message.txt
cat: /data/message.txt: No such file or directory
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-2:- Create Anonymous Volumes

```
# Create anonymous volume (auto-generated name)

docker run -d -v /app/data --name web1 nginx

# Check volume

docker volume ls
```

```
# Shows: anonymous volume with random hash
```

```
# Inspect container to see volume mount
```

```
docker inspect web1 | grep -A 5 Mounts
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d -v /app/data --name web1 nginx
fcfe0204437a044607c8855911678eec8f69b602bed07a8a56557309e7002088
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker volume ls
DRIVER      VOLUME NAME
local       5adec6f1fd1fd7d0d249bfd34529c8d772ba02553d9891060b9ca3e97707289d
local       7df3c4e10ea43120bd4dc69c374ef146853c1d3a23eb0765a7072734fed86973
local       09a2291714933ee6889ffc76de1876c22b3a4155202699955f11bdf6a1444a83
local       a819b7b6209c286bbe583c4b946df9698026b1a600d26f7adfdfe85d2f9622f9
local       assignment-1_pg_data
local       assignment-1_pgdata
local       b550c7299d1b79facb631c0beddaa6181f26ad245d419b9fc3eae7458e5f635
local       c34a011f99407a3e2811198cd00c903baafa9c9dd6806f96e33370f9ae4d8d1d
local       dhairyaVolume
local       e527fb6bd846ae63d87485de9a10379b5908b0025af6e7acfc40a9a7175b93a
local       jenkins_home
local       k3d-mycluster-images
local       lecture-14_mysql_data
local       lecture-14_wp_content
local       my-app_jenkins_home
local       portainer_data
local       wordpress-scale_mysql_data
local       wordpress-scale_wp_content
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker inspect web1 | grep -A 5 Mounts
"Mounts": [
  {
    "Type": "volume"
    "Name": "c34a011f99407a3e2811198cd00c903baafa9c9dd6806f96e33370f9ae4d8d1d",
    "Source": "/var/lib/docker/volumes/c34a011f99407a3e2811198cd00c903baafa9c9dd6806f96e33370f9ae4d8d1d/_data",
    "Destination": "/app/data",
  }
]
```

Step-3:- Create Named Volumes

```
# Create named volume
```

```
docker volume create mydata
```

```
# Use named volume
```

```
docker run -d -v mydata:/app/data --name web2 nginx
```

```
# List volumes
```

```
docker volume ls
```

```
# Shows: mydata
```

```
# Inspect volume
```

```
docker volume inspect mydata
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker volume create mydata
mydata
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d -v mydata:/app/data --name web2 nginx
9a431ab7d6de3777b84c61392f3c2f272e07cd575eca61ea5810d3863678f996
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker volume ls
DRIVER      VOLUME NAME
local       5adec6f1fd1fd7d0d249bfd34529c8d772ba02553d9891060b9ca3e97707289d
local       7df3c4e10ea43120bd4dc69c374ef146853c1d3a23eb0765a7072734fed86973
local       09a2291714933ee6889ffc76de1876c22b3a4155202699955f11bdf6a1444a83
local       a819b7b6209c286bbe583c4b946df9698026b1a600d26f7adfdfe85d2f9622f9
local       assignment-1_pg_data
local       assignment-1_pgdata
local       b550c7299d1b79facb631c0beddaa6181f26ad245d419b9fc3eae7458e5f635
local       c34a011f99407a3e2811198cd00c903baafa9c9dd6806f96e33370f9ae4d8d1d
local       dhairyaVolume
local       e527fb6bd846ae63d87485de9a10379b5908b0025af6e7acfc40a9a7175b93a
local       jenkins_home
local       k3d-mycluster-images
local       lecture-14_mysql_data
local       lecture-14_wp_content
local       my-app-jenkins_home
local       mydata
local       portainer_data
local       wordpress-scale_mysql_data
local       wordpress-scale_wp_content
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker volume inspect mydata
[
  {
    "CreatedAt": "2026-04-06T12:38:42Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/mydata/_data",
    "Name": "mydata",
    "Options": null,
    "Scope": "local"
  }
]
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |

```

Step-4: Bind Mounts

```

# Create directory on host

mkdir ~/myapp-data

# Mount host directory to container

docker run -d -v ~/myapp-data:/app/data --name web3 nginx

# Add file on host

echo "From Host" > ~/myapp-data/host-file.txt

# Check in container

docker exec web3 cat /app/data/host-file.txt

# Shows: From Host

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# mkdir ~/myapp-data
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d -v ~/myapp-data:/app/data --name web3 nginx
8412b0aee762be432b63b011c6cc529b5e342ce0cd344f8117b87377301fcab1
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# echo "From Host" > ~/myapp-data/host-file.txt
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker exec web3 cat /app/data/host-file.txt
From Host
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |

```

Step-5:- Database with Persistent Storage

```

# MySQL with named volume

docker run -d \

    --name mysql-db \

```

```

-v mysql-data:/var/lib/mysql \

-e MYSQL_ROOT_PASSWORD=secret \

mysql:8.0

# Check data persists

docker stop mysql-db

docker rm mysql-db

# New container with same volume

docker run -d \

--name new-mysql \

-v mysql-data:/var/lib/mysql \

-e MYSQL_ROOT_PASSWORD=secret \

mysql:8.0

# Data is preserved!

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d \
--name mysql-db \
-v mysql-data:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=secret \
mysql:8.0
Unable to find image 'mysql:8.0' locally
8.0: Pulling from library/mysql
d4c7048d1cf1: Pull complete
6375ec02106f: Pull complete
771773a1fe7a: Pull complete
c3b8cfb10b41: Pull complete
1378718ee973: Pull complete
5915b0e5580c: Pull complete
60c3210ab1f6: Pull complete
440208b3f29f: Pull complete
10ee8ef3b9fc: Pull complete
cd8097e7c26b: Pull complete
0940ecf5daec: Pull complete
Digest: sha256:64756cc92f707eb504496d774353990bcb0f6999ddf598b6ad188f2da66bd000
Status: Downloaded newer image for mysql:8.0
6ed24b28b1f9ff68d6c6169d0185d87feb922c61ac1ff259d5bef63e3f40da3f
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker stop mysql-db
INFO[2026-04-06T12:42:24.435772393Z] ignoring event container=6ed24b28b1f9ff68d6c6169d0185d87feb922c61ac1ff259d5bef63e3f40da3
f module=libcontainerd namespace=moby topic=/tasks/delete type="events.TaskDelete" id=6ed24b28b1f9ff68d6c6169d0185d87feb922c61ac1ff259d5bef63e3f40da3f names
pace=moby
WARN[2026-04-06T12:42:24.435843397Z] cleaning up after shim disconnected id=6ed24b28b1f9ff68d6c6169d0185d87feb922c61ac1ff259d5bef63e3f40da3f names
pace=moby
INFO[2026-04-06T12:42:24.435873153Z] cleaning up dead shim namespace=moby
mysql-db
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d \
--name new-mysql \
-v mysql-data:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=secret \
mysql:8.0
5a2043f892067d2659968e426335426f2dace3bb1bc899786f685b30226cc4cd
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5#

```

Step-6: Environment Variable

```

docker run -d \

--name app1 \

-e DATABASE_URL="postgres://user:pass@db:5432/mydb" \

-e DEBUG="true" \

-p 3000:3000 \

my-node-app

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --name app1 \
-e DATABASE_URL="postgres://user:pass@host:5432/db" \
-e DEBUG=true \
-p 3000:80 nginx
e297b974a0646a45d8ef728ab63c68da0555ea2a498a3a66ac5db9982ef1358b
INFO[2026-04-30T14:23:21.624732742Z] connecting to shim e297b974a0646a45d8ef728ab63c68da0555ea2a498a3a66ac5db9982ef1358b address="unix:///run/containerd/s/
added8850e6ae59f5e0ef7034b10af0ab1404cd2ecf1885e9a39e914e90d2fffd" namespace=moby protocol=tttrpc version=3
INFO[2026-04-30T14:23:21.714497325Z] sbJoin: gwep4 '-->'bd98c62f0afd', gwep6 '-->' eid=bd98c62f0afd ep=app1 net=bridge nid=9306b317ae52
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |

```

Step-7:- List .env Variables

```

echo "DATABASE_HOST=localhost" > .env

echo "DATABASE_PORT=5432" >> .env

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# echo "DB_HOST=localhost" > .env
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# echo "DB_PORT=5432" >> .env
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |

```

Step-8: Run Container with .env

```

docker run -d \

--env-file .env \

--name app2 \

my-app

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --env-file .env --name app2 nginx
aad9bc1cc154b9f39460e84980c56a2f186135cddc5ecfb304c3c6e23631f402
INFO[2026-04-30T14:25:17.358576575Z] connecting to shim aad9bc1cc154b9f39460e84980c56a2f186135cddc5ecfb304c3c6e23631f402 address="unix:///run/containerd/s/
1c085497d0584d954421b38a9611fb46061238412b784d0492f5f7bf693ecda6" namespace=moby protocol=tttrpc version=3
INFO[2026-04-30T14:25:17.456112866Z] sbJoin: gwep4 '-->'df403e6d85a6', gwep6 '-->' eid=df403e6d85a6 ep=app2 net=bridge nid=9306b317ae52
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |

```

Step-9:- Dockerfile Env

```

nano Dockerfile

cat Dockerfile

```

Paste this:-

```

FROM python:3.9-slim

ENV PORT=5000

ENV DEBUG=false

CMD ["sleep", "3600"]

```



```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# nano Dockerfile
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# cat Dockerfile
FROM python:3.9-slim
ENV PORT=5000
ENV DEBUG=false
CMD ["sleep", "3600"]
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |

```

Step-10:- Build Flask App with the Dockerfile

```

docker build -t flask-app .

docker run -d --name flask-app -p 5000:5000 flask-app

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker build -t flask-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  894.5kB
Step 1/4 : FROM python:3.9-slim
--> 085da638e1b8
Step 2/4 : ENV PORT=5000
--> Running in e62aac022887
INFO[2026-04-30T14:28:05.592969241Z] Cleaned up layer sha256:b483329e076bcb3e44fccad73d86590868c7e9b172a7f759340fd25b3d027f05  chainID="sha256:b483329e076bc
b3e44fccad73d86590868c7e9b172a7f759340fd25b3d027f05"
--> Removed intermediate container e62aac022887
--> 71d62106050d
Step 3/4 : ENV DEBUG=false
--> Running in bc7ef2abaf46
INFO[2026-04-30T14:28:05.634770026Z] Cleaned up layer sha256:b483329e076bcb3e44fccad73d86590868c7e9b172a7f759340fd25b3d027f05  chainID="sha256:b483329e076bc
b3e44fccad73d86590868c7e9b172a7f759340fd25b3d027f05"
--> Removed intermediate container bc7ef2abaf46
--> e986b65af245
Step 4/4 : CMD ["sleep", "3600"]
--> Running in ee7603d10033
INFO[2026-04-30T14:28:05.675959147Z] Cleaned up layer sha256:b483329e076bcb3e44fccad73d86590868c7e9b172a7f759340fd25b3d027f05  chainID="sha256:b483329e076bc
b3e44fccad73d86590868c7e9b172a7f759340fd25b3d027f05"
--> Removed intermediate container ee7603d10033
--> 4bfff12612b52
Successfully built 4bfff12612b52
Successfully tagged flask-app:latest
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --name flask-app -p 5000:5000 flask-app
146df24eee774d56ea074f9c644aa253460b9abb0c5af00b0e3f4a5a67305a79
INFO[2026-04-30T14:28:29.245410821Z] connecting to shim 146df24eee774f9c644aa253460b9abb0c5af00b0e3f4a5a67305a79  address="unix:///run/containerd/s/
60c3066f3ab744f4f1865e6613fe6517a5ee5d0c334acae2c9dc6183b955d830" namespace=moby protocol=tttrpc version=3
INFO[2026-04-30T14:28:29.344249014Z] sbJoin: gwep4 '-->'1804f8b4b729', gwep6 '-->'  eid=1804f8b4b729  ep=flask-app  net=bridge  nid=9306b317ae52
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |

```

Step-11:- Monitoring

```

docker stats --no-stream

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker stats --no-stream
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O       BLOCK I/O     PIDS
146df24eee77   flask-app 0.00%     1.074MiB / 7.57GiB   0.01%     796B / 126B    3.03MB / 0B    1
aad9bc1cc154   app2      0.00%     13.18MiB / 7.57GiB   0.17%     992B / 126B    0B / 12.3kB    17
e297b974a064   app1      0.00%     15.53MiB / 7.57GiB   0.20%     1.41kB / 126B  4.4MB / 12.3kB  17
fa6ca33cf9c6   k3d-mycluster-serverlb  0.01%     30.34MiB / 7.57GiB   0.39%     592kB / 585kB  22.8MB / 16.4kB  27
dc6a4ec53cef   k3d-mycluster-agent-0    1.69%     307.2MiB / 7.57GiB   3.96%     1.64MB / 635kB 289MB / 175MB   162
68a91800571c   k3d-mycluster-server-0   4.52%     666.6MiB / 7.57GiB   8.60%     1.14MB / 2MB   465MB / 238MB   235
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |

```

Step-12: Monitor

```

docker stats --no-stream

```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker stats --no-stream
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O       BLOCK I/O   PIDS
146df24eee77   flask-app     0.00%    1.074MiB / 7.57GiB   0.01%     796B / 126B    3.03MB / 0B   1
aad9bc1cc154   app2         0.00%    13.18MiB / 7.57GiB   0.17%     992B / 126B    0B / 12.3kB   17
e297b974a064   app1         0.00%    15.53MiB / 7.57GiB   0.20%     1.41kB / 126B   4.4MB / 12.3kB 17
fa6ca33cf9c6   k3d-mycluster-serverlb  0.01%    30.34MiB / 7.57GiB   0.39%     592kB / 585kB   22.8MB / 16.4kB 27
dc6a4ec53cef   k3d-mycluster-agent-0    1.69%    307.2MiB / 7.57GiB   3.96%     1.64MB / 635kB  289MB / 175MB  162
68a91800571c   k3d-mycluster-server-0    4.52%    666.6MiB / 7.57GiB   8.60%     1.14MB / 2MB    465MB / 238MB  235
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-13:- Get Stats via Top

```
docker top flask-app
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker top flask-app
UID          PID          PPID         C            STIME        TTY          TIME         CMD
root         11547        11523        0            17:32        ?            00:00:00     sleep 3600
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-14:- Inspect Container details

```
docker inspect flask-app
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker inspect flask-app
[
  {
    "Id": "146df24eee774d56ea074f9c644aa253460b9abb0c5af00b0e3f4a5a67305a79",
    "Created": "2026-04-30T14:28:29.207232531Z",
    "Path": "sleep",
    "Args": [
      "3600"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 11547,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-04-30T14:28:29.239258862Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:4bfff12612b522ed30b6aa0ca38db224fa342f698637483f2263856ffc1b033e6",
    "ResolvConfPath": "/var/lib/docker/containers/146df24eee774d56ea074f9c644aa253460b9abb0c5af00b0e3f4a5a67305a79/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/146df24eee774d56ea074f9c644aa253460b9abb0c5af00b0e3f4a5a67305a79/hostname",
    "HostsPath": "/var/lib/docker/containers/146df24eee774d56ea074f9c644aa253460b9abb0c5af00b0e3f4a5a67305a79/hosts",
    "LogPath": "/var/lib/docker/containers/146df24eee774d56ea074f9c644aa253460b9abb0c5af00b0e3f4a5a67305a79/146df24eee774d56ea074f9c644aa253460b9abb0c5af00b0e3f4a5a67305a79-json.log",
    "Name": "/flask-app",
    "RestartCount": 0,
    "Driver": "overlay2",
    "Platform": "linux",
    "MountLabel": "",
    "ProcessLabel": "",
    "AppArmorProfile": "",
    "ExecIDs": null,
    "HostConfig": {
      "Binds": null,
      "ContainerIDFile": "",
      "LogConfig": {
        "Type": "json-file",
```

Step-15: Inspect Container Status Only

```
docker inspect --format='{{.State.Status}}' flask-app
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker inspect --format='{{.State.Status}}' flask-app
running
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-16:- View Events

docker events

```
harshit@LAPTOP-38KOG53E: ~$ docker events
2026-04-17T14:46:52.582016800Z container exec_create: /bin/sh -c pg_isready -U admin 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffcbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=8e758199b125a5d921f911253d7974cc438503b3b8bcb9a2d39c0134c3d94a97, image=assignment-1-database, name=postgres_db)
2026-04-17T14:46:52.582635800Z container exec_start: /bin/sh -c pg_isready -U admin 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffcbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=8e758199b125a5d921f911253d7974cc438503b3b8bcb9a2d39c0134c3d94a97, image=assignment-1-database, name=postgres_db)
2026-04-17T14:46:52.726667000Z container exec_die 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffcbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=8e758199b125a5d921f911253d7974cc438503b3b8bcb9a2d39c0134c3d94a97, exitCode=0, image=assignment-1-database, name=postgres_db)
```

Step-17:- View Conatiner Events Only

docker events --filter 'type=container'

```
harshit@LAPTOP-38KOG53E: ~$ docker events --filter 'type=container'
2026-04-17T14:47:12.864352300Z container exec_create: /bin/sh -c pg_isready -U admin 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffcbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=a54b514bfcf789947f43dd57288903644af1646a182e15e8b29a4f1be264ae27, image=assignment-1-database, name=postgres_db)
2026-04-17T14:47:12.864743000Z container exec_start: /bin/sh -c pg_isready -U admin 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffcbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=a54b514bfcf789947f43dd57288903644af1646a182e15e8b29a4f1be264ae27, image=assignment-1-database, name=postgres_db)
2026-04-17T14:47:12.983155700Z container exec_die 5576c99e1f844298e37e457411eaf6148683286ee879b3275b25f1d4e2a99f46 (com.docker.compose.config-hash=20acc352eedb28b030a74623e241b85d8e71cffffcbc917b95db2a11be61c6399, com.docker.compose.container-number=1, com.docker.compose.depends_on=, com.docker.compose.image=sha256:0521772ca0e94659b8cad5cf18c1418b3baf3ea44a8bdf4d2477d8f5d526ff74, com.docker.compose.oneoff=False, com.docker.compose.project=assignment-1, com.docker.compose.project.config_files=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1/docker-compose.yml, com.docker.compose.project.working_dir=/mnt/d/Containerization/ASSIGNMENTS/ASSIGNMENT-1, com.docker.compose.replace=postgres_db, com.docker.compose.service=database, com.docker.compose.version=5.0.1, desktop.docker.io/ports.scheme=v2, desktop.docker.io/wsl-distro=Ubuntu, execID=a54b514bfcf789947f43dd57288903644af1646a182e15e8b29a4f1be264ae27, exitCode=0, image=assignment-1-database, name=postgres_db)
```

Step-18:- Create `monitor.sh`

nano monitor.sh

cat monitor.sh

Paste This:-

```
#!/bin/bash

echo "=== Docker Monitoring Dashboard ==="

echo "Time: $(date)"

echo
```

```
echo "1. Running Containers:"

docker ps --format "table \t\t"

echo


echo "2. Resource Usage:"

docker stats --no-stream --format "table \t\t\t"

echo


echo "3. Recent Events:"

docker events --since '5m' --until '0s' --format " " " | tail -5

echo


echo "4. System Info:"

docker system df
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  ∨
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# nano monitor.sh
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# cat monitor.sh
#!/bin/bash

echo "=== Docker Monitoring Dashboard ==="
echo "Time: $(date)"
echo

echo "1. Running Containers:"
docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}"
echo

echo "2. Resource Usage:"
docker stats --no-stream --format "table {{.Name}}\t{{.CPUPerc}}\t{{.MemUsage}}\t{{.NetIO}}"
echo

echo "3. Recent Events:"
docker events --since '5m' --until '0s' --format "{{.Time}} {{.Type}} {{.Action}}}" | tail -5
echo

echo "4. System Info:"
docker system df
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-19:- bash `monitor.sh`

```
bash monitor.sh
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  ▾

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# bash monitor.sh
=== Docker Monitoring Dashboard ===
Time: Thu Apr 30 17:51:34 UTC 2026

1. Running Containers:
NAMES          STATUS      PORTS
flask-app      Up 3 hours  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
app2           Up 3 hours  80/tcp
app1           Up 3 hours  0.0.0.0:3000->80/tcp, [::]:3000->80/tcp
k3d-mycluster-serverlb Up 4 hours  80/tcp, 0.0.0.0:37911->6443/tcp
k3d-mycluster-agent-0 Up 4 hours
k3d-mycluster-server-0 Up 4 hours

2. Resource Usage:
NAME          CPU %      MEM USAGE / LIMIT  NET I/O
flask-app     0.00%     524KiB / 7.57GiB   1.08kB / 126B
app2          0.00%     13.1MiB / 7.57GiB  1.2kB / 126B
app1          0.00%     13.1MiB / 7.57GiB  1.55kB / 126B
k3d-mycluster-serverlb 0.01%     30.34MiB / 7.57GiB 923kB / 914kB
k3d-mycluster-agent-0  1.26%     307.7MiB / 7.57GiB 2.26MB / 1.05MB
k3d-mycluster-server-0 6.36%     666.2MiB / 7.57GiB 1.75MB / 2.82MB

3. Recent Events:

4. System Info:
TYPE          TOTAL    ACTIVE    SIZE      RECLAIMABLE
Images        53       4         12.22GB   4.712GB (38%)
Containers    6         6         77.23kB   0B (0%)
Local Volumes 43        9         7.857GB   4.162GB (52%)
Build Cache   7         0         14.4MB    0B

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-20:- System Info

```
docker system df
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  ▾

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker system d
TYPE          TOTAL    ACTIVE    SIZE      RECLAIMABLE
Images        53       4         12.22GB   4.712GB (38%)
Containers    6         6         77.23kB   0B (0%)
Local Volumes 43        9         7.857GB   4.162GB (52%)
Build Cache   7         0         14.4MB    0B

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-21:- List Networks

```
docker networks ls
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  ▾

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network ls
NETWORK ID    NAME                DRIVER            SCOPE
9306b317ae52  bridge             bridge            local
31d18b044efd  docker_gwbridge    bridge            local
f9d0a2e00f87  host               host              local
op3lgyamknn3  ingress            overlay           swarm
1776f67d8fea  k3d-mycluster      bridge            local
6e205de754b8  none               null              local

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-22:- Remove Network

```
docker network rm my-network
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network rm my-network
my-network
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-23: Create network

```
docker network create my-network
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network create my-network
b3b0e11328bd0bf99b1c7fea950db4aa540368273d17ce331aa27c018d139e4f
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-24:- Run Containers with Bridge Network

```
docker rm -f web1 web2

docker run -d --name web1 --network my-network nginx

docker run -d --name web2 --network my-network nginx

docker exec web1 curl http://web2
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker rm -f web1 web2
Error response from daemon: No such container: web1
Error response from daemon: No such container: web2
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --name web1 --network my-network nginx
114f90cc18a919664a93aec54b0e30bccd18292e285f5bae0b195447bcbc84a4
INFO[2026-04-30T17:59:34.041116067Z] connecting to shim 114f90cc18a919664a93aec54b0e30bccd18292e285f5bae0b195447bcbc84a4 address="unix:///run/containerd/s/
580030c667a96f484a89c57d72ab5794528a9de583799677e5f9718d67cde43e" namespace=moby protocol=ttrpc version=3
INFO[2026-04-30T17:59:34.174350928Z] sbJoin: gwep4 ''->'24173b8e95d3', gwep6 ''->' eid=24173b8e95d3 ep=web1 net=my-network nid=b3b0e11328bd
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --name web2 --network my-network nginx
2b6d6b65107085d86d1d6180d7ba0d954ac94c29561780a155f24ae998830fee
INFO[2026-04-30T17:59:40.416354366Z] connecting to shim 2b6d6b65107085d86d1d6180d7ba0d954ac94c29561780a155f24ae998830fee address="unix:///run/containerd/s/
504e35573979f6b357a9030232fca445a56f6d79930ee0f3141e2ccac9278bdf" namespace=moby protocol=ttrpc version=3
INFO[2026-04-30T17:59:40.548002353Z] sbJoin: gwep4 ''->'09a74d1538b4', gwep6 ''->' eid=09a74d1538b4 ep=web2 net=my-network nid=b3b0e11328bd
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker exec web1 curl http://web2
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 615 100 615 0 0 53520 0 --:--:-- --:--:-- --:--:-- 55909
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color:scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-25:- Run nginx Container

```
docker run -d --network host nginx

curl https://localhost
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --network host nginx
699e746921c125ed5ee8379d77e7f26a48dde39bfb55b599f0d501505fee7679
INFO[2026-04-30T18:00:53.046018138Z] connecting to shim 699e746921c125ed5ee8379d77e7f26a48dde39bfb55b599f0d501505fee7679 address="unix:///run/containerd/s/
5d32abececdf45005af8534c655384a31546506072b772a80ad07adb7d719a979" namespace=moby protocol=tttrpc version=3
INFO[2026-04-30T18:00:53.116062368Z] sbJoin: gwep4 ''->'', gwep6 ''->'' eid=1655eb725815 ep=silly_liskov net=host nid=f9d0a2e00f87
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# curl https://localhost
curl: (7) Failed to connect to localhost port 443 after 0 ms: Couldn't connect to server
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color:scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-26:- Run `alpine` Container

```
docker run -d --network none alpine sleep 3600
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --network none alpine sleep 3600
6d82b173bb89c2adb00577ea5b90354a34e1a00fc78a6aff896eae988fa07b2b
INFO[2026-04-30T18:03:14.900891594Z] connecting to shim 6d82b173bb89c2adb00577ea5b90354a34e1a00fc78a6aff896eae988fa07b2b address="unix:///run/containerd/s/
01c8aa7ccebcb60573f30545bffb251f3637ea2f4178ceaae5b081cbe63cc1c6" namespace=moby protocol=tttrpc version=3
INFO[2026-04-30T18:03:14.964909851Z] sbJoin: gwep4 ''->'', gwep6 ''->'' eid=fdb44c9fbd64 ep=sweet_lederberg net=none nid=6e205de754b8
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-27:- Get Ip Address of Container

```
docker exec <Container ID> ip addr
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker exec 6d82b ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-28:- Create a Network

```
docker network create app-network
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network create app-network
cd9a5762279a0300ec744a0933c1efda668ee732c6c4bfb6165a1030911829be
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```


Step-29:- Connect network with Container

```
docker network connect app-network web1
docker network connect app-network web2
docker network connect app-network flask-app
docker network inspect app-network
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network connect app-network web1
INFO[2026-04-30T18:07:26.968247574Z] sbJoin: gwep4 '24173b8e95d3'-'>'e47909227d53', gwep6 ''->' eid=e47909227d53 ep=web1 net=app-network nid=cd9a5762279a
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network connect app-network web2
INFO[2026-04-30T18:07:35.177497327Z] sbJoin: gwep4 '09a74d1538b4'-'>'fc890ffac2ff', gwep6 ''->' eid=fc890ffac2ff ep=web2 net=app-network nid=cd9a5762279a
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network connect app-network flask-app
INFO[2026-04-30T18:07:40.737109094Z] sbJoin: gwep4 '1804f8b4b729'-'>'7329193920e1', gwep6 ''->' eid=7329193920e1 ep=flask-app net=app-network nid=cd9a5762279a
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network inspect app-network
[
  {
    "Name": "app-network",
    "Id": "cd9a5762279a0300ec744a0933c1efda668ee732c6c4bfff6165a1030911829be",
    "Created": "2026-04-30T18:06:41.767881013Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.21.0.0/16",
          "IPRange": "",
          "Gateway": "172.21.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {
      "114f90cc18a919664a93aec54b0e30bccd18292e285f5bae0b195447bcbc84a4": {
        "Name": "web1",
        "EndpointID": "e47909227d53158b8f33c5d507e24b6a61d4f16f6168d853575ce8022c972918",

```

Step-30:- curl in Container

```
docker exec web1 curl http://web2
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker exec web1 curl http://web2
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           Dload  Upload   Total   Spent    Left     Speed
100    615    100    615     0     0    221k      0 --:--:-- --:--:-- --:--:--   300k
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```


Step-31:- Disconnect Network

```
docker network disconnect app-network web1
docker network disconnect app-network web2
docker network disconnect app-network flask-app
docker network rm app-network
```

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network disconnect app-network web1
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network disconnect app-network web2
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# INFO[2026-04-30T18:10:13.612945516Z] NetworkDB stats LAPT
OP-H7SV3T8B(1ccd12532440) - netID:op31gyamknn34bn2i5dwf2bpm Leaving:false netPeers:1 entries:1 Queue qlen:0+0 netMsg/s:0

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network disconnect app-network flask-app
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network rm app-network
app-network
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-32:- Create a new Network

```
docker network create my-app-network
```

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker network create myapp-network
76c0937b3ad62f4ffa7b03d043e1d5d52c82573a949e96bbdc80f1c3d19f8d52
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-33:- Run `postgres` Container

```
docker run -d --name postgres \
--network myapp-network \
-e POSTGRES_PASSWORD=secret \
postgres:15
```

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --name postgres \
--network myapp-network \
-e POSTGRES_PASSWORD=secret \
postgres:15
237e5d31a365c2508d10801bb236e9c470416b49af6d4b6d97be47291d7d49b8
INFO[2026-04-30T18:11:43.102167691Z] connecting to shim 237e5d31a365c2508d10801bb236e9c470416b49af6d4b6d97be47291d7d49b8 address="unix:///run/containerd/s/
8f4a85fc26ef40be16f34401fff478b2fda3281d3ed1bbf7bc74493960cafdfd" namespace=moby protocol=ttrpc version=3
INFO[2026-04-30T18:11:43.227079679Z] sbJoin: gwep4 ''->'aa80b5f8dbde', gwep6 ''->' /eid=aa80b5f8dbde ep=postgres net=myapp-network nid=76c0937b3ad6
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-34:- Run a `redis` Container

```
docker run -d --name redis \
--network myapp-network \
redis:7-alpine
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --name redis \
--network myapp-network \
redis:7-alpine
Unable to find image 'redis:7-alpine' locally
7-alpine: Pulling from library/redis
897d797d2723: Pull complete
1f0676652ae4: Pull complete
51bf34b872a6: Pull complete
9572acab3f32: Pull complete
47836b8a3274: Pull complete
2da9501c87e4: Pull complete
4f4fb700ef54: Pull complete
6447bc944d85: Pull complete
Digest: sha256:7aec734b2bb298a1d769fd8729f13b8514a41bf90fcd1f38ec52267fbaa8ee6
Status: Downloaded newer image for redis:7-alpine
e1efa3ff4b09ce54675ae10ab65224980b8288febb98c0af1360057d6ff27bf
INFO[2026-04-30T18:12:21.355620122Z] connecting to shim e1efa3ff4b09ce54675ae10ab65224980b8288febb98c0af1360057d6ff27bf address="unix:///run/containerd/s/
e024c6d495bbf650aa4435562570959c8b91b33a251f90644f83c01dfb937e26" namespace=moby protocol=tttrpc version=3
INFO[2026-04-30T18:12:21.458532665Z] sbJoin: gwep4 ''->'a142883a975e', gwep6 ''->' eid=a142883a975e ep=redis net=myapp-network nid=76c0937b3ad6
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-35:- Run a flask app with the previous container

```
docker run -d --name flask-app \

--network myapp-network \

-p 5000:5000 \

-e DATABASE_URL="postgres://postgres:secret@postgres:5432/mydb" \

-e REDIS_URL="redis://redis:6379" \
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker run -d --name flask-app \
--network myapp-network \
-p 5000:5000 \
-e DATABASE_URL="postgres://postgres:secret@postgres:5432/mydb" \
-e REDIS_URL="redis://redis:6379" \
flask-app:latest
fed9c0c6b2035c495b479bbfd45907e798a7da117c7338515474547c5dd53159
INFO[2026-04-30T18:14:06.953409746Z] connecting to shim fed9c0c6b2035c495b479bbfd45907e798a7da117c7338515474547c5dd53159 address="unix:///run/containerd/s/
fc4a694ab2613d19b05611e915b9933ab5cb1b95a48b4f79718b44275a17a3d" namespace=moby protocol=tttrpc version=3
INFO[2026-04-30T18:14:07.046637069Z] sbJoin: gwep4 ''->'83c42462d31a', gwep6 ''->' eid=83c42462d31a ep=flask-app net=myapp-network nid=76c0937b3ad6
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-36:- List Containers

```
docker ps
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAME
fed9c0c6b203   flask-app:late "sleep 3600"            16 seconds ago Up 15 seconds 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp   flas
k-app
e1efa3ff4b09   redis:7-alpine "docker-entrypoint.s..." 2 minutes ago Up 2 minutes   6379/tcp                                     redi
s
237e5d31a365   postgres:15    "docker-entrypoint.s..." 2 minutes ago Up 2 minutes   5432/tcp                                     post
gres
6d82b173bb89   alpine         "sleep 3600"            11 minutes ago Up 11 minutes                                     swee
t_lederberg
699e746921c1   nginx          "/docker-entrypoint..." 13 minutes ago Up 13 minutes                                     sill
y_liskov
2b6d6b651070   nginx          "/docker-entrypoint..." 14 minutes ago Up 14 minutes   80/tcp                                     web2
114f90cc18a9   nginx          "/docker-entrypoint..." 14 minutes ago Up 14 minutes   80/tcp                                     web1
aad9bc1cc154   nginx          "/docker-entrypoint..." 4 hours ago   Up 4 hours     80/tcp                                     app2
e297b974a064   nginx          "/docker-entrypoint..." 4 hours ago   Up 4 hours     0.0.0.0:3000->80/tcp, [::]:3000->80/tcp       app1
fa6ca33cf9c6   ghcr.io/k3d-io/k3d-proxy:5.8.3 "/bin/sh -c nginx-pr..." 25 hours ago   Up 4 hours     80/tcp, 0.0.0.0:37911->6443/tcp              k3d-
mycluster-serverlb
dc6a4ec53cef   rancher/k3s:v1.31.5-k3s1 "/bin/k3d-entrypoint..." 25 hours ago   Up 4 hours                                     k3d-
mycluster-agent-0
60a91800571c   rancher/k3s:v1.31.5-k3s1 "/bin/k3d-entrypoint..." 25 hours ago   Up 4 hours                                     k3d-
mycluster-server-0
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-37:- Monitor

```
docker stats --no-stream
```

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker stats --no-stream
CONTAINER ID   NAME          CPU %     MEM USAGE / LIMIT   MEM %     NET I/O     BLOCK I/O   PIDS
fed9c0c6b203   flask-app     0.00%    444KiB / 7.57GiB    0.01%    726B / 126B  111kB / 0B   1
e1efa3ff4b09   redis        0.38%    3.594MiB / 7.57GiB  0.05%    992B / 126B  0B / 0B      6
237e5d31a365   postgres     0.00%    54.49MiB / 7.57GiB  0.70%    1.64kB / 126B  46.6MB / 41.2MB 6
6d82b172bb89   sweet_ledenb 0.00%    416KiB / 7.57GiB    0.01%    0B / 0B      1.24MB / 0B    1
699e746921c1   silly_liskov 0.00%    13.16MiB / 7.57GiB  0.17%    0B / 0B      4.1kB / 12.3kB 17
2b6d6b651070   web2         0.00%    13.43MiB / 7.57GiB  0.17%    1.63kB / 1.4kB 156kB / 12.3kB 17
114f90cc18a9   web1         0.00%    20.67MiB / 7.57GiB  0.27%    2.93kB / 748B  13.1MB / 12.3kB 17
aad9bc1cc154   app2         0.00%    13.1MiB / 7.57GiB    0.17%    1.27kB / 126B  0B / 12.3kB    17
e297b974a064   app1         0.00%    13.11MiB / 7.57GiB  0.17%    1.62kB / 126B  4.4MB / 12.3kB 17
fa6ca33cf9c6   k3d-mycluster-serverlb 0.01%    30.34MiB / 7.57GiB  0.39%    1.36MB / 1.35MB 22.8MB / 16.4kB 27
dc6a4ec53cef   k3d-mycluster-agent-0  12.50%   311.1MiB / 7.57GiB  4.01%    3.01MB / 1.57MB 308MB / 175MB   166
68a91800571c   k3d-mycluster-server-0 6.50%    670.1MiB / 7.57GiB  8.64%    2.54MB / 3.84MB 496MB / 286MB   238
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-38:- Ping inside container

```
docker exec -it flask-app bash
apt update
apt install -y iputils-ping
```

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# docker exec -it flask-app bash
root@fed9c0c6b203:/# apt update
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [129 kB]
Fetched 10.0 MB in 3s (3967 kB/s)
14 packages can be upgraded. Run 'apt list --upgradable' to see them.
Notice: Repository 'http://deb.debian.org/debian trixie InRelease' changed its 'Version' value from '13.1' to '13.4'
root@fed9c0c6b203:/# apt install -y iputils-ping
Installing:
  iputils-ping

Installing dependencies:
  libidn2-0  libunistring5  linux-sysctl-defaults

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 14
  Download size: 643 kB
  Space needed: 2810 kB / 1001 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 libunistring5 amd64 1.3-2 [477 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 libidn2-0 amd64 2.3.8-2 [109 kB]
Get:3 http://deb.debian.org/debian trixie/main amd64 iputils-ping amd64 3:20240905-3 [51.2 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 linux-sysctl-defaults all 4.12.1 [5724 B]
Fetched 643 kB in 0s (2120 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79, <STDIN> line 4.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8, <STDIN> line 4.)
debconf: falling back to frontend: Teletype
Selecting previously unselected package libunistring5:amd64.
(Reading database ... 5644 files and directories currently installed.)
Preparing to unpack .../libunistring5_1.3-2_amd64.deb ...
Unpacking libunistring5:amd64 (1.3-2) ...
```

Step-39:- Ping postgres Container

```
ping -c 2 postgres
```

```
root@fed9c0c6b203:/# ping -c 2 postgres
PING postgres (172.21.0.2) 56(84) bytes of data.
64 bytes from postgres.myapp-network (172.21.0.2): icmp_seq=1 ttl=64 time=2.57 ms
64 bytes from postgres.myapp-network (172.21.0.2): icmp_seq=2 ttl=64 time=0.062 ms

--- postgres ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 0.062/1.314/2.566/1.252 ms
```

Step-40:- Ping `redis` Container

```
ping -c 2 redis
```

```
root@fed9c0c6b203:/# ping -c 2 redis
PING redis (172.21.0.3) 56(84) bytes of data.
64 bytes from redis.myapp-network (172.21.0.3): icmp_seq=1 ttl=64 time=0.349 ms
64 bytes from redis.myapp-network (172.21.0.3): icmp_seq=2 ttl=64 time=0.047 ms

--- redis ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1020ms
rtt min/avg/max/mdev = 0.047/0.198/0.349/0.151 ms
root@fed9c0c6b203:/# exit
exit
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

Step-41:- Inspect Network

```
docker network inspect myapp-network
```

```
root@fed9c0c6b203:/# ping -c 2 redis
PING redis (172.21.0.3) 56(84) bytes of data.
64 bytes from redis.myapp-network (172.21.0.3): icmp_seq=1 ttl=64 time=0.349 ms
64 bytes from redis.myapp-network (172.21.0.3): icmp_seq=2 ttl=64 time=0.047 ms

--- redis ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1020ms
rtt min/avg/max/mdev = 0.047/0.198/0.349/0.151 ms
root@fed9c0c6b203:/# exit
exit
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-5# |
```

CONCLUSION

Key Takeaways

- 1 **Volumes** persist data beyond container lifecycle
- 2 **Environment variables** configure containers dynamically
- 3 **Monitoring commands** help debug and optimize containers
- 4 **Networks** enable secure container communication
- 5 **Always use named volumes** for production data
- 6 **Custom networks** provide better isolation and DNS
- 7 **Monitor resource usage** to prevent issues
- 8 **Use .env files** for sensitive configuration

This experiment covers essential Docker features for building, configuring, and managing production-ready containerized applications.

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Experiment 6: Comparison of Docker Run and Docker Compose

INTRODUCTION

DOCKER RUN (IMPERATIVE APPROACH)

The `docker run` command is used to create and start a container from an image. It requires explicit flags for:

- Port mapping (`-p`)
- Volume mounting (`-v`)
- Environment variables (`-e`)
- Network configuration (`--network`)
- Restart policies (`--restart`)
- Resource limits (`--memory`, `--cpus`)
- Container name (`--name`)

This approach is **imperative**, meaning you specify step-by-step instructions.

DOCKER COMPOSE (DECLARATIVE APPROACH)

Docker Compose uses a YAML file (`docker-compose.yml`) to define services, networks, and volumes in a structured format.

Instead of multiple `docker run` commands, a single command is used:

```
docker compose up -d
```

Compose is **declarative**, meaning you define the desired state of the application.

Mapping: Docker Run vs Docker Compose

Docker Run Flag	Docker Compose Equivalent
<code>-p 8080:80</code>	<code>ports:</code>
<code>-v host:container</code>	<code>volumes:</code>

Docker Run Flag	Docker Compose Equivalent
<code>-e KEY=value</code>	<code>environment:</code>
<code>--name</code>	<code>container_name:</code>
<code>--network</code>	<code>networks:</code>
<code>--restart</code>	<code>restart:</code>
<code>--memory</code>	<code>deploy.resources.limits.memory</code>
<code>--cpus</code>	<code>deploy.resources.limits.cpus</code>
<code>-d</code>	<code>docker compose up -d</code>

Advantages of Docker Compose

- 1 Simplifies multi-container applications
- 2 Provides reproducibility
- 3 Version controllable configuration
- 4 Unified lifecycle management
- 5 Supports service scaling

HANDS-ON

Step-1:- Run nginx Container

```
docker run -d \
--name lab-nginx \
-p 8081:80 \
-v $(pwd)/html:/usr/share/nginx/html \
nginx:alpine

docker ps
```

```

root@LAPTOP-H7SV3T8B: /m × + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker run -d \
--name lab-nginx \
-p 8081:80 \
-v "$(pwd)/html:/usr/share/nginx/html" \
nginx:alpine
05972be40049c5560895b08fbc136bd1be23e74175fedf765096e3e0d1bdfb3
INFO[2026-04-24T13:37:02.084842144Z] connecting to shim 05972be40049c5560895b08fbc136bd1be23e74175fedf765096e3e0d1bdfb3
address="unix:///run/containerd/s/bee70b896d64f872f27092871acb533cb1ea4b7fbb45778a00bd7f7b2e86de2c" namespace=moby pro
tocol=tttrpc version=3
INFO[2026-04-24T13:37:02.169760106Z] sbJoin: gwep4 ''->'6ca9f3da4625', gwep6 ''->' eid=6ca9f3da4625 ep=lab-nginx net=b
ridge nid=eddbbc0dadfc5
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker ps
CONTAINER ID        IMAGE               COMMAND                  CREATED              STATUS              PORTS
05972be40049        nginx:alpine        "/docker-entrypoint.    10 seconds ago      Up 10 seconds      0.0.0.0:8081->
80/tcp, [::]:8081->80/tcp
e19ed8631e8b        jenkins/jenkins:lts "/usr/bin/tini -- /u    6 days ago          Up 42 seconds      0.0.0.0:8080->
8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
16a9aefbac86        assignment-1_backend "docker-entrypoint.s... 3 weeks ago         Up 42 seconds      project_backend
93764d07652d        assignment-1_db      "docker-entrypoint.s... 3 weeks ago         Up 42 seconds (healthy) project_db

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# |

```

Step-2:- Create a file `index.html` mounting on a conatiner

```

nano html/index.html
cat html/index.html
docker stop lab-nginx
docker rm lab-nginx
docker run -d \
--name lab-nginx \
-p 8081:80 \
-v "$(pwd)/html:/usr/share/nginx/html" \
nginx:alpine

```

`index.html` :-

```

<h1>Experiment 6 Working</h1>
<p>Nginx is running successfully</p>

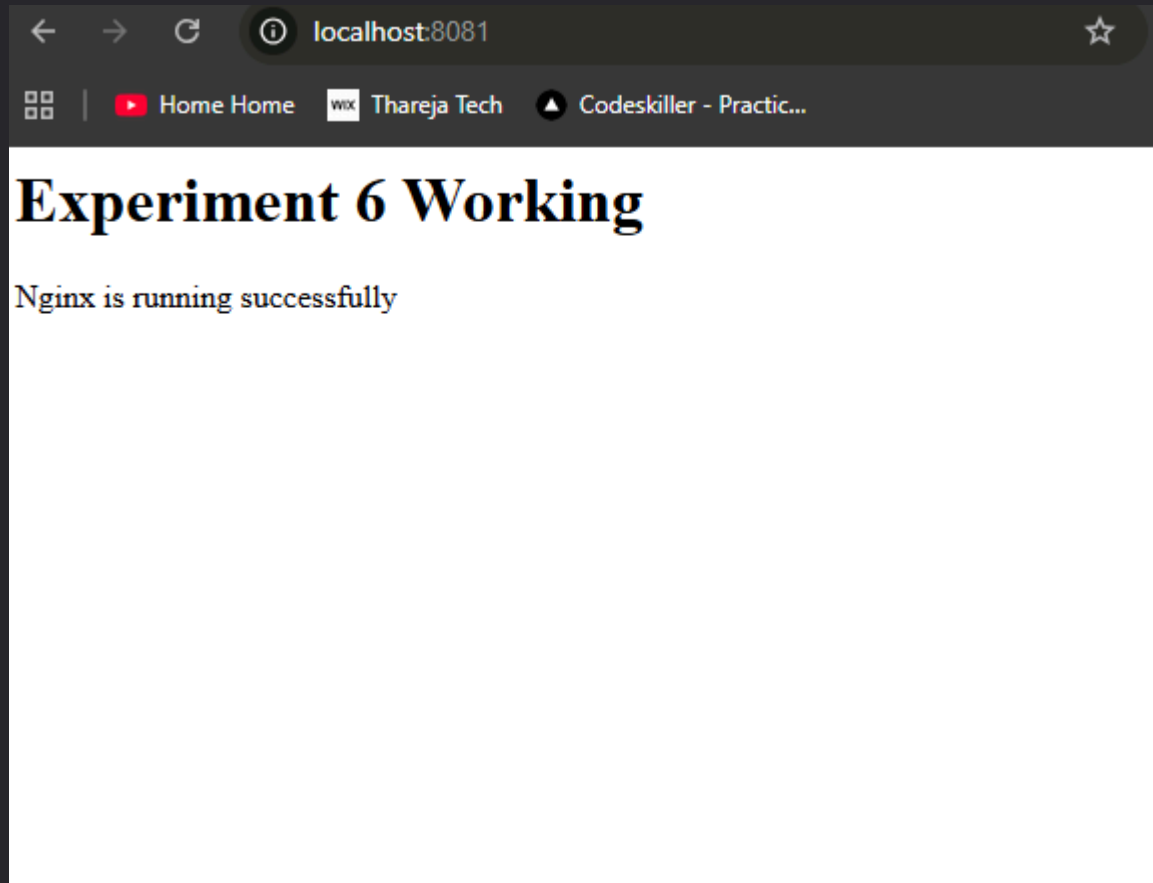
```

```

root@LAPTOP-H7SV3T8B: /m × + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# nano html/index.html
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# cat html/index.html
<h1>Experiment 6 Working</h1>
<p>Nginx is running successfully</p>
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker stop lab-nginx
INFO[2026-04-24T13:41:07.732789899Z] shim disconnected id=05972be40049c5560895b08fbc136bd1b
e23e74175fedf765096e3e0d1bdfb3 namespace=moby
INFO[2026-04-24T13:41:07.732880512Z] cleaning up after shim disconnected id=05972be40049c5560895b08fbc136bd1b
e23e74175fedf765096e3e0d1bdfb3 namespace=moby
INFO[2026-04-24T13:41:07.732888900Z] cleaning up dead shim id=05972be40049c5560895b08fbc136bd1b
e23e74175fedf765096e3e0d1bdfb3 namespace=moby
INFO[2026-04-24T13:41:07.732956760Z] ignoring event container=05972be40049c5560895b08fbc136bd1be23e74175fedf765096e3e0d1bdfb3 module=libcontainerd namespace=moby topic=/tasks/delete type="events.TaskDelete"
lab-nginx
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker rm lab-nginx
lab-nginx
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker run -d \
--name lab-nginx \
-p 8081:80 \
-v "$(pwd)/html:/usr/share/nginx/html" \
nginx:alpine
037e82ee9a2e6694381eecfef36d377c460fdbf242db57fcf10ed7fd7e63fa9

```


Step-3:- View Page on Browser



Step-4:- Create same via compose

```
nano docker-compose.yml
```

```
cat docker-compose.yml
```

```
docker-compose.yml:-
```

```
version: '3.8'

services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
```

```
root@LAPTOP-H7SV3T8B: /m  X + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# nano docker-compose.yml
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# cat docker-compose.yml
services:
  nginx:
    image: nginx:alpine
    container_name: lab-nginx
    ports:
      - "8081:80"
    volumes:
      - ./html:/usr/share/nginx/html
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# |
```

Step-5:- Create Containers via Compose

```
docker-compose up -d

docker-compose ps
```

```
root@LAPTOP-H7SV3T8B: /m  X + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker-compose up -d
Creating lab-nginx ...
INFO[2026-04-24T14:14:06.062907986Z] connecting to shim f8282a00ecaa331390dc68fe069168adac35377e9418a57c9c34b388433e373c address="unix:///run/containerd/s/
d4a2fd5edc4511cce8660d87597254634e50b29ec9807cb823758784f9539821" namespace=moby protocol=ttrpc version=3
Creating lab-nginx ... done
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker-compose ps
Name                Command                                State      Ports
-----
lab-nginx            /docker-entrypoint.sh nginx ...        Up         0.0.0.0:8081->80/tcp, :::8081->80/tcp
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# |
```

Step-6:- Delete Compose Services

```
docker-compose down
```

```
root@LAPTOP-H7SV3T8B: /m  X + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker-compose down
Stopping lab-nginx ...
INFO[2026-04-24T14:25:11.198182436Z] ignoring event                                container=f8282a00ecaa331390dc68fe069168adac35377e9418a57c9c34b388433e373
c module=libcontainerd namespace=moby topic=/tasks/delete type=*events.TaskDelete" id=f8282a00ecaa331390dc68fe069168adac35377e9418a57c9c34b388433e373c names
pace=moby
INFO[2026-04-24T14:25:11.198419936Z] cleaning up after shim disconnected             id=f8282a00ecaa331390dc68fe069168adac35377e9418a57c9c34b388433e373c names
pace=moby
INFO[2026-04-24T14:25:11.198440173Z] cleaning up dead shim                         id=f8282a00ecaa331390dc68fe069168adac35377e9418a57c9c34b388433e373c names
Stopping lab-nginx ... done
Removing lab-nginx ... done
Removing network experiment-6_default
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# |
```

Step-7:- Create Network

```
docker network create wp-net
```

```
root@LAPTOP-H7SV3T8B: /m  X + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker network create wp-net
c5f6539c3cb304377e7d663fb2fa26f44c608bc0f2f34f5e2e0562654980509b
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# |
```

Step-8:- Run `mysql` container

```
docker run -d \  
  
--name mysql \  
  
--network wp-net \  
  
-e MYSQL_ROOT_PASSWORD=secret \  
  
-e MYSQL_DATABASE=wordpress \  
  
mysql:5.7
```

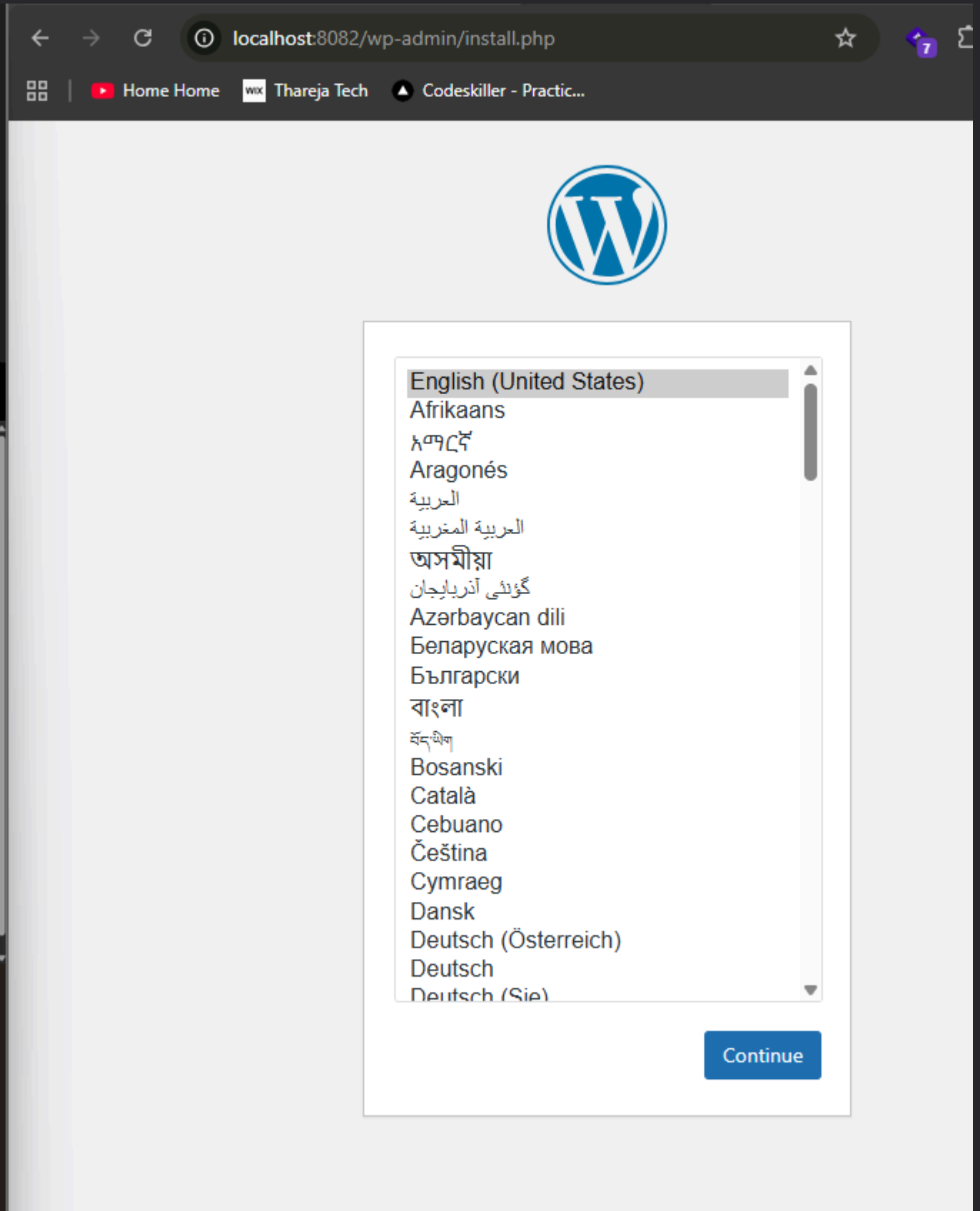
```
root@LAPTOP-H7SV3T8B: /m  x + v  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker run -d \  
--name mysql \  
--network wp-net \  
-e MYSQL_ROOT_PASSWORD=secret \  
-e MYSQL_DATABASE=wordpress \  
mysql:5.7  
a006032916f74c40fe61390514f75f62ed69b16487de94fdf641da7744636c8c  
INFO[2026-04-24T14:26:07.360745210Z] connecting to shim a006032916f74c40fe61390514f75f62ed69b16487de94fdf641da7744636c8c address="unix:///run/containerd/s/  
a305577195fea6fa199a5b04b5d946128ed60138ebd13ed4daf819ee317bbcd" namespace=moby protocol=ttrpc version=3  
INFO[2026-04-24T14:26:07.464375775Z] sbJoin: gwep4 ''->'2a2796094012', gwep6 ''->' eid=2a2796094012 ep=mysql net=wp-net nid=c5f6539c3cb3  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# |
```

Step-9:- Run `wordpress` container

```
docker run -d \  
  
--name wordpress \  
  
--network wp-net \  
  
-p 8082:80 \  
  
-e WORDPRESS_DB_HOST=mysql \  
  
-e WORDPRESS_DB_USER=root \  
  
-e WORDPRESS_DB_PASSWORD=secret \  
  
-e WORDPRESS_DB_NAME=wordpress \  
  
wordpress:latest
```

```
root@LAPTOP-H7SV3T8B: /m  x + v  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker run -d \  
--name wordpress \  
--network wp-net \  
-p 8082:80 \  
-e WORDPRESS_DB_HOST=mysql \  
-e WORDPRESS_DB_USER=root \  
-e WORDPRESS_DB_PASSWORD=secret \  
-e WORDPRESS_DB_NAME=wordpress \  
wordpress:latest  
334f3be51195531bf985a8e208d00042d9b9c157fe11c9b98f0de97b1de67ff6  
INFO[2026-04-24T16:38:49.282572977Z] connecting to shim 334f3be51195531bf985a8e208d00042d9b9c157fe11c9b98f0de97b1de67ff6 address="unix:///run/containerd/s/  
5dbf9acbbcdffa2aed795b798118ab38a836f4d5d99bef8d1e6fe9cb7d947fa7" namespace=moby protocol=ttrpc version=3  
INFO[2026-04-24T16:38:49.372237929Z] sbJoin: gwep4 ''->'c7417946db95', gwep6 ''->' eid=c7417946db95 ep=wordpress net=wp-net nid=6dfe43ad9538  
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# |
```

Step-10:- Verify Page on Browser



Step-11:- Wordpress & MySql via compose

```
mkdir wp-compose
cd wp-compose
```

```
nano docker-compose.yml
```

```
cat docker-compose.yml
```

```
services:
```

```
  mysql:
```

```
    image: mysql:5.7
```

```
    container_name: mysql
```

```
    restart: always
```

```
    environment:
```

```
      MYSQL_ROOT_PASSWORD: secret
```

```
      MYSQL_DATABASE: wordpress
```

```
      MYSQL_USER: wpuser
```

```
      MYSQL_PASSWORD: secret
```

```
    volumes:
```

```
      - mysql_data:/var/lib/mysql
```

```
  wordpress:
```

```
    image: wordpress:latest
```

```
    container_name: wordpress
```

```
    restart: always
```

```
    ports:
```

```
      - "8082:80"
```

```
    environment:
```

```
      WORDPRESS_DB_HOST: mysql
```

```
      WORDPRESS_DB_USER: wpuser
```

```
      WORDPRESS_DB_PASSWORD: secret
```

```
    depends_on:
```

```
      - mysql
```

```
volumes:
```

```
  mysql_data:
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# mkdir wp-compose
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# cd wp-compose
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/wp-compose# nano docker-compose.yml
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/wp-compose# cat docker-compose.yml
services:
  mysql:
    image: mysql:5.7
    container_name: mysql
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: secret
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: secret
    volumes:
      - mysql_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    container_name: wordpress
    restart: always
    ports:
      - "8082:80"
    environment:
      WORDPRESS_DB_HOST: mysql
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: secret
    depends_on:
      - mysql

volumes:
  mysql_data:
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/wp-compose# |

```

Step-12:- Start Services

```
docker-compose up -d
```

```
docker-compose ps
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/wp-compose# docker-compose up -d
Creating mysql ...
INFO[2026-04-24T16:43:13.035407552Z] connecting to shim 632383334c570253c9dbe8dde5fec5399874d3df683c7518c631e0af66282e59 address="unix:///run/containerd/s/
07fc46e1a0aaaa20450bfb6283af3003d50ac45f61d62e246e5f09320df4cb36" namespace=moby protocol=ttrpc version=3
Creating mysql ... done
Creating wordpress ...
INFO[2026-04-24T16:43:13.225873195Z] connecting to shim 013d92e4693cbbd0990d36870cc58550ed28e6af839fc-f152cc1bf937f40aca0 address="unix:///run/containerd/s/
be7729e9ad05bd4ca92a8d8eab6410899c47cdb0bde37d3682c466feeacfe427" namespace=moby protocol=ttrpc version=3
Creating wordpress ... done
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/wp-compose# docker-compose ps
Name                Command                                State      Ports
-----
mysql               docker-entrypoint.sh mysqld            Up         3306/tcp, 33060/tcp
wordpress           docker-entrypoint.sh apache2 ...      Up         0.0.0.0:8082->80/tcp, :::8082->80/tcp
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/wp-compose# |

```

Step-13:- View On Browser

← → ↻ ⓘ localhost:8082/wp-admin/install.php 🔑 ☆ 8 📁 | 📄 All Book

📺 | 📺 Home Home 📺 Thareja Tech 📺 Codeskiller - Practic...



Welcome

Welcome to the famous five-minute WordPress installation process! Just fill in the information below and you'll be on your way to using the most extendable and powerful personal publishing platform in the world.

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title

Username

Usernames can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password [Hide](#)

Strong

Important: You will need this password to log in. Please store it in a secure location.

Your Email

Double-check your email address before continuing.

Search engine visibility ☐ Discourage search engines from indexing this site

It is up to search engines to honor this request.

[Install WordPress](#)

Step-14:- Create container with Node Image

```
docker run -d \  
--name webapp \  
-p 5000:5000 \  
-e APP_ENV=production \  
-e DEBUG=false
```

```
--restart unless-stopped \
```

```
node:18-alpine
```

```
root@LAPTOP-H7SV3T8B: /m  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# docker run -d \
--name webapp \
-p 5000:5000 \
-e APP_ENV=production \
-e DEBUG=false \
--restart unless-stopped \
node:18-alpine
7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729
INFO[2026-04-24T16:52:09.626031574Z] connecting to shim 7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 address="unix:///run/containerd/s/
3c7c90391d2b9b2fced0430ffd42699bc2fd612d97f24170bd65567118f3ae13" namespace=moby protocol=ttrpc version=3
INFO[2026-04-24T16:52:09.713299663Z] sbJoin: gwep4 ''->'c6b6ca874569', gwep6 ''->' eid=c6b6ca874569 ep=webapp net=bridge nid=eddbbc0dadfc5
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# INFO[2026-04-24T16:52:09.902556205Z] ignoring event
container=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 module=libcontainerd namespace=moby topic=/tasks/delete
type=*events.TaskDelete"
INFO[2026-04-24T16:52:09.902734611Z] shim disconnected id=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 names
pace=moby
INFO[2026-04-24T16:52:09.902830095Z] cleaning up after shim disconnected id=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 names
pace=moby
INFO[2026-04-24T16:52:09.902838953Z] cleaning up dead shim id=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 names
pace=moby
INFO[2026-04-24T16:52:10.121965982Z] connecting to shim 7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 address="unix:///run/containerd/s/
3c7c90391d2b9b2fced0430ffd42699bc2fd612d97f24170bd65567118f3ae13" namespace=moby protocol=ttrpc version=3
INFO[2026-04-24T16:52:10.213062083Z] sbJoin: gwep4 ''->'7221bd19418f', gwep6 ''->' eid=7221bd19418f ep=webapp net=bridge nid=eddbbc0dadfc5
INFO[2026-04-24T16:52:10.275911949Z] shim disconnected id=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 names
pace=moby
INFO[2026-04-24T16:52:10.275954356Z] cleaning up after shim disconnected id=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 names
pace=moby
INFO[2026-04-24T16:52:10.275969593Z] cleaning up dead shim id=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 names
pace=moby
INFO[2026-04-24T16:52:10.276066592Z] ignoring event container=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f772
9 module=libcontainerd namespace=moby topic=/tasks/delete type=*events.TaskDelete"
INFO[2026-04-24T16:52:10.501556043Z] connecting to shim 7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 address="unix:///run/containerd/s/
3c7c90391d2b9b2fced0430ffd42699bc2fd612d97f24170bd65567118f3ae13" namespace=moby protocol=ttrpc version=3
INFO[2026-04-24T16:52:10.576021997Z] sbJoin: gwep4 ''->'8d5649071ad0', gwep6 ''->' eid=8d5649071ad0 ep=webapp net=bridge nid=eddbbc0dadfc5
INFO[2026-04-24T16:52:10.638508986Z] shim disconnected id=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 names
pace=moby
INFO[2026-04-24T16:52:10.638561517Z] cleaning up after shim disconnected id=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 names
pace=moby
INFO[2026-04-24T16:52:10.638568670Z] cleaning up dead shim id=7b56b43eb16eb3c48b79b25864f7d451d9a03a3227d7cdc62001f7d0a43f7729 names
```

Step-15:- Create Node container via Compose

```
mkdir task3-webapp
```

```
cd task3-webapp
```

```
nano docker-compose.yml
```

```
cat docker-compose.yml
```

```
docker-compose.yml:-
```

```
services:
```

```
  webapp:
```

```
    image: node:18-alpine
```

```
    container_name: webapp
```

```
    ports:
```

```
      - "5000:5000"
```

```
    environment:
```

```
      APP_ENV: production
```

```
      DEBUG: "false"
```

```
    restart: unless-stopped
```



```

root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# mkdir task3-webapp
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6# cd task3-webapp
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# nano docker-compose.yml
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# cat docker-compose.yml
services:
  webapp:
    image: node:18-alpine
    container_name: webapp
    ports:
      - "5000:5000"
    environment:
      APP_ENV: production
      DEBUG: "false"
    restart: unless-stopped
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# |

```

Step-16:- Start Services

```

docker-compose up -d

docker-compose ps

```

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker-compose up -d
Creating network "task3-webapp_default" with the default driver
Creating webapp ...
INFO[2026-04-24T17:04:57.653659098Z] connecting to shim 9a4b34e0ef14e7357487d81ad50bc29da737fbcd9935499643a6a688948d72e3  address="unix:///run/containerd/s/
9cf7d727cdd498c902b8e82b1c91ea3ec591dee7aa917b3c5672ba0eede06074" namespace=moby protocol=ttrpc version=3
Creating webapp ... done
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker-compose ps
Name                                Command                                State                                Ports
-----
webapp  docker-entrypoint.sh sh -c ...  Up                                0.0.0.0:5000->5000/tcp, :::5000->5000/tcp
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# |

```

Step-17:- Veridy On Browser

localhost:5000

Hello from Compose

Step-18:- Stop Services

```

docker-compose down

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker-compose down
Stopping webapp ...
INFO[2026-04-24T17:08:44.938651779Z] Container failed to exit within 10s of signal 15 - using the force container=9a4b34e0ef14e7357487d81ad50bc29da737fbc9
935499643a6a688948d72e3
INFO[2026-04-24T17:08:44.958445678Z] ignoring event container=9a4b34e0ef14e7357487d81ad50bc29da737fbc9935499643a6a688948d72e
3 module=libcontainerd namespace=moby topic=/tasks/delete type="events.TaskDelete"
INFO[2026-04-24T17:08:44.958558708Z] shim disconnected id=9a4b34e0ef14e7357487d81ad50bc29da737fbc9935499643a6a688948d72e3 names
pace=moby
INFO[2026-04-24T17:08:44.958627949Z] cleaning up after shim disconnected id=9a4b34e0ef14e7357487d81ad50bc29da737fbc9935499643a6a688948d72e3 names
pace=moby
INFO[2026-04-24T17:08:44.958633858Z] cleaning up dead shim id=9a4b34e0ef14e7357487d81ad50bc29da737fbc9935499643a6a688948d72e3 names
pace=moby
WARN[2026-04-24T17:08:44.966774229Z] ShouldRestart failed, container will not be restarted container=9a4b34e0ef14e7357487d81ad50bc29da737fbc9935499643a6a6
88948d72e3 daemonShuttingDown=false error="restart canceled" execDuration=3m47.31931437s exitStatus="{137 2026-04-24 17:08:44.951144008 +0000 UTC}" hasBeenM
Stopping webapp ... done
Removing webapp ... done
Removing network task3-webapp_default
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp#

```

Step-19:- Create Network

```
docker network create app-net
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker network create app-net
3a9baa83d43de952f6acb3ce01bb24e8e57810994132b17c4274a4bb96a06c79
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp#

```

Step-20:- Create Python & Postgres Conatiners via Terminal

```

docker run -d \
--name postgres-db \
--network app-net \
-e POSTGRES_USER=admin \
-e POSTGRES_PASSWORD=secret \
-v pgdata:/var/lib/postgresql/data \
postgres:15

```

```

docker run -d \
--name backend \
--network app-net \
-p 8000:8000 \
-e DB_HOST=postgres-db \
-e DB_USER=admin \
-e DB_PASS=secret \
python:3.11-slim

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker run -d \
--name postgres-db \
--network app-net \
-e POSTGRES_USER=admin \
-e POSTGRES_PASSWORD=secret \
-v pgdata:/var/lib/postgresql/data \
postgres:15
Unable to find image 'postgres:15' locally
15: Pulling from library/postgres
3531af2bc2a9: Pull complete
be67b9cdfb60: Pull complete
c375454b4ff0: Pull complete
ff8482cd03b7: Pull complete
ac28d232c7f3: Pull complete
fa77b44f96d2: Pull complete
dd1cc88a000a: Pull complete
58e6420a0782: Pull complete
4c7f2d7f4f80: Pull complete
3e0c43622abe: Pull complete
0892560966bb: Pull complete
9fa0198235u9: Pull complete
00da1877cea6: Pull complete
26803cb302a4: Pull complete
Digest: sha256:29342cb52157b098821961d2c14eec3c019071f56a5d559e990cf07cf541ea9b
Status: Downloaded newer image for postgres:15
081756c3e24d1cca5ab3fafc225452953a49bc539e85d4f0eb9d08b5bd805289
INFO[2026-04-24T17:12:58.631154633Z] connecting to shim 081756c3e24d1cca5ab3fafc225452953a49bc539e85d4f0eb9d08b5bd805289 address="unix:///run/containerd/s/
6c52bdb33f6631cdf2b723c7ae18fb0afe89ebbac57ce6ce8d24eab7bf97aab2" namespace=moby protocol=ttrpc version=3
INFO[2026-04-24T17:12:58.721081430Z] sbJoin: gwep4 '-->'9861806b0389', gwep6 '-->' eid=9861806b0389 ep=postgres-db net=app-net nid=3a9baa83d43d
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker run -d \
--name backend \
--network app-net \
-p 8000:8000 \
-e DB_HOST=postgres-db \
-e DB_USER=admin \
-e DB_PASS=secret \
python:3.11-slim
c1966fcddc1c9319eb5428b68260a9465f40b35fd4077098f31be5228396ad45
INFO[2026-04-24T17:13:06.881723274Z] connecting to shim c1966fcddc1c9319eb5428b68260a9465f40b35fd4077098f31be5228396ad45 address="unix:///run/containerd/s/
ebd27ef9a070e558d9a19ef9bb1e98b34f91da3b3f31f85eb34bbf7b474c5b1b" namespace=moby protocol=ttrpc version=3
INFO[2026-04-24T17:13:06.985518490Z] sbJoin: gwep4 '-->'7bb3a73f4b55', gwep6 '-->' eid=7bb3a73f4b55 ep=backend net=app-net nid=3a9baa83d43d

```

Step-21:- List Containers

```
docker ps
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
081756c3e24d   postgres:15    "docker-entrypoint.s..." 32 seconds ago Up 31 seconds 5432/tcp
013d92e4693c   wordpress:latest "docker-entrypoint.s..." 30 minutes ago Up 30 minutes 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
632383334c57   mysql:5.7      "docker-entrypoint.s..." 30 minutes ago Up 30 minutes 3306/tcp, 33060/tcp
e19ed8631e8b   jenkins/jenkins:lts "/usr/bin/tini -- /u..." 6 days ago    Up 4 hours    0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:
50000->50000/tcp, [::]:50000->50000/tcp
16a9aefbac86   assignment-1_backend "docker-entrypoint.s..." 3 weeks ago   Up 4 hours
93764d07652d   assignment-1_db "docker-entrypoint.s..." 3 weeks ago   Up 4 hours (healthy)
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp#

```

Step-22:- List Networks

```
docker network ls
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker network ls
NETWORK ID     NAME                DRIVER    SCOPE
3a9baa83d43d   app-net             bridge    local
270fd966565e   assignment-1_project_ipvlan_net  ipvlan    local
eddbcb0dadfc5   bridge              bridge    local
59ef1b01feab   experiment-7_default bridge    local
2199230a6a27   experiment-10_sonarqube-lab      bridge    local
f9d0a2e00f87   host                host      local
48f719b2f044   my-app-default      bridge    local
6e205de754b8   none                null      local
edfd9da8a952   wp-compose_default  bridge    local
6dfe43ad9538   wp-net              bridge    local
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp#

```

Step-23:- List Volumes

```
docker volumes ls
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker volume ls
DRIVER      VOLUME NAME
local       5adec6f1fd1fd7d0d249bfd34529c8d772ba02553d9891060b9ca3e97707289d
local       7df3c4e10ea43120bd4dc69c374ef146853c1d3a23eb0765a7072734fed86973
local       09a2291714933ee6889ffc76de1876c22b3a4155202699955f11bdf6a1444a83
local       86be085b5455eee60c38d3b54f4d4347b8bbf91118ea3d8effeb5939abe3681d
local       96d5aac1be625e3afd0d1db89c5112f2e1fc50296192ff49a066d5b0ae93ce5
local       97abc2469bbbae8e9e8b62c571869002f2de49417dc0a8ba52e725e8444bd465d
local       977ec5890ad30e336b04679d0ea385c24ab78029c4e625c56ecd8924fba02c79
local       97251bc64de6294f5cbea3e83185e937cb990dd5a8569bd7caa5e8d33f88dba4
local       a819b7b6209c286bbe583c4b946df9698026b1a600d26f7adfdfe85d2f9622f9
local       app-volume
local       assignment-1_pg_data
local       assignment-1_pgdata
local       b550c7299d1b79facb631c0beddaa6181f26ad245d419b9fc3eae7458e5f635
local       c34a011f99407a3e2811198cd00c903baafa9c9dd6806f96e33370f9ae4d8d1d
local       c872746c05ccabc409df2943119227b6c4da156a2ed34e813fd6bb462c860bb
local       dhainyaVolume
local       e527fb6bd846ae63d87485de9a10379b5908b0025af6e7acfd40a9a7175b93a
local       experiment-7_jenkins_home
local       experiment-10_sonar-data
local       experiment-10_sonar-db-data
local       experiment-10_sonar-extensions
local       jenkins_home
local       k3d-mycluster-images
local       lecture-14_mysql_data
local       lecture-14_wp_content
local       my-app_jenkins_home
local       mydata
local       mysql-data
local       pgdata
local       portainer_data
local       wordpress-scale_mysql_data
local       wordpress-scale_wp_content
local       wp-compose_mysql_data
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# |
```

Step-24:- Stop Containers & Remove Network

```
docker stop backend postgres-db
```

```
docker rm backend postgres-db
```

```
docker network rm app-net
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker stop backend postgres-db
backend
INFO[2026-04-24T17:14:49.536028937Z] ignoring event                                container=081756c3e24d1cca5ab3fac225452953a49bc539e85d4f0eb9d08b5bd80528
9 module=libcontainerd namespace=moby topic=/tasks/delete type="*events.TaskDelete"
INFO[2026-04-24T17:14:49.536298055Z] shim disconnected                                id=081756c3e24d1cca5ab3fac225452953a49bc539e85d4f0eb9d08b5bd805289 names
pace=moby
INFO[2026-04-24T17:14:49.536465998Z] cleaning up after shim disconnected              id=081756c3e24d1cca5ab3fac225452953a49bc539e85d4f0eb9d08b5bd805289 names
pace=moby
INFO[2026-04-24T17:14:49.536475243Z] cleaning up dead shim                          id=081756c3e24d1cca5ab3fac225452953a49bc539e85d4f0eb9d08b5bd805289 names
pace=moby
postgres-db
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker rm backend postgres-db
backend
postgres-db
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# docker network rm app-net
app-net
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task3-webapp# |
```

Step-25:- Create same services via Compose

```
nano docker-compose.yml
```

```
cat docker-compose.yml
```

```
docker-compose.yml:-
```

```
services:
```

```
  postgres-db:
```

```
image: postgres:15
container_name: postgres-db
environment:
  POSTGRES_USER: admin
  POSTGRES_PASSWORD: secret
volumes:
  - pgdata:/var/lib/postgresql/data
networks:
  - app-net
```

backend:

```
image: python:3.11-slim
container_name: backend
ports:
  - "8000:8000"
```

environment:

```
DB_HOST: postgres-db
DB_USER: admin
DB_PASS: secret
```

depends_on:

```
- postgres-db
```

networks:

```
- app-net
```

volumes:

```
pgdata:
```

networks:

```
app-net:
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# nano docker-compose.yml
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# cat docker-compose.yml
services:
  postgres-db:
    image: postgres:15
    container_name: postgres-db
    environment:
      POSTGRES_USER: admin
      POSTGRES_PASSWORD: secret
    volumes:
      - pgdata:/var/lib/postgresql/data
    networks:
      - app-net

  backend:
    image: python:3.11-slim
    container_name: backend
    ports:
      - "8000:8000"
    environment:
      DB_HOST: postgres-db
      DB_USER: admin
      DB_PASS: secret
    depends_on:
      - postgres-db
    networks:
      - app-net

volumes:
  pgdata:

networks:
  app-net:
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# |

```

Step-26:- Start Services

```
docker-compose up -d
```

```
docker-compose ps
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# docker-compose up -d
Creating network "task4-app_app-net" with the default driver
Creating volume "task4-app_pgdata" with default driver
Creating postgres-db ...
INFO[2026-04-24T17:17:22.703632687Z] connecting to shim 450ecd2d4f2cd0fb2475f11b176d0120dcee2003da52013f5d02a8badefee12c address="unix:///run/containerd/s/
a54f0d0441c0455b06e9f5c05b4d6978d61b41dc2ca9d2d1c8b7581db5bab4b8" namespace=moby protocol=ttrpc version=3
Creating postgres-db ... done
Creating backend ...
INFO[2026-04-24T17:17:22.879895091Z] connecting to shim c147cc2cd879dd432cb33d2204266bdb2048f1080c3f778e41cb71cc6840cf57 address="unix:///run/containerd/s/
9be541c7203a5ca86ea90a37e965029cd5827a7b3259d641d5e6f173a4de8480" namespace=moby protocol=ttrpc version=3
Creating backend ... done
INFO[2026-04-24T17:17:23.040829048Z] ignoring event                                container=c147cc2cd879dd432cb33d2204266bdb2048f1080c3f778e41cb71cc6840cf5
7 module=libcontainerd namespace=moby topic=/tasks/delete type="events.TaskDelete"
INFO[2026-04-24T17:17:23.040906430Z] shim disconnected                                id=c147cc2cd879dd432cb33d2204266bdb2048f1080c3f778e41cb71cc6840cf57 names
pace=moby
INFO[2026-04-24T17:17:23.040962211Z] cleaning up after shim disconnected              id=c147cc2cd879dd432cb33d2204266bdb2048f1080c3f778e41cb71cc6840cf57 names
pace=moby
INFO[2026-04-24T17:17:23.040967984Z] cleaning up dead shim                          id=c147cc2cd879dd432cb33d2204266bdb2048f1080c3f778e41cb71cc6840cf57 names
pace=moby
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# docker-compose ps
   Name                       Command                                State      Ports
-----
backend      python3                                Up          5032/tcp
postgres-db  docker-entrypoint.sh postgres         Up          5432/tcp
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# |

```

Step-27:- List Volumes

```
docker volume ls
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# docker volume ls
DRIVER      VOLUME NAME
local       5adec6f1fd1fd7d0d249bfd34529c8d772ba02553d9891060b9ca3e97707289d
local       7df3c4e10ea43120bd4dc69c374ef146853c1d3a23eb0765a7072734fed86973
local       09a2291714933ee6889ffc76de1876c22b3a4155202699955f11bdf6a1444a83
local       86be085b5455eee60c38d3b54f4d4347b8bbf91118ea3d8effeb5939abe3681d
local       96d5aac1be625e3afd0d1dbe89c5112f2e1fc50296192ff49a066d5b0ae93ce5
local       97abc2469bbae8e9e8b62c571869002f2de49417dc0a8ba52e725e8444bd465d
local       973ec5890ad30e336b04679d0ea385c24ab78029c4e625c56ecd8924fba92c79
local       97251bc64de6294f5cbea3e83185e937cb990dd5a8569bd7caa5e8d33f88dba4
local       a819b7b6209c286bbe583c4b946df9698026b1a600d26f7adfdfe85d2f9622f9
local       app-volume
local       assignment-1_pg_data
local       assignment-1_pgdata
local       b550c7299d1b79facb631c0beddaa6181f26ad245d419b9fc3eaed7458e5f635
local       c34a011f99407a3e2811198cd00c903baafa9c9dd6806f96e33370f9ae4d8d1d
local       c872746c05ccabc409df29443119227b6c4da156a2ed34e813fd6bb462c860bb
local       dhairyaVolume
local       e527fb6bd846ae63d87485de9a10379b5908b0025af6e7acfc40a9a7175b93a
local       experiment-7_jenkins_home
local       experiment-10_sonar-data
local       experiment-10_sonar-db-data
local       experiment-10_sonar-extensions
local       jenkins_home
local       k3d-mycluster-images
local       lecture-14_mysql_data
local       lecture-14_wp_content
local       my-app_jenkins_home
local       mydata
local       mysql-data
local       pgdata
local       portainer_data
local       task4-app_pgdata
local       wordpress-scale_mysql_data
local       wordpress-scale_wp_content
local       wp-compose_mysql_data
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# |
```

Step-28:- Stop Services

```
docker-compose down
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# docker-compose down
Stopping postgres-db ...
INFO[2026-04-24T17:18:19.464143547Z] ignoring event          container=450ecd2d4f2cd0fb2475f11b176d0120dcee2003da52013f5d02a8badefee12
c module=libcontainerd namespace=moby topic=/tasks/delete type="*events.TaskDelete"
INFO[2026-04-24T17:18:19.464318360Z] shim disconnected      id=450ecd2d4f2cd0fb2475f11b176d0120dcee2003da52013f5d02a8badefee12c names
pace=moby
INFO[2026-04-24T17:18:19.464388848Z] cleaning up after shim disconnected id=450ecd2d4f2cd0fb2475f11b176d0120dcee2003da52013f5d02a8badefee12c names
pace=moby
INFO[2026-04-24T17:18:19.464406860Z] cleaning up dead shim id=450ecd2d4f2cd0fb2475f11b176d0120dcee2003da52013f5d02a8badefee12c names
Stopping postgres-db ... done
Removing backend ... done
Removing postgres-db ... done
Removing network task4-app_app-net
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task4-app# s|
```

Step-29:- Create Node app

```
nano app.js

cat app.js
```

app.js:-

```
const http = require('http');

http.createServer((req, res) => {

    res.end("Docker Compose Build Lab");

}).listen(3000);
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# nano app.js
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# cat app.js
const http = require('http');

http.createServer((req, res) => {
  res.end("Docker Compose Build Lab");
}).listen(3000);
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# |
```

Step-30:- Create Dockerfile

```
nano Dockerfile
cat Dockerfile
```

Dockerfile:-

```
FROM node:18-alpine

WORKDIR /app

COPY app.js .

EXPOSE 3000

CMD ["node", "app.js"]
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# nano Dockerfile
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# cat Dockerfile
FROM node:18-alpine

WORKDIR /app

COPY app.js .

EXPOSE 3000

CMD ["node", "app.js"]
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# |
```

Step-31:- Create Compose File

```
nano docker-compose.yml
cat docker-compose.yml
```

docker-compose.yml:-

```
services:
  nodeapp:
```



```
build:

  context: .

  dockerfile: Dockerfile

container_name: custom-node-app

ports:

  - "3000:3000"
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# nano docker-compose.yml
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# cat docker-compose.yml
services:
  nodeapp:
    build:
      context: .
      dockerfile: Dockerfile
    container_name: custom-node-app
    ports:
      - "3000:3000"
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# |
```

Step-32:- Start Compose Services

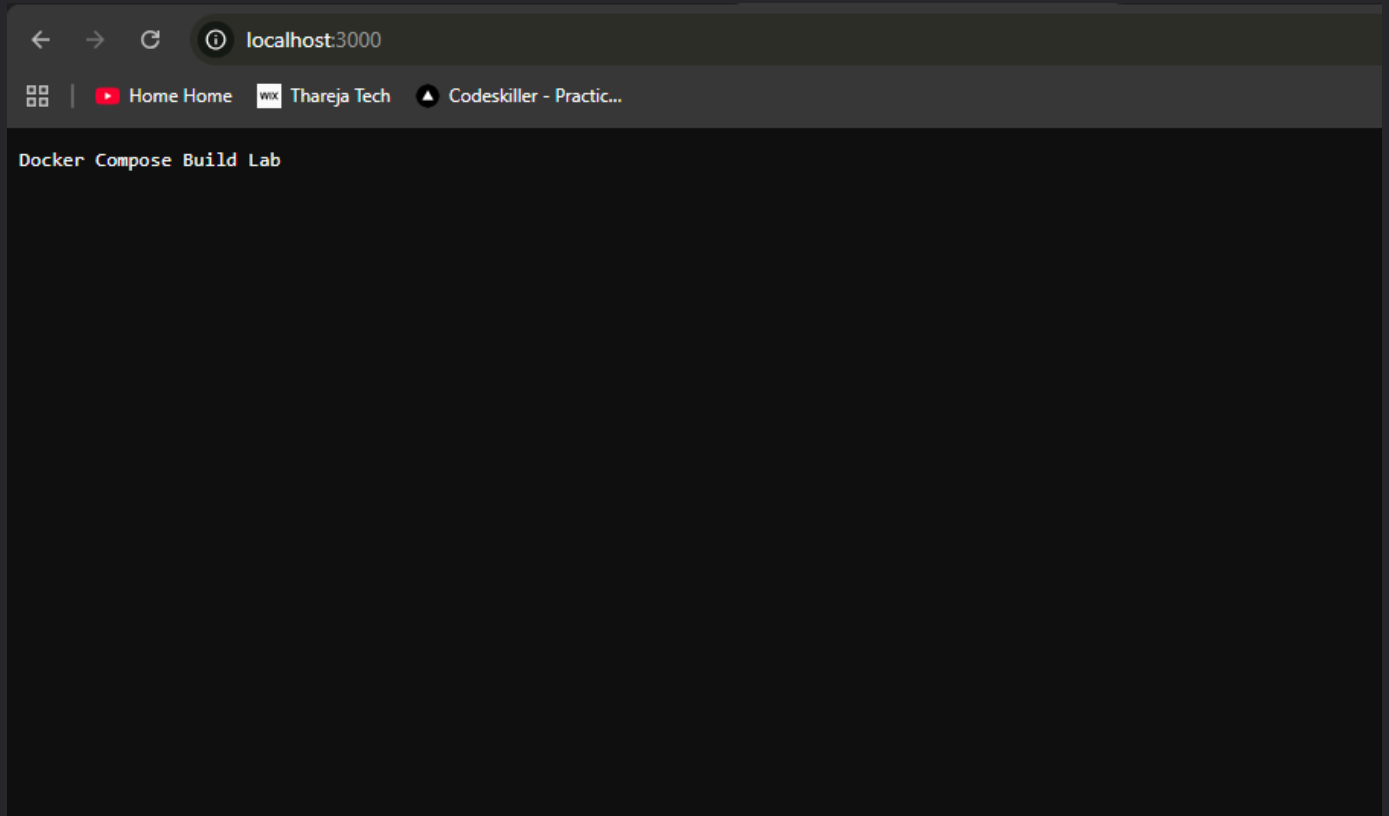
```
docker-compose up --build -d

docker-compose ps
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# docker-compose up --build -d
Creating network "task5-nodeapp_default" with the default driver
Building nodeapp
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 4.096kB
Step 1/5 : FROM node:18-alpine
----> ee77c6cd7c18
Step 2/5 : WORKDIR /app
----> Using cache
----> b44ffa3cb89e9
Step 3/5 : COPY app.js .
----> 8c833ef5abc3
Step 4/5 : EXPOSE 3000
----> Running in 12e9f773af64
INFO[2026-04-24T17:22:31.611134876Z] Cleaned up layer sha256:a70c2428cccf2c209cd4c3688c5763092f6b7f053667cdd03a2d0c0c7c890208 chainID="sha256:a70c2428cccf2
c209cd4c3688c5763092f6b7f053667cdd03a2d0c0c7c890208"
----> Removed intermediate container 12e9f773af64
----> 24b4c4f4b53c
Step 5/5 : CMD ["node", "app.js"]
----> Running in f5a8c78fcba4
INFO[2026-04-24T17:22:31.656378748Z] Cleaned up layer sha256:a70c2428cccf2c209cd4c3688c5763092f6b7f053667cdd03a2d0c0c7c890208 chainID="sha256:a70c2428cccf2
c209cd4c3688c5763092f6b7f053667cdd03a2d0c0c7c890208"
----> Removed intermediate container f5a8c78fcba4
----> 0c4ed2e10261
Successfully built 0c4ed2e10261
Successfully tagged task5-nodeapp_nodeapp:latest
Creating custom-node-app ...
INFO[2026-04-24T17:22:31.719773727Z] connecting to shim ef0ccf22cf6ae2b316e871b6b8a845e9ede3d55c9174796a43ed2ee6ec80430a address="unix:///run/containerd/s/
19ea40b218059f890b969b6e05e9a9185121eeb47211deb9712f45195d276c4" namespace=moby protocol=ttrpc version=3
INFO[2026-04-24T17:22:31.813417556Z] sbJoin: gwep4 ''->'393048dcd2e3', gwep6 ''->' eid=393048dcd2e3 ep=custom-node-app net=task5-nodeapp_default nid=829db
Creating custom-node-app ... done
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# docker-compose ps
-----
Name                                Command                                State      Ports
-----
custom-node-app  docker-entrypoint.sh node ... Up         0.0.0.0:3000->3000/tcp,:::3000->3000/tcp
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-6/task5-nodeapp# |
```

Step-33:- Verify on Browser



CONCLUSION

This experiment demonstrated:

- How to deploy **WordPress + MySQL** using Docker Compose.
- How containers communicate using internal networking.
- Importance of **volumes for persistence**.

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Built by Dhairya Thareja 🚀

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Experiment 7: CI/CD using Jenkins, GitHub and Docker Hub

INTRODUCTION

WHAT IS JENKINS?

Jenkins is a **web-based GUI automation server** used to:

- Build applications
- Test code
- Deploy software

It provides:

- Dashboard (browser-based UI)
- Plugin ecosystem (GitHub, Docker, etc.)
- Pipeline as Code using `Jenkinsfile`

WHAT IS CI/CD ??

- **Continuous Integration (CI):** Code is automatically built and tested after each commit
- **Continuous Deployment (CD):** Built artifacts (Docker images) are automatically delivered/deployed

WORKFLOW OVERVIEW

Developer → GitHub → Webhook → Jenkins → Build → Docker Hub

PREREQUISITES

Create a repository on GitHub:

```
[my-app](https://github.com/dhairiyathareja/my-app.git)
```

Project Structure

my-app/

├─ app.py

├─ requirements.txt

- └─ Dockerfile
- └─ Jenkinsfile

HANDS-ON

Step-1:- Create a `app.py`

```
nano app.py
```

Paste This:

```
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    # return "Hello from CI/CD Pipeline!"
    return "Hello from CI/CD Pipeline!, my sapid is 500118984"

app.run(host="0.0.0.0", port=80)
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# cat app.py
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    # return "Hello from CI/CD Pipeline!"
    return "Hello from CI/CD Pipeline!, my sapid is 500118984"

app.run(host="0.0.0.0", port=80)
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# |
```

Step-2:- Create `requirements.txt`

```
nano requirements.txt
```

Paste this:

flask

```
root@LAPTOP-H7SV3T8B: /m × + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# cat requirements.txt
flask
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# |
```

Step-3:- Create Dockerfile

nano Dockerfile

Paste this:

```
FROM python:3.10-slim

WORKDIR /app

COPY . .

RUN pip install -r requirements.txt

EXPOSE 80

CMD ["python", "app.py"]
```

```
root@LAPTOP-H7SV3T8B: /m × + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# cat Dockerfile
FROM python:3.10-slim

WORKDIR /app
COPY . .

RUN pip install -r requirements.txt

EXPOSE 80
CMD ["python", "app.py"]
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# |
```

Step-4:- Create Jenkinsfile

nano Jenkinsfile

Paste this:

```
pipeline {
  agent any

  environment {
    IMAGE_NAME = "dhairyathareja/myapp"
  }

  stages {

    stage('Build Docker Image') {
      steps {
        s++h 'docker build -t $IMAGE_NAME:latest .'
      }
    }

    stage('Login to Docker Hub') {
      steps {
        withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')]) {
          sh 'echo $DOCKER_TOKEN | docker login -u dhairyathareja --password-stdin'
        }
      }
    }

    stage('Push to Docker Hub') {
      steps {
        sh 'docker push $IMAGE_NAME:latest'
      }
    }
  }
}
```

```
root@LAPTOP-H7SV3T8B: /m × + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# cat Jenkinsfile
pipeline {
  agent any

  environment {
    IMAGE_NAME = "dhairyathareja/myapp"
  }

  stages {
    stage('Clone Source') {
      steps {
        git 'https://github.com/dhairyathareja/my-app.git'
      }
    }

    stage('Build Docker Image') {
      steps {
        sh 'docker build -t $IMAGE_NAME:latest .'
      }
    }

    stage('Login to Docker Hub') {
      steps {
        withCredentials([string(credentialsId: 'dockerhub-token', variable: 'DOCKER_TOKEN')]) {
          sh 'echo $DOCKER_TOKEN | docker login -u dhairyathareja --password-stdin'
        }
      }
    }

    stage('Push to Docker Hub') {
      steps {
        sh 'docker push $IMAGE_NAME:latest'
      }
    }
  }
}
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# |
```

Step-5:- Create `docker-compose.yml`

version: '3.8'

services:

jenkins:

image: jenkins/jenkins:lts

container_name: jenkins

restart: always

ports:

- "8080:8080"

- "50000:50000"

volumes:

- jenkins_home:/var/jenkins_home

- /var/run/docker.sock:/var/run/docker.sock

user: root

volumes:

jenkins_home:

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ∨

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# cat docker-compose.yml
version: '3.8'

services:
  jenkins:
    image: jenkins/jenkins:lts
    container_name: jenkins
    restart: always
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock
    user: root

volumes:
  jenkins_home:
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7#

```

Step-6:- Start Up Compose

```
docker-compose up -d
```

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ∨

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# docker-compose up -d
Creating jenkins ...
INFO[2026-04-18T06:17:34.779262285Z] connecting to shim 804b25a0c62caed64a3c6afb22a215004187d12f76a174380668047aa6310a26 address="unix:///run/containerd/s/
28aba6d1b736b2d6c71acd3675788b7775ad7f50db75e689bdde464718153cb5" namespace=moby protocol=ttrpc version=3
Creating jenkins ... done
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# docker ps
CONTAINER ID   IMAGE          COMMAND                  NAMES                  CREATED        STATUS              PORTS
804b25a0c62c   jenkins/jenkins:lts   "/usr/bin/tini -- /u_    jenkins                10 seconds ago   Up 9 seconds        0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0
.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
16a9aefbac86   assignment-1_backend   "docker-entrypoint.s_    project_backend        2 weeks ago     Up About a minute
93764d07652d   assignment-1_db        "docker-entrypoint.s_    project_db              2 weeks ago     Up About a minute (healthy)
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# |

```

Step-7:- Exec Jenkins Contrainer to get Password

```
docker exec -it jenkins cat /var/jenkins_home/secrets/initialAdminPassword
```

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ∨

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/Experiment-7# docker exec -it jenkins bash
root@804b25a0c62c:/# cat /var/jenkins_home/secrets/initialAdminPassword
f7aaefc791d4a7c9abd93e2adb1916c
root@804b25a0c62c:/# |

```

Step-8:- Login to Jenkins Dashboard on Browser

```
http://localhost:8080
```


Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

Continue

Step-9:- Install Sugeested Plugins

Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

Step-10:- Plugins List

Getting Started

Getting Started

✓ Folders	✓ OWASP Markup Formatter	✓ Build Timeout	✓ Credentials Binding
✓ Timestampers	✓ Workspace Cleanup	✓ Ant	✓ Gradle
🔗 Pipeline	🔗 GitHub Branch Source	🔗 Pipeline: GitHub Groovy Libraries	🔗 Pipeline Graph View
🔗 Git	🔗 SSH Build Agents	🔗 Matrix Authorization Strategy	🔗 LDAP
🔗 Email Extension	🔗 Mailer	🔗 Dark Theme	

Workspace Cleanup

Ant

Gradle

Jenkins 2.541.3

Step-11:- Create Admin User

Getting Started

Create First Admin User

Username

dhairythareja

Password

Confirm password

Full name

Dhairya Thareja

E-mail address

Jenkins 2.541.3

[Skip and continue as admin](#)[Save and Continue](#)

Step-12:- Admin Dashboard Interface

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

Search

Settings

Help

Add description

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

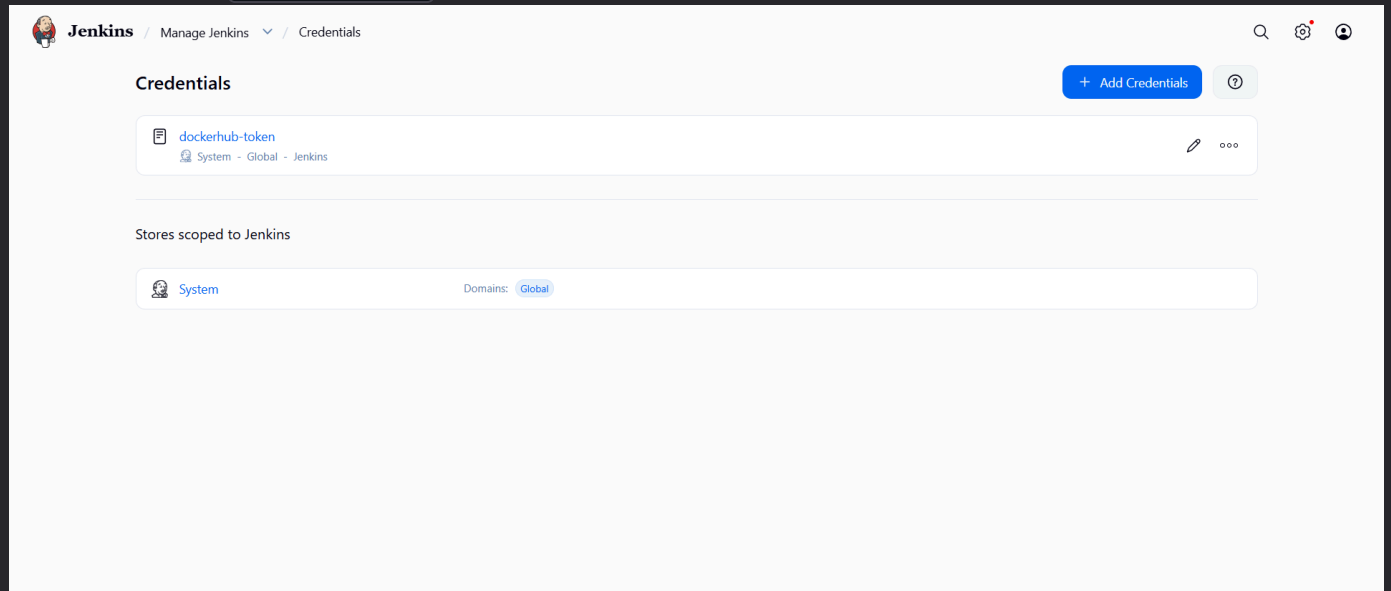
Set up a distributed build

Set up an agent

Configure a cloud

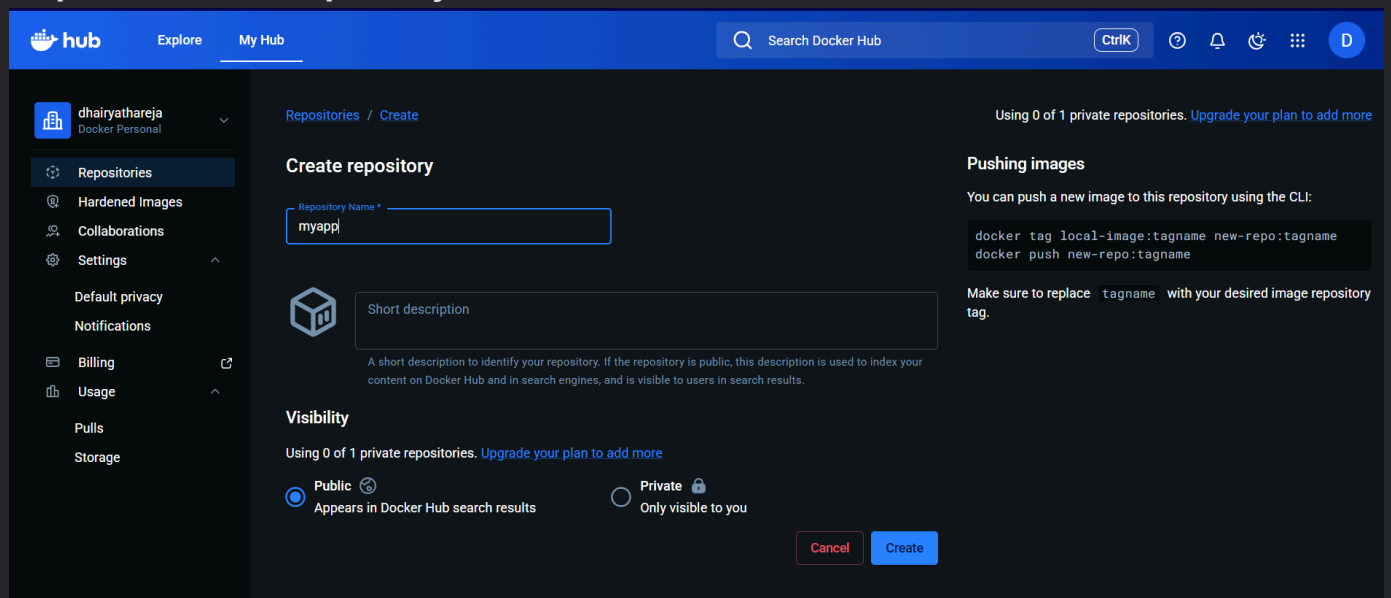
Learn more about distributed builds

Step-13:- Add DockerHub Token in Credentials



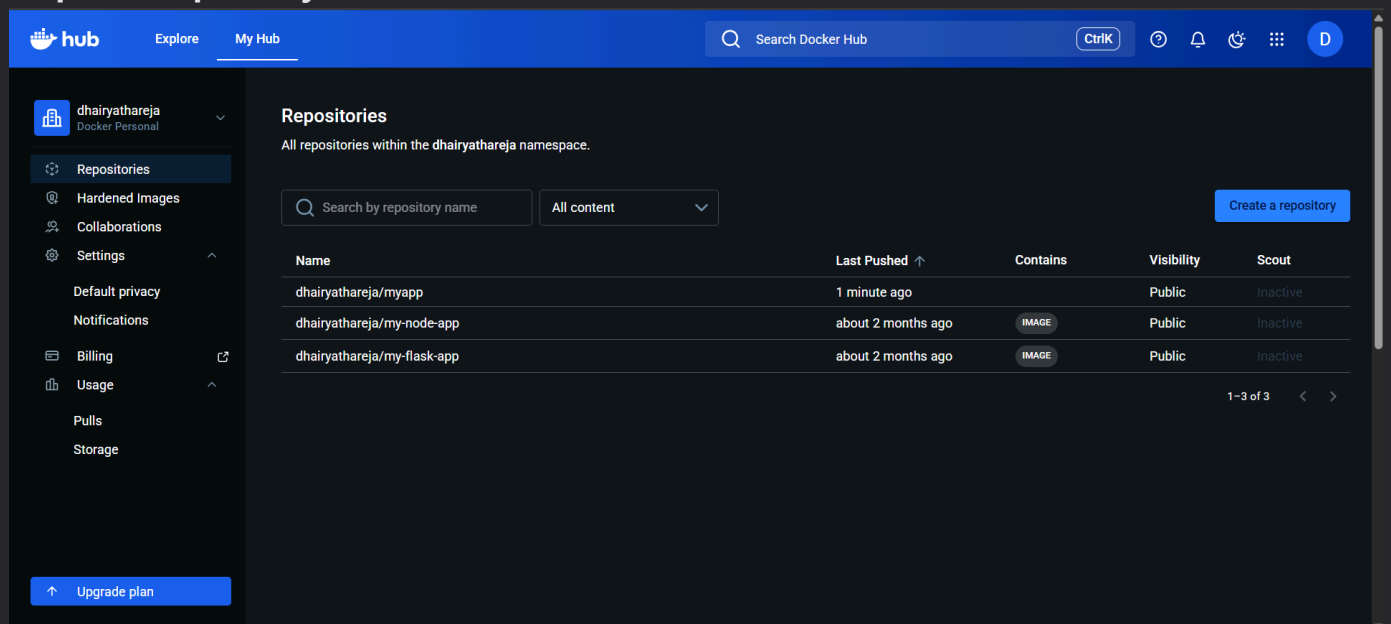
The Jenkins interface shows the 'Credentials' page. At the top, there's a breadcrumb trail: 'Jenkins / Manage Jenkins / Credentials'. A blue button '+ Add Credentials' is in the top right. Below it, a list of credentials is shown, with one entry: 'dockerhub-token' (System - Global - Jenkins). Below the list, it says 'Stores scoped to Jenkins' and shows 'System' as the selected store. The 'Domains' section shows 'Global' as the selected domain.

Step-14:- Create a Repository in Dockerhub



The Docker Hub 'Create repository' page. The left sidebar shows the user 'dhairythareja' and navigation options: Repositories, Hardened Images, Collaborations, Settings, Default privacy, Notifications, Billing, Usage, Pulls, and Storage. The main content area is titled 'Create repository' and shows a form with 'Repository Name' set to 'myapp'. Below the name is a 'Short description' field. The 'Visibility' section shows 'Public' selected, with a note 'Appears in Docker Hub search results'. The 'Pushing images' section provides CLI commands: 'docker tag local-image:tagname new-repo:tagname' and 'docker push new-repo:tagname'. A note says 'Make sure to replace tagname with your desired image repository tag.' At the bottom are 'Cancel' and 'Create' buttons.

Step-15:- Repository will be listed

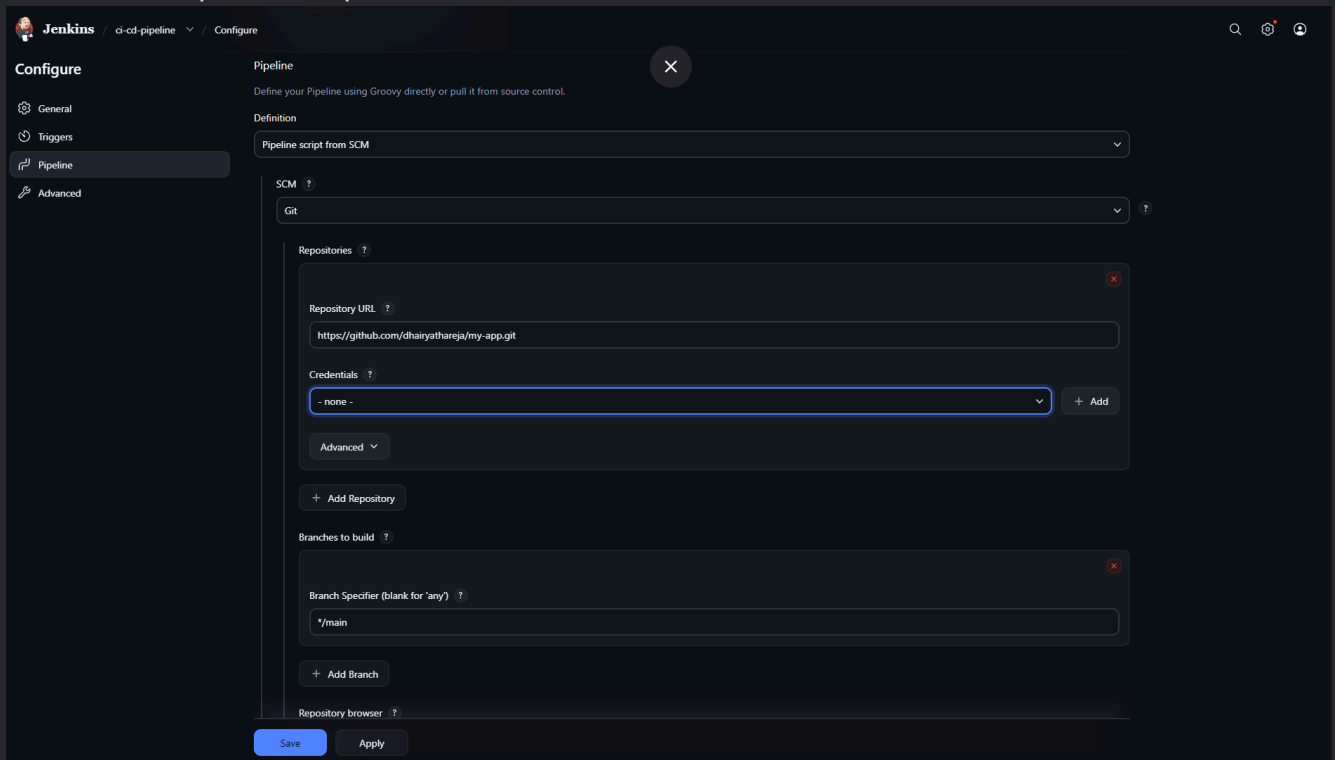


The Docker Hub 'Repositories' page for the user 'dhairythareja'. It shows a list of repositories within the namespace. The table has columns: Name, Last Pushed, Contains, Visibility, and Scout. The repositories listed are 'dhairythareja/myapp', 'dhairythareja/my-node-app', and 'dhairythareja/my-flask-app'. A 'Create a repository' button is in the top right. At the bottom, there's a pagination bar showing '1-3 of 3'.

Name	Last Pushed	Contains	Visibility	Scout
dhairythareja/myapp	1 minute ago		Public	Inactive
dhairythareja/my-node-app	about 2 months ago	IMAGE	Public	Inactive
dhairythareja/my-flask-app	about 2 months ago	IMAGE	Public	Inactive

Step-16:- Create Pipeline Job

- New-Item:- Pipeline
- Name:- `ci-cd-pipeline`
- Definition:- Pipeline Script from SCM



The screenshot shows the Jenkins configuration page for a new pipeline job named 'ci-cd-pipeline'. The 'Definition' is set to 'Pipeline script from SCM'. Under 'SCM', 'Git' is selected. A repository is added with the URL 'https://github.com/dhairiyathareja/my-app.git' and no credentials. The 'Branches to build' section shows the branch specifier '*/main'. The 'Repository browser' is set to 'Git'. The 'Save' button is highlighted in blue.

Jenkins / ci-cd-pipeline / Configure

Configure

General
Triggers
Pipeline
Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script from SCM

SCM

Git

Repositories

Repository URL

https://github.com/dhairiyathareja/my-app.git

Credentials

- none -

+ Add

Advanced

+ Add Repository

Branches to build

Branch Specifier (blank for 'any')

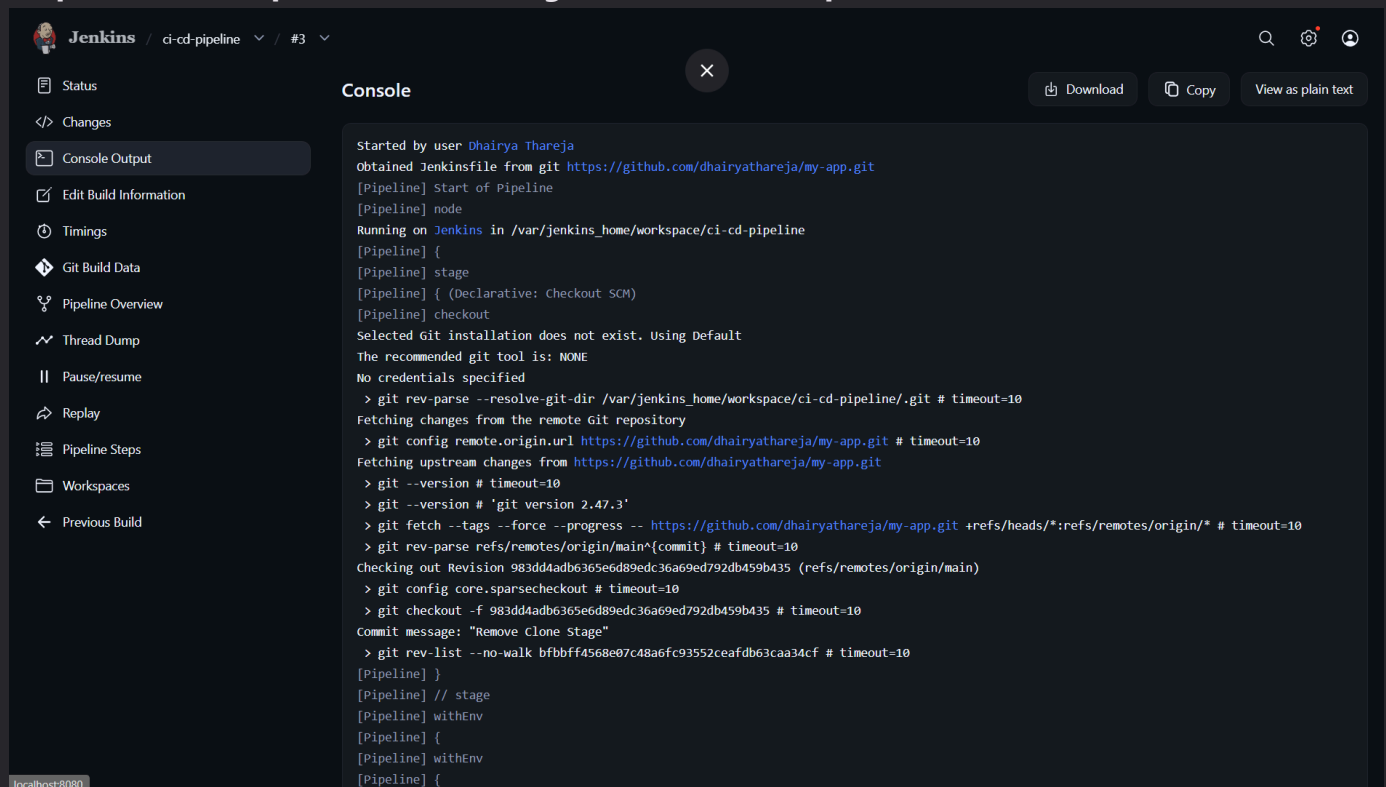
*/main

+ Add Branch

Repository browser

Save Apply

Step-17:- Build Pipeline & Track Stages in Console Output



The screenshot shows the Jenkins console output for the 'ci-cd-pipeline' job. The console output displays the pipeline execution steps, including cloning the repository, checking out the code, and running the pipeline script. The 'Console Output' tab is selected in the left sidebar.

Jenkins / ci-cd-pipeline / #3

Status
Changes
Console Output
Edit Build Information
Timings
Git Build Data
Pipeline Overview
Thread Dump
Pause/resume
Replay
Pipeline Steps
Workspaces
Previous Build

Console

Download Copy View as plain text

```
Started by user dhairya Thareja
obtained Jenkinsfile from git https://github.com/dhairiyathareja/my-app.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/jenkins_home/workspace/ci-cd-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/ci-cd-pipeline/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/dhairiyathareja/my-app.git # timeout=10
Fetching upstream changes from https://github.com/dhairiyathareja/my-app.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/dhairiyathareja/my-app.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 983dd4adb6365e6d89edc36a69ed792db459b435 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 983dd4adb6365e6d89edc36a69ed792db459b435 # timeout=10
Commit message: "Remove Clone Stage"
> git rev-list --no-walk bfbff4568e07c48a6fc93552ceaafdb63caa34cf # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
```

Step-18:- Build Status Success will be shown with Green Tick

 Jenkins / ci-cd-pipeline

Status

</> Changes

▶ Build Now

⚙️ Configure

🗑️ Delete Pipeline

📁 Stages

✎ Rename

🔍 Pipeline Syntax

✓ ci-cd-pipeline

Permalinks

- Last build (#3), 2 min 24 sec ago
- Last stable build (#3), 2 min 24 sec ago
- Last successful build (#3), 2 min 24 sec ago
- Last failed build (#2), 13 min ago
- Last unsuccessful build (#2), 13 min ago
- Last completed build (#3), 2 min 24 sec ago

Builds >

Q Filter

/

Today

✓ #3 08:56

✗ #2 08:45

✗ #1 08:44

Step-19:- Repo will be consisting Image Now !!!

 dhairyathareja
Docker Personal

Repositories

🔍 Hardened Images

👤 Collaborations

⚙️ Settings

Default privacy

Notifications

💰 Billing

📄 Usage

Pulls

Storage

Upgrade plan

Repositories

All repositories within the dhairyathareja namespace.

Q Search by repository name

All content

Create a repository

Name	Last Pushed ↑	Contains	Visibility	Scout
dhairyathareja/myapp	2 minutes ago	IMAGE	Public	Inactive
dhairyathareja/my-node-app	about 2 months ago	IMAGE	Public	Inactive
dhairyathareja/my-flask-app	about 2 months ago	IMAGE	Public	Inactive

1-3 of 3 < >

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Built by Dhairya Thareja 🚀

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Experiment 9: Working with Ansible

INTRODUCTION

WHAT IS ANSIBLE?

- Ansible is an open-source **automation tool** for **configuration management**, **application deployment**, and **orchestration**.
- It follows an **agentless** architecture, using SSH for Linux and WinRM for Windows.
- Uses YAML-based **playbooks** to define automation tasks.

Ansible is software that enables cross-platform automation and orchestration at scale and has become the standard choice among enterprise automation solutions.

KEY CONCEPTS

Component	Description
Control Node	Machine with Ansible installed
Managed Nodes	Target servers (no Ansible agent needed)
Inventory	Defines the list of managed nodes (EC2 instances, servers, etc.). <code>inventory.ini</code>
Playbooks	YAML files containing a sequence of automation steps.
Tasks	Individual actions in playbooks (e.g., installing a package).
Modules	Built-in functionality to perform tasks (e.g., <code>yum</code> , <code>apt</code> , <code>service</code>).
Roles	Pre-defined reusable automation scripts.

BENEFITS OF USING ANSIBLE

- A free and open-source community project with a huge audience.
- Battle-tested over many years as the preferred tool of IT wizards.
- Easy to start and use from day one, without the need for any special coding skills.
- Simple deployment workflow without any extra agents.
- Includes some sophisticated features around modularity and reusability that come in handy as users become more proficient.

- Extensive and comprehensive official documentation that is complemented by a plethora of online material produced by its community.

To sum up, Ansible is simple yet powerful, agentless, community-powered, predictable, and secure.

HANDS-ON

Step-1:- Install Ansible in WSL

```
sudo apt install ansible -y
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# sudo apt install ansible -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libllvm19 libwayland-server0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  ansible-core ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos python3-libcloud
  python3-lockfile python3-netaddr python3-ntlm-auth python3-passlib python3-requests-ntlm python3-resolverlib
  python3-selinux python3-simplejson python3-winrm python3-xmlltodict
Suggested packages:
  cowsay sshpass python3-trio python3-aiquic python3-h2 python3-httpx python3-httpcore python-lockfile-doc ipython3
  python-netaddr-docs
The following NEW packages will be installed:
  ansible-core ieee-data python3-argcomplete python3-dnspython python3-jmespath python3-kerberos
  python3-libcloud python3-lockfile python3-netaddr python3-ntlm-auth python3-passlib python3-requests-ntlm
  python3-resolverlib python3-selinux python3-simplejson python3-winrm python3-xmlltodict
0 upgraded, 18 newly installed, 0 to remove and 0 not upgraded.
Need to get 22.0 MB of archives.
After this operation, 330 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-resolverlib all 1.0.1-1 [25.7 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-dnspython all 2.6.1-1ubuntu1 [163 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 ieee-data all 20220827.1 [2113 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-netaddr all 0.8.0-2ubuntu1 [319 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/universe amd64 ansible-core all 2.16.3-0ubuntu2 [1280 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 ansible all 9.2.0+dfsg-0ubuntu5 [16.4 MB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 python3-argcomplete all 3.1.4-1ubuntu0.1 [33.8 kB]
Get:8 http://archive.ubuntu.com/ubuntu noble/main amd64 python3-jmespath all 1.0.1-1 [21.3 kB]
Get:9 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-kerberos amd64 1.1.14-3.1build9 [21.2 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/universe amd64 python3-lockfile all 1:0.12.2-3 [13.7 kB]
```

Step-2:- Verify Installation

```
ansible --version
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/root/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /root/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Mar 3 2026, 12:15:18) [GCC 13.3.0] (/usr/bin/python3)
  jinja version = 3.1.2
  libyaml = True
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9#
```

Step-3:- Ping Localhost

```
ansible localhost -m ping
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ansible localhost -m ping
[WARNING]: No inventory was parsed, only implicit localhost is available
localhost | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9#

```

Step-4:- Generate `ssh` Keys

```
ssh -keygen -t rsa -b 4096
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:QurPAW7leoXtqpdxiWY9yQVPOCtgiAGCDwE6JYMSo/U root@LAPTOP-H7SV3T8B
The key's randomart image is:
+----[RSA 4096]-----+
|&+o . .          |
|O*.. o  + .       |
|=o E... *         |
|.. o . . . o      |
|.. o o SB +       |
|.. o o* X         |
|+ . o . * .       |
|. o o + .         |
|+..              |
+----[SHA256]-----+
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9#

```

Step-5:- Copy keys in `id_rsa`

```

#Copy in current Directory

cp ~/.ssh/id_rsa.pub .

cp ~/.ssh/id_rsa .

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# cp ~/.ssh/id_rsa.pub .
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# cp ~/.ssh/id_rsa .
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9#

```

Step-6:- Create Dockerfile

```

FROM ubuntu

RUN apt update -y

RUN apt install -y python3 python3-pip openssh-server

RUN mkdir -p /var/run/sshd

# Configure SSH

RUN mkdir -p /run/sshd && \
    echo 'root:password' | chpasswd && \

```



```
sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config && \  
sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config && \  
sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config
```

Create .ssh directory and set proper permissions

```
RUN mkdir -p /root/.ssh && \  
    chmod 700 /root/.ssh
```

Copy SSH keys (note: this is not secure for production!)

```
COPY id_rsa /root/.ssh/id_rsa  
COPY id_rsa.pub /root/.ssh/authorized_keys
```

Set proper permissions for keys

```
RUN chmod 600 /root/.ssh/id_rsa && \  
    chmod 644 /root/.ssh/authorized_keys
```

Fix for SSH login

```
RUN sed -i 's@session\s*required\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc/pam.d/sshd
```

Expose SSH port

```
EXPOSE 22
```

Start SSH service when container starts

```
CMD ["/usr/sbin/sshd", "-D"]
```

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# nano Dockerfile
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# cat Dockerfile
FROM ubuntu

RUN apt update -y
RUN apt install -y python3 python3-pip openssh-server
RUN mkdir -p /var/run/ssh

# Configure SSH
RUN mkdir -p /run/ssh && \
  echo 'root:password' | chpasswd && \
  sed -i 's/#PermitRootLogin prohibit-password/PermitRootLogin yes/' /etc/ssh/sshd_config && \
  sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config && \
  sed -i 's/#PubkeyAuthentication yes/PubkeyAuthentication yes/' /etc/ssh/sshd_config

# Create .ssh directory and set proper permissions
RUN mkdir -p /root/.ssh && \
  chmod 700 /root/.ssh

# Copy SSH keys (note: this is not secure for production!)
COPY id_rsa /root/.ssh/id_rsa
COPY id_rsa.pub /root/.ssh/authorized_keys

# Set proper permissions for keys
RUN chmod 600 /root/.ssh/id_rsa && \
  chmod 644 /root/.ssh/authorized_keys

# Fix for SSH login
RUN sed -i 's@session\s*required\s*pam_loginuid.so@session optional pam_loginuid.so@g' /etc/pam.d/ssh

# Expose SSH port
EXPOSE 22

# Start SSH service when container starts
CMD ["/usr/sbin/sshd", "-D"]
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# |

```

Step-7:- Build Image

```
docker build -t ubuntu-server
```

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# cp ~/.ssh/id_rsa.pub .
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# docker build -t ubuntu-server .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
  https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  497.2kB
Step 1/12 : FROM ubuntu
--> 493218ed0f40
Step 2/12 : RUN apt update -y
--> Running in 673ec897953c
INFO[2026-04-11T09:38:23.822356479Z] connecting to shim 673ec897953c0c8b319cc6c2948cc6fa25ba4396b6fdb91b44c9bcbdc2cfffca  address="unix:///run/containerd/s/
757a85b40b09e2c2b99705de91b7e12d03e4712ecf93aed9c2f4644a71cb82c" namespace=moby protocol=ttrpc version=3
INFO[2026-04-11T09:38:23.908864691Z] sbJoin: gwep4 ''->'9a0b8d1010cd', gwep6 ''->'  eid=9a0b8d1010cd ep=funny_cerf net=bridge nid=1fc2207311fe

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1989 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [3505 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1508 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [34.8 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]

```

Step-8:- Run container from build Image

```
docker run -d -p --rm 2222:22 -p 8221:8221 --name ssh-test-server ubuntu-server
```

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# docker run -d --rm -p 2222:22 -p 8221:8221 --name ssh-test-server ubuntu-server
47c375822e46241f471ae2859c69dbe17e8bb952d4c89a44bf251a39739bb21
INFO[2026-04-11T09:41:00.087528313Z] connecting to shim 47c375822e46241f471ae2859c69dbe17e8bb952d4c89a44bf251a39739bb21  address="unix:///run/containerd/s/
941f709057b5450b78298d1d14123cfd39ecdb7ee263b11f0425acd047514d9d" namespace=moby protocol=ttrpc version=3
INFO[2026-04-11T09:41:00.183085205Z] sbJoin: gwep4 ''->'8e2e61c41bc9', gwep6 ''->'  eid=8e2e61c41bc9 ep=ssh-test-server net=bridge nid=1fc2207311fe
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# |

```

Step-9:- Find Container IP Address

```
docker inspect -f '{{.IPAddress}}' ssh-test-server
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' ssh-test-server
172.17.0.2
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# |
```

Step-10:- Test SSH connection via Conatiner IP

```
# with key

ssh -i ~/.ssh/id_rsa root@172.17.0.2
```

```
root@47c375822e46: ~
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' ssh-test-server
172.17.0.2
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ssh -i ~/.ssh/id_rsa root@172.17.0.2
The authenticity of host '172.17.0.2 (172.17.0.2)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIk0K0EEeuZKu7xnIIJpYQ4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.17.0.2' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@47c375822e46:~# |
```

Step-11:- Start Multiple Conatiners

```
for i in {1..4}; do
    echo -e "\n Creating server${i}\n"
    docker run -d --rm -p 220${i}:22 --name server${i} ubuntu-server
    echo -e "IP of server${i} is $(docker inspect -f '{{.IPAddress}}' server${i})"
done
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# for i in {1..4}; do
echo -e "\n Creating server${i}\n"
docker run -d --rm -p 220${i}:22 --name server${i} ubuntu-server
echo -e "IP of server${i} is $(docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i})"
done

Creating server1
00560ebb3a1c5486de3c377b3973425581edf9cde769ebd4290d1f421cf34b74
INFO[2026-04-11T10:04:06.189364843Z] connecting to shim 00560ebb3a1c5486de3c377b3973425581edf9cde769ebd4290d1f421cf34b74 address="unix:///run/containerd/s/
15260699979606e9eda0eacd723a39fc7b9006df16eb3f3407f8c71f2a5a01d0" namespace=moby protocol=ttrpc version=3
INFO[2026-04-11T10:04:06.339834719Z] sbJoin: gwep4 '-->'90e80ce6748c', gwep6 '-->' eid=90e80ce6748c ep=server1 net=bridge nid=1fc2207311fe
IP of server1 is 172.17.0.3

Creating server2
c69fdf74c895ee081922f86788c69f29d7f8228e08554076f4fc39268301ad62
INFO[2026-04-11T10:04:06.452933277Z] connecting to shim c69fdf74c895ee081922f86788c69f29d7f8228e08554076f4fc39268301ad62 address="unix:///run/containerd/s/
4f17510f4960395afed66f9fa73c79ca4b7fd48c70a58cc95eabfcc4dad0d685" namespace=moby protocol=ttrpc version=3
INFO[2026-04-11T10:04:06.574953190Z] sbJoin: gwep4 '-->'9133238c9958', gwep6 '-->' eid=9133238c9958 ep=server2 net=bridge nid=1fc2207311fe
IP of server2 is 172.17.0.4

Creating server3
f86936d4aef1fd6f971da24554d4e9ce6aba4ffb5f6c985285ccf8aec0d81cb0
INFO[2026-04-11T10:04:06.698464033Z] connecting to shim f86936d4aef1fd6f971da24554d4e9ce6aba4ffb5f6c985285ccf8aec0d81cb0 address="unix:///run/containerd/s/
ffa353ff3ee9ee0a178733ad6ec7e5a0f7c03a7a3786892342c126af49a4e675" namespace=moby protocol=ttrpc version=3
INFO[2026-04-11T10:04:06.803279831Z] sbJoin: gwep4 '-->'c8ecb10ffaf2', gwep6 '-->' eid=c8ecb10ffaf2 ep=server3 net=bridge nid=1fc2207311fe
IP of server3 is 172.17.0.5

Creating server4
1194c80b8334ffe02a1ecc64c2665ffc251ac6332fb3ebc2deb050b20b03174a
INFO[2026-04-11T10:04:06.915195502Z] connecting to shim 1194c80b8334ffe02a1ecc64c2665ffc251ac6332fb3ebc2deb050b20b03174a address="unix:///run/containerd/s/
0c08830b1d0fc4e3261f6859250cd3ef39ff6ca329faf1064ec99a393d3fe7ed" namespace=moby protocol=ttrpc version=3
INFO[2026-04-11T10:04:07.016301377Z] sbJoin: gwep4 '-->'55d281108244', gwep6 '-->' eid=55d281108244 ep=server4 net=bridge nid=1fc2207311fe
IP of server4 is 172.17.0.6
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# |

```

Step-12:- Create `inventory.ini`

```

# Get container IPs

echo "[servers]" > inventory.ini

for i in {1..4}; do

    docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i} >> inventory.ini

done

# Add inventory variables

cat << EOF >> inventory.ini

[servers:vars]

ansible_user=root

; ansible_ssh_private_key_file=./id_rsa

ansible_ssh_private_key_file=~/.ssh/id_rsa

ansible_python_interpreter=/usr/bin/python3

EOF

```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# echo "[servers]" > inventory.ini
for i in {1..4}; do
docker inspect -f '{{range.NetworkSettings.Networks}}{{.IPAddress}}{{end}}' server${i} >> inventory.ini
done

# Add inventory variables
cat << EOF >> inventory.ini

[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=./id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
EOF
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# |

```

Step-13:- Verify `inventory.ini`

```
cat inventory.ini
```

```
root@LAPTOP-H7SV3T8B: /m  × + ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# cat inventory.ini
[servers]
172.17.0.3
172.17.0.4
172.17.0.5
172.17.0.6

[servers:vars]
ansible_user=root
; ansible_ssh_private_key_file=~/id_rsa
ansible_ssh_private_key_file=~/.ssh/id_rsa
ansible_python_interpreter=/usr/bin/python3
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# |
```

Step-14:- Test Connectivity

```
ssh -i ~/.ssh/id_rsa root@172.17.0.3
```

```
root@LAPTOP-H7SV3T8B: /m  × + ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ssh -i ~/.ssh/id_rsa root@172.17.0.3
The authenticity of host '172.17.0.3 (172.17.0.3)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIkOK0EEeuZKu7xnIIJpYQ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.17.0.3' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

root@00560ebb3alc:~# exit
logout
Connection to 172.17.0.3 closed.
```

**

Step-15:- Ansible Ping Test

```
ansible all -i inventory.ini -m ping
```

```

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ansible all -i inventory.ini -m ping
The authenticity of host '172.17.0.5 (172.17.0.5)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIkOK0EEeuZKu7xnIIJpYQ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:7: [hashed name]
The authenticity of host '172.17.0.6 (172.17.0.6)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIkOK0EEeuZKu7xnIIJpYQ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:7: [hashed name]
The authenticity of host '172.17.0.4 (172.17.0.4)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIkOK0EEeuZKu7xnIIJpYQ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:7: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? 172.17.0.3 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
yes
172.17.0.5 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}

172.17.0.6 | UNREACHABLE! => {
  "changed": false,
  "msg": "Failed to connect to the host via ssh: Host key verification failed.",
  "unreachable": true
}
yes
172.17.0.4 | SUCCESS => {
  "changed": false,
  "ping": "pong"
}
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9#

```

Step-16:- Create Playbook `playbook1.yml`

```

--- # it should start with three dash only

- name: Update and configure servers

  hosts: all

  become: yes


  tasks:

    - name: Update apt packages

      apt:

        update_cache: yes

        upgrade: dist


    - name: Install required packages

      apt:

        name: ["vim", "htop", "wget"]

        state: present


    - name: Create test file

      copy:

        dest: /root/ansible_test.txt

        content: "Configured by Ansible on "

```

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# nano playbook1.yml
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# cat playbook1.yml
--- # it should start with three dash only
- name: Update and configure servers
  hosts: all
  become: yes

  tasks:
    - name: Update apt packages
      apt:
        update_cache: yes
        upgrade: dist

    - name: Install required packages
      apt:
        name: ["vim", "htop", "wget"]
        state: present

    - name: Create test file
      copy:
        dest: /root/ansible_test.txt
        content: "Configured by Ansible on {{ inventory_hostname }}"
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# |

```

Step-17:- Run Playbook

```
ansible-playbook -i inventory.ini playbook1.yml
```

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ansible-playbook -i inventory.ini playbook1.yml

PLAY [Update and configure servers] *****

TASK [Gathering Facts] *****
The authenticity of host '172.17.0.6 (172.17.0.6)' can't be established.
ED25519 key fingerprint is SHA256:wMFVV4BleU9UF7sTduY3wIkOK0EEeuZKu7xnIIjpYQ4.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:4: [hashed name]
  ~/.ssh/known_hosts:7: [hashed name]
  ~/.ssh/known_hosts:8: [hashed name]
  ~/.ssh/known_hosts:9: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? ok: [172.17.0.4]
ok: [172.17.0.3]
ok: [172.17.0.5]
yes
ok: [172.17.0.6]

TASK [Update apt packages] *****
changed: [172.17.0.4]
changed: [172.17.0.6]
changed: [172.17.0.5]
changed: [172.17.0.3]

TASK [Install required packages] *****
changed: [172.17.0.6]
changed: [172.17.0.5]
changed: [172.17.0.3]
changed: [172.17.0.4]

TASK [Create test file] *****
changed: [172.17.0.5]
changed: [172.17.0.3]
changed: [172.17.0.6]
changed: [172.17.0.4]

PLAY RECAP *****
172.17.0.3      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.17.0.4      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.17.0.5      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
172.17.0.6      : ok=4    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0

```

Step-18:- Verify Changes

```
ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
```

```

root@LAPTOP-H7SV3T8B: /m  ×  +  ▾
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# ansible all -i inventory.ini -m command -a "cat /root/ansible_test.txt"
172.17.0.6 | CHANGED | rc=0 >>
Configured by Ansible on 172.17.0.6
172.17.0.4 | CHANGED | rc=0 >>
Configured by Ansible on 172.17.0.4
172.17.0.5 | CHANGED | rc=0 >>
Configured by Ansible on 172.17.0.5
172.17.0.3 | CHANGED | rc=0 >>
Configured by Ansible on 172.17.0.3
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# |

```

Step-19:- CleanUp

```
for i in {1..4}; do docker rm -f server${i}; done
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-9# for i in {1..4}; do docker rm -f server${i}; done
INFO[2026-04-11T10:12:38.035336243Z] ignoring event
4 module=libcontainerd namespace=moby topic=/tasks/delete type=*events.TaskDelete" container=00560ebb3a1c5486de3c377b3973425581edf9cde769ebd4290d1f421cf34b7
INFO[2026-04-11T10:12:38.035497850Z] shim disconnected id=00560ebb3a1c5486de3c377b3973425581edf9cde769ebd4290d1f421cf34b74 names
pace=moby
INFO[2026-04-11T10:12:38.035563995Z] cleaning up after shim disconnected id=00560ebb3a1c5486de3c377b3973425581edf9cde769ebd4290d1f421cf34b74 names
pace=moby
INFO[2026-04-11T10:12:38.035585247Z] cleaning up dead shim id=00560ebb3a1c5486de3c377b3973425581edf9cde769ebd4290d1f421cf34b74 names
pace=moby
ERROR[2026-04-11T10:12:38.319911677Z] error removing container container=00560ebb3a1c5486de3c377b3973425581edf9cde769ebd4290d1f421cf34b7
4 error="removal of container 00560ebb3a1c5486de3c377b3973425581edf9cde769ebd4290d1f421cf34b74 is already in progress"
server1
INFO[2026-04-11T10:12:38.404502417Z] shim disconnected id=c69fdf74c895ee081922f86788c69f29d7f8228e08554076f4fc39268301ad62 names
pace=moby
INFO[2026-04-11T10:12:38.404655905Z] cleaning up after shim disconnected id=c69fdf74c895ee081922f86788c69f29d7f8228e08554076f4fc39268301ad62 names
pace=moby
INFO[2026-04-11T10:12:38.404695225Z] cleaning up dead shim id=c69fdf74c895ee081922f86788c69f29d7f8228e08554076f4fc39268301ad62 names
pace=moby
INFO[2026-04-11T10:12:38.404693467Z] ignoring event container=c69fdf74c895ee081922f86788c69f29d7f8228e08554076f4fc39268301ad6
2 module=libcontainerd namespace=moby topic=/tasks/delete type=*events.TaskDelete"
ERROR[2026-04-11T10:12:38.664187484Z] error removing container container=c69fdf74c895ee081922f86788c69f29d7f8228e08554076f4fc39268301ad6
2 error="removal of container c69fdf74c895ee081922f86788c69f29d7f8228e08554076f4fc39268301ad62 is already in progress"
server2
INFO[2026-04-11T10:12:38.755353527Z] shim disconnected id=f86936d4aef1fd6f971da24554d4e9ce6aba4ffb5f6c985285ccf8aec0d81cb0 names
pace=moby
INFO[2026-04-11T10:12:38.755416616Z] cleaning up after shim disconnected id=f86936d4aef1fd6f971da24554d4e9ce6aba4ffb5f6c985285ccf8aec0d81cb0 names
pace=moby
INFO[2026-04-11T10:12:38.755422602Z] cleaning up dead shim id=f86936d4aef1fd6f971da24554d4e9ce6aba4ffb5f6c985285ccf8aec0d81cb0 names
pace=moby
```


Experiment 10: Working with SonarQube

INTRODUCTION

WHAT IS SONARQUBE?

SonarQube is an open-source platform for continuous inspection of code quality. It performs automatic reviews with static analysis to detect bugs, code smells & security vulnerabilities.

HOW SONARQUBE SOLVES THE PROBLEM:

- **Continuous Inspection:** Scans code with every commit, providing immediate feedback.
- **Quality Gates:** Defines pass/fail criteria for code quality.
- **Technical Debt Quantification:** Measures effort needed to fix issues.
- **Multi-language Support:** Supports 20+ programming languages.
- **Visual Analytics:** Dashboard showing code quality metrics and trends.

HANDS-ON

Step-1:- Create a `docker-compose.yml`

```
nano docker-compose.yml
```

Paste This:

```
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
```

networks:

- sonarqube-lab

sonarqube:

image: sonarqube:lts-community

container_name: sonarqube

ports:

- "9000:9000"

environment:

SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube

SONAR_JDBC_USERNAME: sonar

SONAR_JDBC_PASSWORD: sonar

volumes:

- sonar-data:/opt/sonarqube/data
- sonar-extensions:/opt/sonarqube/extensions

depends_on:

- sonar-db

networks:

- sonarqube-lab

volumes:

sonar-db-data:

sonar-data:

sonar-extensions:

networks:

sonarqube-lab:

driver: bridge

```
root@LAPTOP-H7SV3T8B: /m  ×  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10# nano docker-compose.yml
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10# cat docker-compose.yml
version: '3.8'

services:
  sonar-db:
    image: postgres:13
    container_name: sonar-db
    environment:
      POSTGRES_USER: sonar
      POSTGRES_PASSWORD: sonar
      POSTGRES_DB: sonarqube
    volumes:
      - sonar-db-data:/var/lib/postgresql/data
    networks:
      - sonarqube-lab

  sonarqube:
    image: sonarqube:lts-community
    container_name: sonarqube
    ports:
      - "9000:9000"
    environment:
      SONAR_JDBC_URL: jdbc:postgresql://sonar-db:5432/sonarqube
      SONAR_JDBC_USERNAME: sonar
      SONAR_JDBC_PASSWORD: sonar
    volumes:
      - sonar-data:/opt/sonarqube/data
      - sonar-extensions:/opt/sonarqube/extensions
    depends_on:
      - sonar-db
    networks:
      - sonarqube-lab

volumes:
  sonar-db-data:
  sonar-data:
  sonar-extensions:

networks:
  sonarqube-lab:
```

Step-2:- Start Compose

```
docker-compose up -d
```

```
root@LAPTOP-H7SV3T8B: /m  ×  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10# docker-compose up -d
Creating network "experiment-10_sonarqube-lab" with driver "bridge"
Creating volume "experiment-10_sonar-db-data" with default driver
Creating volume "experiment-10_sonar-data" with default driver
Creating volume "experiment-10_sonar-extensions" with default driver
Creating sonar-db ... done
Creating sonarqube ... done
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10# |
```

Step-3:- Verify SonarQube is running via Logs

```
docker-compose logs -f sonarqube
```

```
root@LAPTOP-H7SV3T8B: ~ % docker-compose logs -f sonarqube
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10# docker-compose logs -f sonarqube
Attaching to sonarqube
sonarqube | 2026.04.18 20:15:50 INFO app[[o.s.a.AppFileSystem] Cleaning or creating temp directory /opt/sonarqube/temp
sonarqube | 2026.04.18 20:15:50 INFO app[[o.s.a.es.EsSettings] Elasticsearch listening on [HTTP: 127.0.0.1:9001, TCP: 127.0.0.1:44171]
sonarqube | 2026.04.18 20:15:50 INFO app[[o.s.a.ProcessLauncherImpl] Launch process[ELASTICSEARCH] from [/opt/sonarqube/elasticsearch]: /opt/sonarqube/
elasticsearch/bin/elasticsearch
sonarqube | 2026.04.18 20:15:50 INFO app[[o.s.a.SchedulerImpl] Waiting for Elasticsearch to be up and running
sonarqube | 2026.04.18 20:15:51 INFO es[[o.e.n.Node] version[7.17.15], pid[29], build[default/tar/0b8ecfb4378335f4689c4223d1f115f16bef3ba/2023-11-10T2
2:03:46.987399016Z], OS[linux/6.6.87.2-microsoft-standard-WSL2/amd64], JVM[Eclipse Adoptium/OpenJDK 64-Bit Server VM/17.0.16/17.0.16+8]
sonarqube | 2026.04.18 20:15:51 INFO es[[o.e.n.Node] JVM home [/opt/java/openjdk]
sonarqube | 2026.04.18 20:15:51 INFO es[[o.e.n.Node] JVM arguments [-XX:+UseG1GC, -Djava.io.tmpdir=/opt/sonarqube/temp, -XX:ErrorFile=/opt/sonarqube/lo
ges/es_hs_err_pid%p.log, -Des.networkaddress.cache.ttl=60, -Des.networkaddress.cache.negative.ttl=10, -XX:+AlwaysPreTouch, -Xss1m, -Djava.awt.headless=true,
-Dfile.encoding=UTF-8, -Djava.nosyms=true, -Djna.tmpdir=/opt/sonarqube/temp, -XX:-OmitStackTraceInFastThrow, -Dio.netty.noUnsafe=true, -Dio.netty.noKeySetOpti
mization=true, -Dio.netty.recycler.maxCapacityPerThread=0, -Dio.netty.allocation.numDirectArenas=0, -Dlog4j.shutdownHookEnabled=false, -Dlog4j2.disable.jmx=tr
ue, -Dlog4j2.formatMsgNoLookups=true, -Djava.locale.providers=COMPAT, -Dcom.redhat.fips=false, -Des.enforce.bootstrap.checks=true, -Xmx512m, -Xms512m, -XX:
MaxDirectMemorySize=256m, -XX:+HeapDumpOnOutOfMemoryError, -Des.path.home=/opt/sonarqube/elasticsearch, -Des.path.conf=/opt/sonarqube/temp/conf/es, -Des.dis
tribution.flavor=default, -Des.distribution.type=tar, -Des.bundled_jdk=false]
sonarqube | 2026.04.18 20:15:52 INFO es[[o.e.p.PluginsService] loaded module [analysis-common]
sonarqube | 2026.04.18 20:15:52 INFO es[[o.e.p.PluginsService] loaded module [lang-painless]
sonarqube | 2026.04.18 20:15:52 INFO es[[o.e.p.PluginsService] loaded module [parent-join]
sonarqube | 2026.04.18 20:15:52 INFO es[[o.e.p.PluginsService] loaded module [reindex]
sonarqube | 2026.04.18 20:15:52 INFO es[[o.e.p.PluginsService] loaded module [transport-netty4]
sonarqube | 2026.04.18 20:15:52 INFO es[[o.e.p.PluginsService] no plugins loaded
sonarqube | 2026.04.18 20:15:52 INFO es[[o.e.e.NodeEnvironment] using [1] data paths, mounts [/opt/sonarqube/data (/dev/sdd)], net usable_space [938.
8gb], net total_space [1006.8gb], types [ext4]
sonarqube | 2026.04.18 20:15:52 INFO es[[o.e.e.NodeEnvironment] heap size [512mb], compressed ordinary object pointers [true]
sonarqube | 2026.04.18 20:15:52 INFO es[[o.e.n.Node] node name [sonarqube], node ID [kXpwas5vS6yG6fQfuzg-qA], cluster name [sonarqube], roles [data_fro
zen_master, remote_cluster_client, data, data_content, data_hot, data_warm, data_cold, ingest]
sonarqube | 2026.04.18 20:15:53 INFO es[[o.e.t.NettyAllocator] creating NettyAllocator with the following configs: [name=unpooled, suggested_max_allocat
ion_size=256kb, factors={es.unsafe.use_unpooled_allocator=null, g1gc_enabled=true, g1gc_region_size=1mb, heap_size=512mb}]
sonarqube | 2026.04.18 20:15:54 INFO es[[o.e.i.r.RecoverySettings] using rate limit [40mb] with [default=0mb, read=0b, write=0b, max=0b]
sonarqube | 2026.04.18 20:15:54 INFO es[[o.e.d.DiscoveryModule] using discovery type [zen] and seed hosts providers [settings]
sonarqube | 2026.04.18 20:15:54 INFO es[[o.e.g.DanglingIndicesState] gateway.auto_import_dangling_indices is disabled, dangling indices will not be aut
omatically detected or imported and must be managed manually
sonarqube | 2026.04.18 20:15:54 INFO es[[o.e.n.Node] initialized
sonarqube | 2026.04.18 20:15:54 INFO es[[o.e.n.Node] starting ...
sonarqube | 2026.04.18 20:15:54 INFO es[[o.e.t.TransportService] publish_address {127.0.0.1:44171}, bound_addresses {127.0.0.1:44171}
sonarqube | 2026.04.18 20:15:54 INFO es[[o.e.b.BootstrapChecks] explicitly enforcing bootstrap checks
sonarqube | 2026.04.18 20:15:54 INFO es[[o.e.c.c.Coordinator] setting initial configuration to VotingConfiguration{kXpwas5vS6yG6fQfuzg-qA}
sonarqube | 2026.04.18 20:15:54 INFO es[[o.e.c.s.MasterService] elected-as-master [{1} nodes joined]{sonarqube:[kXpwas5vS6yG6fQfuzg-qA]{129h8q9IRHuTep
7oGdRy2A}[127.0.0.1:127.0.0.1:44171]{cdfhmsr} elect leader, _BECOME_MASTER_TASK_, _FINISH_ELECTION_}. term: 1, version: 1. delta: master node changed [pr
```

Step-4:- Login SonarQube

```
http://localhost:9000
```

- username: admin
- password: admin

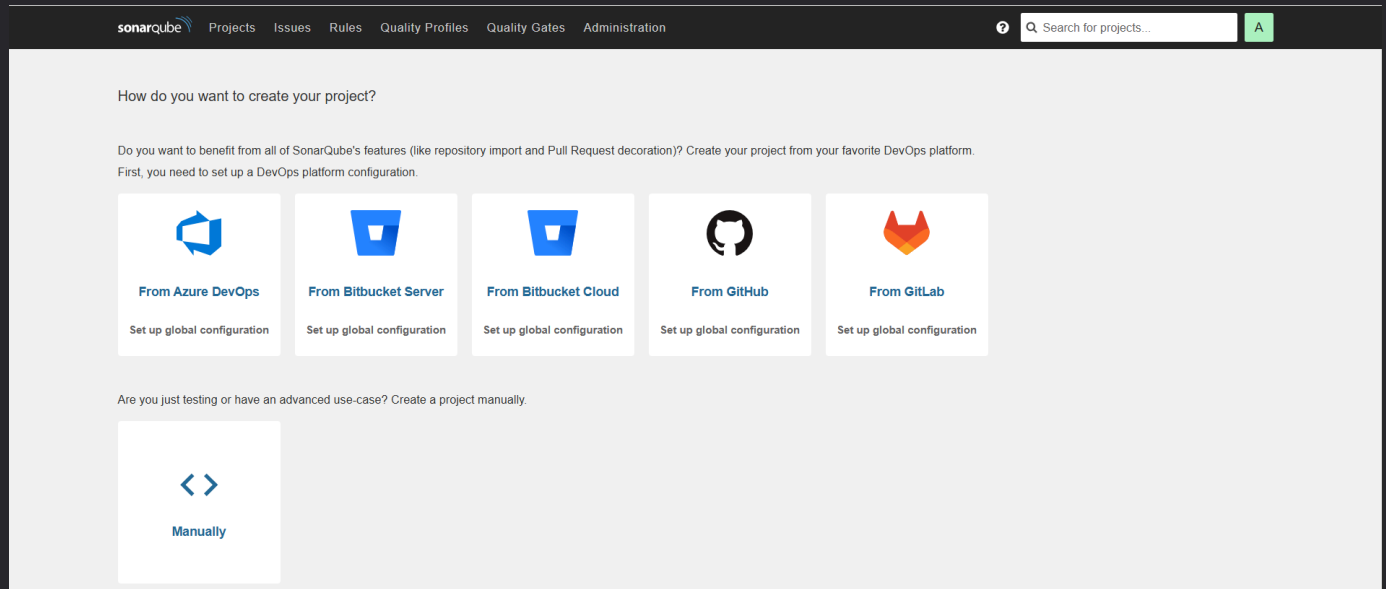
Log in to SonarQube

admin

Log in

Cancel

Step-5:- HomePage



The screenshot shows the SonarQube homepage. At the top, there is a navigation bar with links: sonarqube, Projects, Issues, Rules, Quality Profiles, Quality Gates, and Administration. A search bar on the right says "Search for projects..." with a green button labeled "A".

The main content area starts with the question "How do you want to create your project?". Below this, it says: "Do you want to benefit from all of SonarQube's features (like repository import and Pull Request decoration)? Create your project from your favorite DevOps platform. First, you need to set up a DevOps platform configuration."

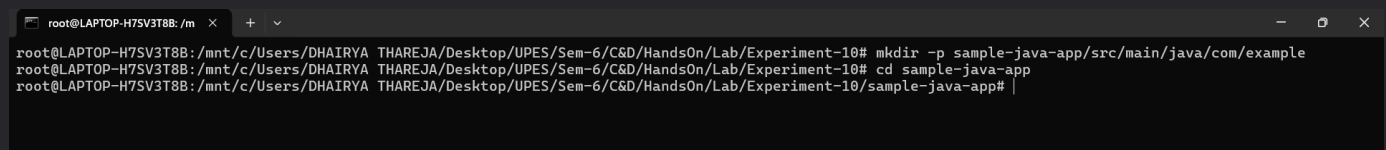
There are five cards representing different DevOps platforms:

- From Azure DevOps**: Set up global configuration
- From Bitbucket Server**: Set up global configuration
- From Bitbucket Cloud**: Set up global configuration
- From GitHub**: Set up global configuration
- From GitLab**: Set up global configuration

Below these cards, it says: "Are you just testing or have an advanced use-case? Create a project manually." and there is a card labeled **Manually** with a blue double arrow icon.

Step-6:- Create Sample Java App

```
mkdir -p sample-java-app/src/main/java/com/example  
cd sample-java-app
```



The screenshot shows a terminal window with the following commands and output:

```
root@LAPTOP-H7SV3T8B: /m  x  +  v  
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10# mkdir -p sample-java-app/src/main/java/com/example  
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10# cd sample-java-app  
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10/sample-java-app# |
```

Step-7:- Create Calculator.java

```
nano src/main/java/com/example/Calculator.java
```

```
package com.example;  
  
public class Calculator {  
  
    public int divide(int a, int b) {  
        return a / b;  
    }  
  
    public int add(int a, int b) {  
        int result = a + b;  
        int unused = 100;  
        return result;  
    }  
}
```

```
public String getUser(String userId) {  
    String query = "SELECT * FROM users WHERE id = " + userId;  
    return query;  
}  
  
public int multiply(int a, int b) {  
    int result = 0;  
    for (int i = 0; i < b; i++) {  
        result = result + a;  
    }  
    return result;  
}  
  
public int multiplyAlt(int a, int b) {  
    int result = 0;  
    for (int i = 0; i < b; i++) {  
        result = result + a;  
    }  
    return result;  
}  
  
public String getName(String name) {  
    return name.toUpperCase();  
}  
  
public void riskyOperation() {  
    try {  
        int x = 10 / 0;  
    } catch (Exception e) {  
    }  
}  
}
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10/sample-java-app# nano src/main/java/com/example/Calculator.java
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10/sample-java-app# cat src/main/java/com/example/Calculator.java
package com.example;

public class Calculator {

    public int divide(int a, int b) {
        return a / b;
    }

    public int add(int a, int b) {
        int result = a + b;
        int unused = 100;
        return result;
    }

    public String getUser(String userId) {
        String query = "SELECT * FROM users WHERE id = " + userId;
        return query;
    }

    public int multiply(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public int multiplyAlt(int a, int b) {
        int result = 0;
        for (int i = 0; i < b; i++) {
            result = result + a;
        }
        return result;
    }

    public String getName(String name) {
        return name.toUpperCase();
    }
}
```

Step-8:- Create `pom.xml`

```
nano pom.xml
```

```
<project xmlns="http://maven.apache.org/POM/4.0.0">

    <modelVersion>4.0.0</modelVersion>

    <groupId>com.example</groupId>
    <artifactId>sample-app</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>

        <sonar.projectKey>sample-java-app</sonar.projectKey>

        <sonar.host.url>http://localhost:9000</sonar.host.url>

        <sonar.login>YOUR_TOKEN</sonar.login>
    </properties>

    <build>

        <plugins>

            <plugin>

                <groupId>org.sonarsource.scanner.maven</groupId>

                <artifactId>sonar-maven-plugin</artifactId>

                <version>3.9.1.2184</version>

            </plugin>

        </plugins>

    </build>

</project>
```

```
</build>

</project>
```

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA.THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10/sample-java-app# nano pom.xml
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA.THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10/sample-java-app# cat pom.xml
<project xmlns="http://maven.apache.org/POM/4.0.0">
  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>
  <artifactId>sample-app</artifactId>
  <version>1.0-SNAPSHOT</version>

  <properties>
    <sonar.projectKey>sample-java-app</sonar.projectKey>
    <sonar.host.url>http://localhost:9000</sonar.host.url>
    <sonar.login>YOUR_TOKEN</sonar.login>
  </properties>

  <build>
    <plugins>
      <plugin>
        <groupId>org.sonarsource.scanner.maven</groupId>
        <artifactId>sonar-maven-plugin</artifactId>
        <version>3.9.1.2184</version>
      </plugin>
    </plugins>
  </build>
</project>
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA.THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-10/sample-java-app# |
```

Step-9:- Create Tokens Profile -> My Account -> Security -> TokenName: `scanner-token` -> Generate

Tokens

If you want to enforce security by not providing credentials of a real SonarQube user to run your code scan or to invoke web services, you can provide a User Token as a replacement of the user login. This will increase the security of your installation by not letting your analysis user's password going through your network.

Generate Tokens

Name	Type	Expires in	
Enter Token Name	Select Token Type	30 days	Generate

 New token "scanner-token" has been created. Make sure you copy it now, you won't be able to see it again!

 `squ_e0e40c377e34bd3b352aadfed54d1e4686996ac9`

Name	Type	Project	Last use	Created	Expiration	
scanner-token	User		Never	April 19, 2026	May 19, 2026	Revoke

Step-10:- Run Scanner

```
mvn sonar:sonar -Dsonar.login=YOUR_TOKEN
```


Step-13:- View Report

sample-app

main

×

Last analysis of this Branch had 1 warning

April 19, 2026 at 2:22 AM

Version 1.0-SNAPSHOT

🏠

Overview

Issues

Security Hotspots

Measures

Code

Activity

Project Settings

Project Information

🔔 To benefit from more of SonarQube's features, set up analysis in your favorite CI.

×

QUALITY GATE STATUS

Passed

All conditions passed.

MEASURES

New Code

Overall Code

1🐛 Bugs

Reliability

D

0🔒 Vulnerabilities

Security

A

1🔥 Security Hotspots

0.0%

Reviewed

Security Review

E

1h 2minDebt

7🧴 Code Smells

Maintainability

A

0.0%

Coverage on 21 Lines to cover

Unit Tests

0.0%

Duplications on 58 Lines

Duplicated Blocks

0

Experiment 11: Orchestration via Docker Swarm

INTRODUCTION

THE PROGRESSION PATH

docker run	→	Docker Compose	→	Docker Swarm	→	Kubernetes
Single container		Multi-container		Orchestration		Advanced
		(single host)		(basic)		orchestration

This experiment focuses on: Moving from Compose → Swarm

HANDS-ON

Step-1:- Create a `docker-compose.yml`

```
nano docker-compose.yml
```

Paste This:

```
version: '3.9'

services:
  db:
    image: mysql:5.7
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql
```

```
wordpress:

  image: wordpress:latest

  depends_on:

    - db

  ports:

    - "8080:80"

  restart: always

  environment:

    WORDPRESS_DB_HOST: db:3306

    WORDPRESS_DB_USER: wpuser

    WORDPRESS_DB_PASSWORD: wppass

    WORDPRESS_DB_NAME: wordpress

  volumes:

    - wp_data:/var/www/html
```

```
volumes:

  db_data:

  wp_data:
```

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# cat docker-compose.yml
# docker-compose.yml (from Experiment 6)
version: '3.9'

services:
  db:
    image: mysql:5.7
    restart: always
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    depends_on:
      - db
    ports:
      - "8080:80"
    restart: always
    environment:
      WORDPRESS_DB_HOST: db:3306
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html

volumes:
  db_data:
  wp_data:
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-2:- Initialize Swarm

```
docker swarm init
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker swarm init --advertise-addr 172.19.98.109
INFO[2026-04-29T08:46:44.701033188Z] Listening for connections addr="[:]:2377" module=node node.id=xk430bloba99f2deiy7u5b14s proto=tcp
INFO[2026-04-29T08:46:44.7071101208Z] Listening for local connections addr=/var/run/docker/swarm/control.sock module=node node.id=xk430bloba99f2deiy7u5b14s proto=unix
INFO[2026-04-29T08:46:44.707198377Z] 56e69b553799b79 switched to configuration voters={} module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.707268451Z] 56e69b553799b79 became follower at term 0 module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.707274638Z] newRaft 56e69b553799b79 [peers: [], term: 0, commit: 0, applied: 0, lastindex: 0, lastterm: 0] module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.707280070Z] 56e69b553799b79 became follower at term 1 module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.707320897Z] 56e69b553799b79 switched to configuration voters=(391366445138287481) module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.710917350Z] 56e69b553799b79 switched to configuration voters=(391366445138287481) module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.710995730Z] 56e69b553799b79 is starting a new election at term 1 module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.711007010Z] 56e69b553799b79 became candidate at term 2 module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.711814203Z] 56e69b553799b79 received MsgVoteResp from 56e69b553799b79 at term 2 module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.711845974Z] 56e69b553799b79 has received 1 MsgVoteResp votes and 0 vote rejections module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.711883675Z] 56e69b553799b79 became leader at term 2 module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.711902201Z] raft.node: 56e69b553799b79 elected leader 56e69b553799b79 at term 2 module=raft node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.713499803Z] Creating default ingress network module=node node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.716352332Z] leadership changed from not yet part of a raft cluster to xk430bloba99f2deiy7u5b14s module=node node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:44.716421050Z] dispatcher starting module=dispatcher node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:45.211669017Z] manager selected by agent for new session: { } module=node/agent node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:45.211929745Z] waiting 0s before registering session module=node/agent node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:45.230134954Z] worker xk430bloba99f2deiy7u5b14s was successfully registered method="(*Dispatcher).register"
INFO[2026-04-29T08:46:45.235020027Z] initialized VXLAN UDP port to 4789 module=node node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:46:45.235083750Z] Initializing Libnetwork Agent advertise-addr=172.19.98.109 data-path-addr= listen-addr=0.0.0.0 local-ad
dr=172.19.98.109 network-control-plane-mtu=1500 remote-addr-list=""
INFO[2026-04-29T08:46:45.235331118Z] New memberlist node - Node:LAPTOP-H7SV3T8B will use memberlist nodeID:a409a3bde0ee with config:&{NodeID:a409a3bde0ee Ho
stname:LAPTOP-H7SV3T8B BindAddr:0.0.0.0 AdvertiseAddr:172.19.98.109 BindPort:0 Keys:[[146 61 198 206 65 226 74 86 167 2 148 191 163 25 76 1] [109 177 13 196
197 181 49 80 73 18 134 214 9 254 186 27] [66 216 36 124 112 137 111 138 159 161 216 81 161 191 120 226]] PacketBufferSize:1400 reapEntryInterval:1800000000
0000 reapNetworkInterval:18250000000000 rejoinClusterDuration:1000000000000 rejoinClusterInterval:600000000000 StatsPrintPeriod:5m0s HealthPrintPeriod:1m0s}
INFO[2026-04-29T08:46:45.235828239Z] Node a409a3bde0ee/172.19.98.109, joined gossip cluster
INFO[2026-04-29T08:46:45.235869733Z] Node a409a3bde0ee/172.19.98.109, added to nodes list
INFO[2026-04-29T08:46:45.247641734Z] cluster update event module=dispatcher node.id=xk430bloba99f2deiy7u5b14s
Swarm initialized: current node (xk430bloba99f2deiy7u5b14s) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-4226w11z12m5h5k7ddxwlswgme9o1kz3d5i9m62d0hszh1jujn-422a1nwudoald50uiklw2pf3 172.19.98.109:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

Step-3:- Verify Swarm is Active

```
docker node ls
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker node ls
ID HOSTNAME STATUS AVAILABILITY MANAGER STATUS ENGINE VERSION
xk430bloba99f2deiy7u5b14s * LAPTOP-H7SV3T8B Ready Active Leader 29.1.3
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-4:- Deploy as a stack

```
docker stack deploy -c docker-compose.yml wpstack
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker stack deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_db
Creating service wpstack_wordpress
INFO[2026-04-29T08:55:55.560804321Z] initialized VXLAN UDP port to 4789 module=node node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T08:55:55.5677972663Z] sbJoin: gwep4 '-->', gwep6 '-->' eid=1a837c8ee44b ep=wpstack_default-endpoint net=wpstack_default nid=yym9
7uzzjhsj
INFO[2026-04-29T08:55:55.915785782Z] connecting to shim ddb64edfc3d5f9e2f57e2d05f433ff32a79a6a099fde6b0e51e3d0621b20a760 address="/unix:///run/containerd/s/
f10d46142c61d7f1ba33b10806d0264238dac7a842b6752bc15c7c906bebb8b1f" namespace=moby protocol=tttrpc version=3
INFO[2026-04-29T08:55:56.146775229Z] sbJoin: gwep4 '-->'63717c4d9f26', gwep6 '-->' eid=63717c4d9f26 ep=gateway_77d65ab09558 net=docker_gwbridge nid=31d18b
044efd
INFO[2026-04-29T08:55:56.155384102Z] sbJoin: gwep4 '-->'63717c4d9f26', gwep6 '-->' eid=635176651ef6 ep=wpstack_db.1.dy39hdkr2gdze74azunf9l8xj net=wpstack_
default nid=yym97uzzjhsj
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# INFO[2026-04-29T08:55.701748063Z] connecting to shim
34ea7885596576761ce2992b673fd9e9c1c9fa81f759493e541bc793550218393 address="/unix:///run/containerd/s/199e77e5bb96a4aa1b0412c2ba75e6187359bf90171b985a4319e6fd9
e95af9c9" namespace=moby protocol=tttrpc version=3
INFO[2026-04-29T08:55:58.953345798Z] sbJoin: gwep4 '-->'fdb5585d214d', gwep6 '-->' eid=fdb5585d214d ep=gateway_c9907e76b209 net=docker_gwbridge nid=31d18b
044efd
INFO[2026-04-29T08:55:58.963888232Z] sbJoin: gwep4 '-->'fdb5585d214d', gwep6 '-->' eid=b66123ef3a0b ep=wpstack_wordpress.1.lglgw5zbdgiol6x5tgr68xi1 net=w
pstack_default nid=yym97uzzjhsj
INFO[2026-04-29T08:55:59.067279069Z] sbJoin: gwep4 'fdb5585d214d'-->'fdb5585d214d', gwep6 '-->' eid=9635e27f9f6c ep=wpstack_wordpress.1.lglgw5zbdgiol6x5tgr
68xi1 net=ingress nid=op31gyamknn3

root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-5:- Verify by listing Services

```
root@LAPTOP-H7SV3T8B: /m  x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker service ls
ID            NAME          MODE          REPLICAS  IMAGE          PORTS
h99num7tg136  wpstack_db    replicated    1/1        mysql:5.7
0eeod69jf355  wpstack_wordpress replicated    0/1        wordpress:latest *:8080->80/tcp
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-6:- Details for container service

```
docker service ps wpstack_wordpress
```

```
root@LAPTOP-H7SV3T8B: /m  x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker service ps wpstack_wordpress
ID            NAME          IMAGE          NODE          DESIRED STATE  CURRENT STATE          ERROR          PORTS
z94qpw8sp5j  wpstack_wordpress.1  wordpress:latest  LAPTOP-H7SV3T8B  Running        Running 10 seconds ago
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

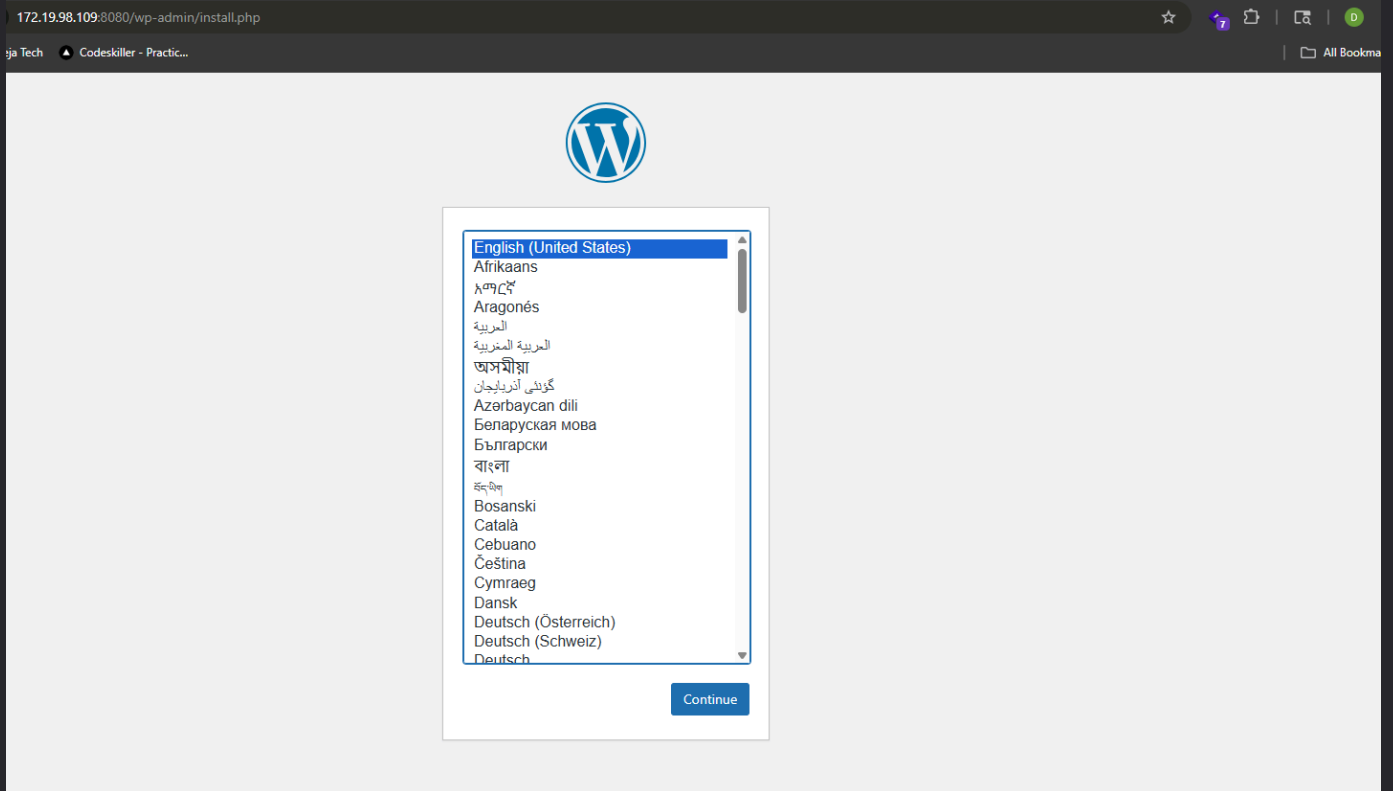
Step-7:- List Containers

```
docker ps
```

```
root@LAPTOP-H7SV3T8B: /m  x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
cd4ed1de5e5d   wordpress:latest  "docker-entrypoint.s..."  48 seconds ago  Up 29 seconds  80/tcp
cde5f090efc7   mysql:5.7        "docker-entrypoint.s..."  About a minute ago  Up About a minute  3306/tcp, 33060/tcp
013d92e4693c   wordpress:latest  "docker-entrypoint.s..."  4 days ago     Up 4 minutes   0.0.0.0:8082->80/tcp, [::]:8082->80/tcp
632383334c57   mysql:5.7        "docker-entrypoint.s..."  4 days ago     Up 4 minutes   3306/tcp, 33060/tcp
e19ed8631e8b   jenkins/jenkins:lts  "/usr/bin/tini -- /u..."  11 days ago    Up 4 minutes   0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp
16a9aefbac86   assignment-1_backend  "docker-entrypoint.s..."  3 weeks ago    Up 4 minutes   project_backend
93764d07652d   assignment-1_db     "docker-entrypoint.s..."  3 weeks ago    Up 4 minutes (healthy)  project_db
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-8:- Verify application on Containers

```
http://localhost:8080/
```



Step-9:- Scale application Containers

```
docker service scale wpstack_wordpress=3
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker service scale wpstack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 1 out of 3 tasks
1/3: running [=====]
2/3: assigned [=====]
3/3: assigned [=====]
overall progress: 1 out of 3 tasks
1/3: running [=====]
2/3: starting [=====]
3/3: starting [=====]
overall progress: 1 out of 3 tasks
1/3: running [=====]
2/3: starting [=====]
3/3: starting [=====]
INFO[2026-04-29T09:05:02.058562717Z] sbJoin: gwep4 '-->'bcc7a071de22', gwep6 '-->' eid=bcc7a071de22 ep=gateway_8a24810be88c net=docker_gwbridge nid=31d18b044efb
INFO[2026-04-29T09:05:02.068503457Z] sbJoin: gwep4 '-->'bcc7a071de22', gwep6 '-->' eid=03a398aad907 ep=wpstack_wordpress.3.a5u8cqjwe53mrrc8ourzh0b96 net=i
ngress nid=op3lgyamknn3
overall progress: 3 out of 3 tasks
1/3: running [=====]
2/3: running [=====]
3/3: running [=====]
verify: Service wpstack_wordpress converged
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-10:- Verify Scalling

```
docker service ls
```

```
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker service ls
ID            NAME          MODE          REPLICAS  IMAGE          PORTS
1w68qt6miu57 wpstack_db    replicated    1/1        mysql:5.7
hls39yjhri03 wpstack_wordpress replicated    3/3        wordpress:latest *:8080->80/tcp
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-11:- List Wordpress Containers

```
docker ps | grep wordpress
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker ps | grep wordpress
f6139bf29516 wordpress:latest "docker-entrypoint.s..." 53 seconds ago Up 52 seconds 80/tcp wpstack_wordpress.2.jsymwlbk
tw6y46qo9eofrlxpk
4574c5e9ac5c wordpress:latest "docker-entrypoint.s..." 53 seconds ago Up 52 seconds 80/tcp wpstack_wordpress.3.a5u8cqjw
e53mrrc8ourzh0b96
01b1db652b89 wordpress:latest "docker-entrypoint.s..." 5 minutes ago Up 5 minutes 80/tcp wpstack_wordpress.1.vlamza6i
7n086ipsw5c8adbxl
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-12:- Kill any Container

```
docker kill <container-id>
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker kill f613
INFO[2026-04-29T09:08:08.227646707Z] shim disconnected id=f6139bf295160738c6ad68b924ad6b1f4209213ee7f22461dcb4236e4732dd9a names
pace=moby
INFO[2026-04-29T09:08:08.227720034Z] cleaning up after shim disconnected id=f6139bf295160738c6ad68b924ad6b1f4209213ee7f22461dcb4236e4732dd9a names
pace=moby
INFO[2026-04-29T09:08:08.227733411Z] cleaning up dead shim id=f6139bf295160738c6ad68b924ad6b1f4209213ee7f22461dcb4236e4732dd9a names
pace=moby
INFO[2026-04-29T09:08:08.227735754Z] ignoring event container=f6139bf295160738c6ad68b924ad6b1f4209213ee7f22461dcb4236e4732dd9
a module=libcontainerd namespace=moby topic=/tasks/delete type="*events.TaskDelete"
ERROR[2026-04-29T09:08:08.584499003Z] fatal task error error="task: non-zero exit (137)" module=node/agent/taskmanager node.id=x
k430b1oba99f2dei7u5b14s service.id=hls39yjhri03mmmkc0snzvh3 task.id=jsymwlbktw6y46qo9eofrlxpk
f613
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-13:- Watch Swarm Recreating Contrainer

```
docker service ps wpstack_wordpress
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker service ps wpstack_wordpress
ID NAME IMAGE NODE DESIRED STATE CURRENT STATE ERROR PORTS
vlamza6i7n08 wpstack_wordpress.1 wordpress:latest LAPTOP-H7SV3T8B Running Running 8 minutes ago
vymkb2a7olce wpstack_wordpress.2 wordpress:latest LAPTOP-H7SV3T8B Running Running 24 seconds ago
jsymwlbktw6y \_ wpstack_wordpress.2 wordpress:latest LAPTOP-H7SV3T8B Shutdown Failed 30 seconds ago "task: non-zero exit (137)"
a5u8cqjwe53m wpstack_wordpress.3 wordpress:latest LAPTOP-H7SV3T8B Running Running 3 minutes ago
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-14:- Verify Container running

```
docker ps | grep wordpress
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker ps | grep wordpress
18b16f5fb594 wordpress:latest "docker-entrypoint.s..." 52 seconds ago Up 46 seconds 80/tcp wpstack_wordpress.2.vymkb2a7
olceviy9rg6dfrzvh
4574c5e9ac5c wordpress:latest "docker-entrypoint.s..." 3 minutes ago Up 3 minutes 80/tcp wpstack_wordpress.3.a5u8cqjw
e53mrrc8ourzh0b96
01b1db652b89 wordpress:latest "docker-entrypoint.s..." 8 minutes ago Up 8 minutes 80/tcp wpstack_wordpress.1.vlamza6i
7n086ipsw5c8adbxl
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# |
```

Step-15:- Remove Stack

docker stack rm wpstack

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker stack rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# ERROR[2026-04-29T09:09:31.182329140Z] Failed to allocate
network resources for node xk430bloba99f2deiy7u5b14s error="could not find network allocator state for network n2cbqbsvw2vecscmbu2swgb6n" module=node node.
id=xk430bloba99f2deiy7u5b14s
WARN[2026-04-29T09:09:31.182643614Z] network n2cbqbsvw2vecscmbu2swgb6n should be removed, but still has active attachments module=node/agent node.id=xk430b
loba99f2deiy7u5b14s
INFO[2026-04-29T09:09:31.182773897Z] initialized VXLAN UDP port to 4789 module=node node.id=xk430bloba99f2deiy7u5b14s
ERROR[2026-04-29T09:09:31.306599158Z] Failed to allocate network resources for node xk430bloba99f2deiy7u5b14s error="could not find network allocator state
for network n2cbqbsvw2vecscmbu2swgb6n" module=node node.id=xk430bloba99f2deiy7u5b14s
ERROR[2026-04-29T09:09:31.306677307Z] Failed to allocate network resources for node xk430bloba99f2deiy7u5b14s error="could not find network allocator state
for network n2cbqbsvw2vecscmbu2swgb6n" module=node node.id=xk430bloba99f2deiy7u5b14s
ERROR[2026-04-29T09:09:31.306736130Z] Failed to allocate network resources for node xk430bloba99f2deiy7u5b14s error="could not find network allocator state
for network n2cbqbsvw2vecscmbu2swgb6n" module=node node.id=xk430bloba99f2deiy7u5b14s
ERROR[2026-04-29T09:09:31.306766920Z] Failed to allocate network resources for node xk430bloba99f2deiy7u5b14s error="could not find network allocator state
for network n2cbqbsvw2vecscmbu2swgb6n" module=node node.id=xk430bloba99f2deiy7u5b14s
ERROR[2026-04-29T09:09:31.410880522Z] Failed to allocate network resources for node xk430bloba99f2deiy7u5b14s error="could not find network allocator state
for network n2cbqbsvw2vecscmbu2swgb6n" module=node node.id=xk430bloba99f2deiy7u5b14s
ERROR[2026-04-29T09:09:31.431675679Z] Failed to allocate network resources for node xk430bloba99f2deiy7u5b14s error="could not find network allocator state
for network n2cbqbsvw2vecscmbu2swgb6n" module=node node.id=xk430bloba99f2deiy7u5b14s
INFO[2026-04-29T09:09:35.331456375Z] shim disconnected id=4574c5e9ac5c46c0c773607b121824fa6baf3ea8916aef70ccd67b93b0f8c240 names
pace=moby
INFO[2026-04-29T09:09:35.331515396Z] cleaning up after shim disconnected id=4574c5e9ac5c46c0c773607b121824fa6baf3ea8916aef70ccd67b93b0f8c240 names
pace=moby
INFO[2026-04-29T09:09:35.331520983Z] cleaning up dead shim id=4574c5e9ac5c46c0c773607b121824fa6baf3ea8916aef70ccd67b93b0f8c240 names
pace=moby
```

Step-16:- Verify Service Removal

docker service ls

docker ps

```
root@LAPTOP-H7SV3T8B: /m  x  +  v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11# docker service ps wpstack_wordpress
ID NAME IMAGE NODE DESIRED STATE CURRENT STATE ERROR PORTS
v1amza6i7n08 wpstack_wordpress.1 wordpress:latest LAPTOP-H7SV3T8B Running Running 8 minutes ago
vymkb2a7olce wpstack_wordpress.2 wordpress:latest LAPTOP-H7SV3T8B Running Running 24 seconds ago
jsymwlkbtw6y \_ wpstack_wordpress.2 wordpress:latest LAPTOP-H7SV3T8B Shutdown Failed 30 seconds ago "task: non-zero exit (137)"
a5u8cqjwe53m wpstack_wordpress.3 wordpress:latest LAPTOP-H7SV3T8B Running Running 3 minutes ago
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-11#
```

CONCLUSION

SUMMARY

You started with	You can now do
Single container (docker run)	Multi-container (Compose)
Manual scaling	One-command scaling (scale)
Manual recovery	Automatic self-healing
Single host	Multi-host cluster ready

FINAL TAKEAWAY

Compose defines the application. Swarm runs it reliably.

QUICK REFERENCE CARD

Initialize Swarm

```
docker swarm init
```

Deploy stack

```
docker stack deploy -c docker-compose.yml <stack-name>
```

List services

```
docker service ls
```

Scale service

```
docker service scale <stack-name_service-name>=<replicas>
```

See service tasks

```
docker service ps <service-name>
```

Remove stack

```
docker stack rm <stack-name>
```

Leave Swarm (if needed)

```
docker swarm leave --force
```



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Experiment 12: Kubernetes

INTRODUCTION

WHY KUBERNETES OVER DOCKER SWARM?

Reason	Explanation
Industry standard	Most companies use Kubernetes
Powerful scheduling	Automatically decides where to run your app
Large ecosystem	Many tools and plugins available
Cloud-native support	Works on AWS, Google Cloud, Azure, etc.

CORE KUBERNETES CONCEPTS (SIMPLE EXPLANATION)

Docker Concept	Kubernetes Equivalent	What it means
Container	Pod	A pod is a group of one or more containers. Smallest unit in K8s.
Compose service	Deployment	Describes how your app should run (e.g., 2 copies, which image to use)
Load balancing	Service	Exposes your app to the outside world or other pods
Scaling	ReplicaSet	Ensures a certain number of pod copies are always running

HANDS-ON

Step-1:- Create a `wordpress-deployment.yaml`

```
nano wordpress-deployment.yaml
```

Paste This:

```
apiVersion: apps/v1

kind: Deployment

metadata:
  name: wordpress

spec:
  replicas: 2
  selector:
    matchLabels:
      app: wordpress
  template:
    metadata:
      labels:
        app: wordpress
    spec:
      containers:
      - name: wordpress
        image: wordpress:latest
        ports:
        - containerPort: 80
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# nano wordpress-deployment.yaml
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# cat wordpress-deployment.yaml

apiVersion: apps/v1
kind: Deployment
metadata:
  name: wordpress
spec:
  replicas: 2
  selector:
    matchLabels:
      app: wordpress
  template:
    metadata:
      labels:
        app: wordpress
    spec:
      containers:
      - name: wordpress
        image: wordpress:latest
        ports:
        - containerPort: 80
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# |
```

Step-2:- Apply Deployment

```
kubectl apply -f wordpress-deployment.yaml
```

```
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# kubectl apply -f wordpress-deployment.yaml
deployment.apps/wordpress created
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# INFO[2026-04-29T17:04:35.689145247Z] NetworkDB stats LAP
TOP-H7SV3T8B(cce4c3d70f4f) - netID:op31gyamknn34bn2i5dwf2bpm leaving:false netPeers:1 entries:1 Queue qLen:0+0 netMsg/s:0
|
```

Step-3:- Create `wordpress-service.yaml`

```
nano wordpress-service.yaml
```

Paste this:-

```
apiVersion: v1
kind: Service
metadata:
  name: wordpress-service
spec:
  type: NodePort
  selector:
    app: wordpress
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```

```
root@LAPTOP-H7SV3T8B: /m  x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# nano wordpress-service.yaml
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# cat wordpress-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: wordpress-service
spec:
  type: NodePort
  selector:
    app: wordpress
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# |
```

Step-4:- Deploy Service

```
kubectl apply -f wordpress-service.yaml
```

```
root@LAPTOP-H7SV3T8B: /m  x + v
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# kubectl apply -f wordpress-service.yaml
service/wordpress-service created
root@LAPTOP-H7SV3T8B:/mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# |
```

Step-5:- Get pods

```
kubectl get pods
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
wordpress-7d6f6db8d8-mckkj         1/1     Running   0           2m35s
wordpress-7d6f6db8d8-qvds8         1/1     Running   0           2m35s
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# |

```

Step-6:- Check service

```
kubectl get svc
```

```

root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# kubectl get svc
NAME            TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes      ClusterIP     10.43.0.1    <none>        443/TCP          3m48s
wordpress-service NodePort       10.43.142.195 <none>        80:30007/TCP     35s
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/HandsOn/Lab/Experiment-12# |

```

Step-7:- Verify On Browser

```
http://localhost:8080/
```

localhost:8080/wp-admin/setup-config.php

Thareja Tech Codeskiller - Practic...



English (United States)

Afrikaans

አማርኛ

Aragonés

العربية

العربية المغربية

অসমীয়া

گۆنئی آذربایجان

Azərbaycan dili

Беларуская мова

Български

বাংলা

བོད་སྐད་

Bosanski

Català

Cebuano

Čeština

Cymraeg

Dansk

Deutsch (Österreich)

Deutsch (Schweiz)

Deutsch

Continue

Step-8:- Scale Deployment

```
kubectl scale deployment wordpress --replicas=4
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Hands0n/Lab/Experiment-12# kubectl scale deployment wordpress --replicas=4
deployment.apps/wordpress scaled
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Hands0n/Lab/Experiment-12# |
```

Step-9:- Get Pods

```
kubectl get pods
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Hands0n/Lab/Experiment-12# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
wordpress-7d6f6db8d8-b677g         1/1     Running   0           22s
wordpress-7d6f6db8d8-mckkj         1/1     Running   0           11m
wordpress-7d6f6db8d8-qvds8         1/1     Running   0           11m
wordpress-7d6f6db8d8-s9kgL         1/1     Running   0           22s
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Hands0n/Lab/Experiment-12# |
```

Step-10:- Delete Pods to check Self-healing

```
kubectl delete pod <pod-name>

kubectl get pods
```

```
root@LAPTOP-H7SV3T8B: /m x + v
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Hands0n/Lab/Experiment-12# kubectl delete pod wordpress-7d6f6db8d8-b677g
pod "wordpress-7d6f6db8d8-b677g" deleted from default namespace
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Hands0n/Lab/Experiment-12# kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
wordpress-7d6f6db8d8-gfx6w         1/1     Running   0           5s
wordpress-7d6f6db8d8-mckkj         1/1     Running   0           12m
wordpress-7d6f6db8d8-qvds8         1/1     Running   0           12m
wordpress-7d6f6db8d8-s9kgL         1/1     Running   0           90s
root@LAPTOP-H7SV3T8B: /mnt/c/Users/DHAIRYA THAREJA/Desktop/UPES/Sem-6/C&D/Hands0n/Lab/Experiment-12# |
```

INSIGHTS

WHEN TO USE WHICH TOOL

Tool	Best for
k3d	Quick learning on your laptop
Minikube	Single-node cluster testing
kubeadm	Real, production-style cluster

SUMMARY OF COMMANDS (CHEAT SHEET)

Goal	Command
Apply a YAML file	<code>kubectl apply -f file.yaml</code>
See all pods	<code>kubectl get pods</code>
See all services	<code>kubectl get svc</code>
Scale a deployment	<code>kubectl scale deployment <name> --replicas=N</code>
Delete a pod	<code>kubectl delete pod <pod-name></code>
See all nodes	<code>kubectl get nodes</code>

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